OECD SME and Entrepreneurship Papers

Data and methods for entrepreneurial ecosystem diagnostics

By Roberto Crotti, Jonathan Potter, Pablo Shah and Erik Stam

OECD

OECD SME and Entrepreneurship Papers

Data and methods for entrepreneurial ecosystem diagnostics

By Roberto Crotti, Jonathan Potter, Pablo Shah and Erik Stam

The levels of impactful entrepreneurship generated in different economies depend on the quality of a range of supporting conditions, which can be analysed using the entrepreneurial ecosystem concept. This sees entrepreneurship as being embedded in a system involving various institutional conditions, resource endowments, interactions and feedbacks. To support policy makers diagnose the vitality of entrepreneurial ecosystems and identify policy priorities, the OECD has developed an *Entrepreneurial Ecosystems Diagnostics* report with comparative statistics on entrepreneurship levels and drivers for OECD countries. This companion paper presents details of the report's methodological approach. It covers the conceptual framework, the choice of indicators, data normalisation, aggregation, missing value imputation, and sensitivity analyses and robustness tests with respect to alternative methodologies. The paper concludes by outlining the methodological limitations and directions for future development of the diagnostics.

JEL codes: L22, L26, M21, C38
Keywords: Entrepreneurial ecosystem, Entrepreneurship, Economic development, Economic policy, Entrepreneurship policy

ABOUT THE OECD

The OECD is a multi-disciplinary inter-governmental organisation with member countries which engages in its work an increasing number of non-members from all regions of the world. The Organisation's core mission today is to help governments work together towards a stronger, cleaner, fairer global economy. Through its network of specialised committees and working groups, the OECD provides a setting where governments compare policy experiences, seek answers to common problems, identify good practice, and co-ordinate domestic and international policies. More information available: www.oecd.org.

ABOUT THE SMEs AND ENTREPRENEURS PAPERS

The series provides comparative evidence and analysis on SME and entrepreneurship performance and trends and on a broad range of policy areas, including SME financing, innovation, productivity, skills, internationalisation, and others.

OECD Working Papers should not be reported as representing the official views of the OECD or of its member countries. The opinions expressed and arguments employed are those of the author(s).

Working Papers describe preliminary results or research in progress by the author(s) and are published to stimulate discussion on a broad range of issues on which the OECD works. Comments on Working Papers are welcome, and may be sent to the Centre for Entrepreneurship, SMEs, Regions and Cities, OECD, 2 rue André-Pascal, 75775 Paris Cedex 16, France.

This paper is authorised for publication by Lamia Kamal-Chaoui, Director, Centre for Entrepreneurship, SMEs, Regions and Cities, OECD.

Cover image: Orbon Alija / Getty Images Plus.

Acknowledgements

This working paper was prepared by the OECD Centre for Entrepreneurship, SMEs, Regions and Cities (CFE) led by Lamia Kamal-Chaoui (Director). The report contributes to the item "Developing ecosystems and policies for startups" of the programme of work of the OECD Committee on SMEs and Entrepreneurship (CSMEE).

The lead authors of this working paper are Roberto Crotti (Policy Analyst, CFE), Jonathan Potter (Head of unit, CFE), Pablo Shah (Policy Analyst, CFE), and Erik Stam (Professor, Utrecht University). Jonathan Potter directed the project. Lucia Cusmano (Head of the SMEs and Entrepreneurship Division, CFE) supervised the work.

The paper benefitted from comments and inputs from delegates at the 6th and 7th sessions of the CSMEE and from the experts and policymakers who intervened at the CSMEE workshops on entrepreneurial ecosystem benchmarking in December 2023 and December 2024. We are particularly grateful to the workshops' speakers, which included Mark Hart (University of Warwick), Cesare Riillo, (STATEC), Arnis Sauka (Stockholm School of Economics), Rolf Sternberg (Leibniz University of Hannover), Roy Thurik (Erasmus School of Economics), Lucien Vijverberg (Ministry of Economy of the Netherlands), Nataliya Wright (Columbia Business School), Bernd Wurth (University of Glasgow).

Further productive comments were provided by Nadim Ahmad (Deputy Director, CFE), Rudiger Ahrend (Head of Economic Analysis, Data and Statistics Division, CFE), Annabelle Mourougane (Head of Division, Statistics and Data Directorate) and Gueram Sargsyan (Statistician, Statistics and Data Directorate).

The authors are also grateful to Heather Mortimer-Charoy (Assistant, CFE) and Juan Mendez (Assistant, CFE) for technical support, and to Alix Philouze (Communications Co-ordinator, CFE) and Jack Waters (Product and Project Co-ordinator, CFE) for preparation of the publication.

Table of contents

FIGURES

Executive Summary

The OECD has produced comparative international entrepreneurship data for almost two decades, however, a comprehensive, comparable set of metrics to benchmark entrepreneurial ecosystem drivers and outcomes across all OECD countries is still lacking. To fill this gap, the OECD has launched a pilot version of the Entrepreneurial Ecosystem Diagnostics Report, which provides measurements of inputs (elements), output and variation of entrepreneurial ecosystems of all 38 OECD economies.

Concept and framework

This working paper details the methodology used to build the Entrepreneurial Ecosystem Diagnostics Report's framework and data. The framework, drawing from the existing literature, defines the concept of entrepreneurial ecosystem as the set of factors and conditions that encourage 'productive entrepreneurship', that is, the creation of new ventures that contribute to the national economy's productivity, innovation, and/or employment.

The framework defines 10 elements, grounded in the literature, that constitute they main inputs in determining strong ecosystems. These elements are: 1. Institutions, 2. Entrepreneurial Culture, 3. Networks, 4. Infrastructure, 5. Markets, 6. Finance, 7. Knowledge, 8. Talent, 9. Leadership, 10. Intermediate services. There is ongoing feedback between inputs and outputs (e.g. startup rates, startup growth and exit). However, for the purpose of this framework inputs and outputs are separated for analytical clearness.

Measurement

The Entrepreneurial Ecosystem Diagnostics provides a dashboard of indicators to track entrepreneurship outputs and social, regional variation. These are single indicators collected from OECD and other sources.

Inputs, instead, are measured through ten composite indices, which combine data from multiple sources. The main purpose of this paper is to describe the metrologies and rationales for building such set of composite indices. The main features of the methodology are:

- **Selection.** The selection of indicators composing each element is guided by the principles set out in the OECD Statistical Quality Framework, informed by the key concepts identified by literature and constrained by the availability of data for at least 80% of OECD countries, for more than 1 year. The principal component analysis (PCA) of the indicators composing each element has been run to validate the selection.
- **Normalisation.** Indicators are normalised in two steps. In the first step, moving averages of the raw data for the periods 2020-2023, 2018-2022, 2016-2020 are computed. In the second step these moving averages are transformed into 0-100 scores using a clipped min-max normalisation where the minimum value is the sample mean minus two times the standard deviation of the 2020-

2023 sample, and the maximum value is the sample mean plus two times the standard deviation of the 2020-2023 sample. By doing so, scores are anchored to 2020-2023 period.

- **Imputation of missing values**. Missing values are imputed first, through time-series interpolation, and next, for few indicators, a linear regression with an external explanatory variable and countries fixed effects is used to predict the missing data points.
- **Aggregation**. Indicators are combined into element scores using geometric means. A sensitivity using arithmetic mean and weighted averages alternatives is presented.

Future work

The present Entrepreneurial Ecosystem Diagnostics framework should be regarded as a first, pilot edition. The following areas will be explored in future development of this work.

- Data refinement. Identify new and better metrics and improve imputation methodologies.
- Expand geography. Explore the possibility to cover non-OECD countries with good data coverage. Also, explore possible computation of sub-national data for ecosystems elements, output and variation indicators.
- Clubs of countries. Develop a methodology to identify relevant peer-countries for ecosystem benchmarking purposes, rather than comparing to general OECD average.
- Analysis of relationships in entrepreneurial ecosystems. Better research the relative importance of each of the ten input elements in driving entrepreneurship outputs.

1 Introduction

The OECD entrepreneurial ecosystem diagnostics exercise

This paper discusses the methodological approach used to produce the OECD report on Entrepreneurial Ecosystem Diagnostics. It covers the nature of the diagnostics, the conceptual approach, the selection, normalisation and aggregation of indicators, the sensitivity and robustness test results, and potential future directions for the work.

The aim of the diagnostics is to assemble and present comparative international data on levels and systemic conditions of entrepreneurship in OECD countries. This can help policymakers to identify ecosystem bottlenecks and policy reform priorities in their countries and to monitor the evolution of their entrepreneurial ecosystems over time. It also offers a unified source of key international entrepreneurship data to support research.

Developing comparative international entrepreneurship data has been on the OECD agenda for many years. An important milestone was in 2008, when the OECD produced a foundational paper setting out a conceptual framework, definitions, and proposed indicators on entrepreneurship (Ahmad and Hoffman, 2008[1]). The OECD set out to assemble these data, recognising that it would take a long time to deliver some of the indicators proposed and to establish a comprehensive database. The paper served as the basis for the OECD-Eurostat Entrepreneurship Indicators Programme, and relevant indicators were published in the OECD Entrepreneurship at a Glance series (2011-2017).

The OECD entrepreneurial ecosystem diagnostics project provides new momentum to the entrepreneurship data work. It assembles many of the earlier key indicators and extends them with new data sources. It also adapts the conceptual framework to put it in line with new academic and policy thinking on the importance of entrepreneurial ecosystems and presents the evidence using this new conceptual framework.

The entrepreneurial ecosystems concept

Entrepreneurship in the form of business start-ups and scale-ups has been shown to be a driver of economic growth (Bosma et al., 2018[2]). This occurs in various ways. Start-ups and scale-ups can be a key source of innovation through introducing and diffusing new products, services and business models to the economy (OECD, 2023[3]; Kolev et al., 2023[4]). They also increase competition, compelling incumbents to adapt or cede market share. And they create jobs. Scale-ups in particular make disproportionate contributions to job creation, often representing most new job creation from businesses in the economy (OECD, 2021[5]). Not all forms of entrepreneurship are equal, however, and to generate economic benefits policy makers are encouraged to target 'productive entrepreneurship' with innovation, employment and survival prospects (OECD, 2020[6]).

The entrepreneurial ecosystems concept has been applied to understanding the influences on levels of productive entrepreneurship in an economy. The framework focuses attention on the various aspects of the business environments of places and how together they impact entrepreneurship. This is seen in particular in terms of the resources, institutions, and actors that interact with entrepreneurs and potential entrepreneurs.

Obstacles in the entrepreneurial ecosystems of places can significantly hinder the development of start-ups and scale-ups. These bottlenecks could be present in various parts of an entrepreneurial ecosystem, such as cultural attitudes towards entrepreneurship, the availability of talent, knowledge and finance, or the conduciveness of formal institutions (business regulations and laws), informal institutions (societal norms and networks) and infrastructure. They can impact both directly on productive entrepreneurship levels and indirectly, through their impact on each other.

The concept of the entrepreneurial ecosystem has become established in the core of the entrepreneurship literature in recent years, as witnessed by contributions such as (Audretsch, Cruz and Torres, 2022[7]; Isenberg, 2010[8]; Huggins et al., 2024[9]; Brown and Mason, 2017[10]; Qian and Acs, 2023[11]; Feld, 2012[12]; Spigel, Kitagawa and Mason, 2020[13]) and is increasingly used in the development of policy. One of the milestones was an influential OECD workshop with the Dutch Ministry of Economic Affairs on Entrepreneurial Ecosystems and Growth Oriented Entrepreneurship in 2011, which explored the concept, the role of policy, and metrics for entrepreneurial ecosystems and led to one of the most highly cited papers on the concept (Mason, Colin; Brown, 2014[14]), building on the work of others such as (Isenberg, 2010[8]) and (Feld, 2012[12]) . Over the past decade, this early literature has expanded, proposing a wider and more complete set of business environment conditions that increase the likelihood of entrepreneurship and providing substantial empirical evidence on the relevance of different aspects of entrepreneurial ecosystems in propelling entrepreneurship (Wurth, Stam and Spigel, 2023[15]; Queissner, Stolz and Weiss, 2022[16]).

This paper discusses how we have operationalised the entrepreneurial ecosystem concept for a structured diagnostic system of entrepreneurship conditions across OECD countries. It takes into account feedback from experts received during the two technical workshops organised by the OECD in December 2024 and December 2023 on benchmarking entrepreneurial ecosystems. It also takes into account comments from CSMEE delegates received at the CSMEE meeting in November 2024 on a previous version of this paper.

2 Description of the conceptual model

Although different researchers have used different formulations to define the entrepreneurial ecosystem, a broadly accepted definition is the following:

> *"The entrepreneurial ecosystem is a set of interdependent actors and factors coordinated in such a way as that they enable productive entrepreneurship." (Stam, 2015[17])*

The conceptual model we adopt for the OECD entrepreneurial ecosystem diagnostics builds on this definition and reflects the entrepreneurial ecosystem framework put forward by (Stam, 2015[17]; Stam and Spiegel, 2018[18]) and (Stam and van de Ven, 2021[19]), as shown in Figure 2.1.

We define productive entrepreneurship as the types of entrepreneurial activity that generate innovation, productivity growth or employment growth (Baumol, Litan and Schramm, 2009[20]), ranging from high growth scale-ups to employer enterprise start-ups. Our framework draws on a substantial body of accumulated academic evidence (e.g. (Queissner, Stolz and Weiss, 2022[16]); (Wurth, Stam and Spigel, 2023[15]) and OECD work on the drivers and impacts of start-ups and scale-ups (OECD, 2022[21]).

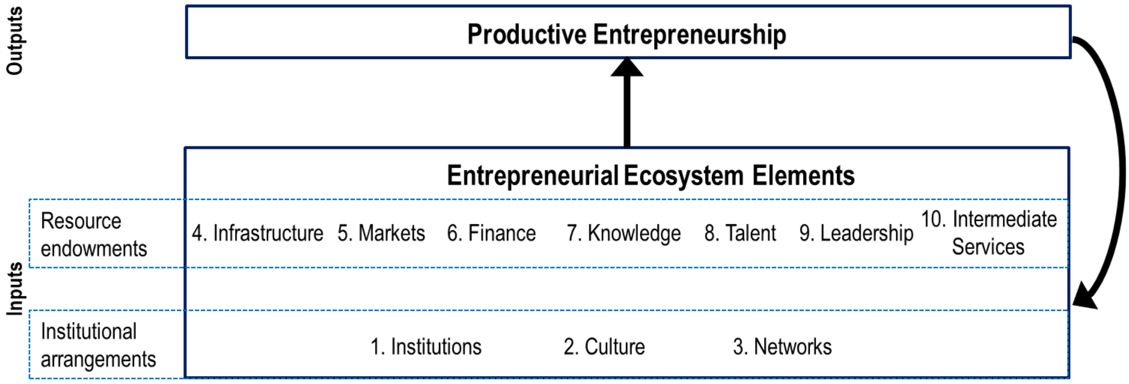
The concept of the entrepreneurial ecosystem is based on the idea that entrepreneurship emerges within systems composed of interdependent elements and outputs that can reinforce one another or hold each other back. A key focus of the operationalisation of this concept is therefore the identification of bottlenecks (i.e. the weakest links in the system) which can hinder entrepreneurship outcomes either directly or indirectly by weakening other components.

The model identifies two types of entrepreneurial ecosystem elements that affect productive entrepreneurship: institutional arrangements and resource endowments. Institutional arrangements are broader, cross-cutting enabling conditions that affect the degree to which actors are motivated by entrepreneurship and how stakeholders interact. Resource endowments are more specific sets of assets and services that determine the success of an ecosystem. Within institutional arrangements and resource endowments are a range of elements that directly affect the level and quality of entrepreneurship. They also interact with each other, so that a bottleneck in any element can have adverse consequences on others, reducing the overall capacity of the ecosystem to generate entrepreneurship. Our framework identifies the ten ecosystem elements shown in Figure 2.1, based on empirical research findings that show causal relationships between each element and productive entrepreneurship (see (Stam, 2015[17]; Stam and Spiegel, 2018[18]) and (Stam and van de Ven, 2021[19])).

Our conceptual model also identifies feedbacks from entrepreneurship outputs to the entrepreneurial ecosystem elements. These feedbacks are likely to involve substantial but varying time lags. For instance, entrepreneurial recycling (i.e. the recycling of the wealth and learnings generated by entrepreneurs in earlier entrepreneurial ventures as they exit and seek new opportunities) typically happens after events such as the sale of a successful business (Mason and Harrison, 2006[22]). This may augment the financial capital available in the ecosystem, thus contributing to an increase in its overall quality. The time from the initial idea of the venture to its success, and then to visible effects on the entrepreneurial ecosystem, may range from a few years (e.g. demonstration effect of a start-up) to a few decades (e.g. successful entrepreneurs becoming successful venture capitalists).

Figure 2.1. The Entrepreneurial Ecosystem Model

Representation of interactions between ecosystem element inputs and productive entrepreneurship outputs



Source: Adapted from (Stam and van de Ven, 2021[19])

The OECD diagnostics develops composite indexes to measure each of the ten entrepreneurial ecosystems elements. These indexes combine indicators covering different aspects of the element. Descriptions of the intended concepts covered by each element are provided below, following (Stam, 2015[17]) and (Wurth, Stam and Spigel, 2023[15]):

1. **Institutions.** This covers the quality of formal institutions (e.g. rule of law, absence of corruption), business regulations (e.g. the ease of registering a start-up, or the ease of exit for entrepreneurs and investors through liquidation), competition, and the favourability of taxation to entrepreneurship.
2. **Culture.** This covers entrepreneurial culture in terms of how far entrepreneurship is valued in society. Another aspect of this element is the degree of trust in other people, which is important for economic interactions.
3. **Networks.** This covers the existence of informal and formal networks for entrepreneurs, such as entrepreneur clubs or cluster organisations, as well as connections with other stakeholders (e.g. universities and corporates) that enable entrepreneurs to access knowledge or innovation.
4. **Infrastructure.** This covers digital connectivity and transport infrastructure, which enables entrepreneurs to access markets, suppliers, and sources of innovation and to exchange information with others.
5. **Markets.** This covers domestic market size, access to export markets and competitive barriers to entrepreneurs in accessing markets. It also covers the sophistication of demand, in terms of levels of consumer and business demand for innovative products and services.
6. **Finance.** This covers access to credit and equity and other forms of capital for the creation, operation and growth of start-ups and scale-ups. Particularly important are access to credit and access to venture capital.
7. **Knowledge.** This covers the availability of scientific and technological knowledge in universities and corporations, as a source of new business opportunities.
8. **Talent.** This covers talent in the sense of motivated and capable entrepreneurs possessing entrepreneurial skills. Another aspect of this element is the availability of skilled workers to work in start-up or scale-up enterprises in potential skill shortage areas such as digital skills or marketing skills.

9. **Leadership.** This covers the presence of public or private actors who can develop a shared vision and collective action among the range of ecosystem actors who can help develop the ecosystem. Highly successful entrepreneurs may be important players in such leadership roles.
10. **Intermediate Services.** This covers the presence of entrepreneurship-targeted business services offering information, advice, consultancy and mentoring to entrepreneurs. It includes business services such as legal support and accountancy, as well as consultancy and advice.

In terms of productive entrepreneurship outputs, the conceptual model points to both business start-ups and scale-ups. The types of start-ups and scale-ups envisaged by the model are those that are expected to generate innovation, employ workers, anticipate growth, or have good survival prospects.

We also consider the issue of variation within an entrepreneurial ecosystem, recognising that there may be considerable heterogeneity in entrepreneurship conditions within a country by region and social group.

Metrics are required to operationalise these concepts for benchmarking. The process used for selecting indicators is discussed in the next section.

3 Selection of indicators

This section discusses the selection of indicators for the diagnostic tool. This includes the criteria for indicator selection, the choices made in selection of individual indicators within each entrepreneurial ecosystem element, the construction of composite indexes with summary scores for each element, and the choice of productive entrepreneurship output indicators, and the treatment of variation or heterogeneity in a country-level entrepreneurial ecosystem.

Criteria for indicator selection

The selection of indicators for the inputs and outputs of the entrepreneurial ecosystem is guided by the principles set out in the OECD Statistical Quality Framework (OECD, 2012[23]). We set out below how we seek to conform to the principles in the selection of indicators:

- **Coverage and accessibility**: Only indicators available for significant share of OECD member countries are considered. An indicator is considered viable if a least one data point over the past 10 years is available for at least 80% of OECD countries.
- **Timeliness:** Only indicators updated regularly and for which time series exist are considered. This is to enable the measurement of progress over time and to offer a current view of the situation in each country.
- **Reliability and comparability**: Only data computed according to high quality standards produced by reputable sources are considered. To make sure that statistics are internationally comparable, data must be collected, to the extent possible, following international standards, and applying consistent definitions and methods to all countries.
- **Relevance and accuracy**: Only data that well capture the theoretical concept are considered. This involves a qualitative assessment of which metric is a reasonable representation of the concept among those available that match the other criteria above. Both direct and indirect measures of the underlying concepts are used. Abstract concepts are more easily measured indirectly or using proxies. Countable concepts are better measured with direct metrics. Some composite elements are more complex than others and require multiple indicators to capture different (non-overlapping) parts of a concept.
- **Evidence base on "congruence"**: When possible, indicators should be selected based on credible studies that have shown the relevance of the measure for understanding entrepreneurial ecosystems and their effect on entrepreneurship outputs.

These principles are applied to the selection of both entrepreneurial ecosystem element (input) indicators and productive entrepreneurship (output) indicators. There is however, one important difference. The coverage principle is a more stringent condition for the selection of indicators used to compute composite indexes. Since, as discussed in greater detail in section 4, the presence of missing values is particularly problematic for the construction of composite indexes, input indicators that do not have widespread data across different regions of the OECD area were discarded. All data points older than 2016 are removed from the dataset and not considered for the analysis or calculations. As a result, all indicators used must have data for at least one year in the period 2016-2023 in 80% of OECD countries. With respect to output indicators, since they are not used for the computation of composite indexes, the coverage principle has been applied less strictly to include indicators that capture relevant output or variation dimensions.

Use of indicators from non-official sources

National statistical offices and international governmental organisations produce a wide range of indicators related to entrepreneurship. For instance, the OECD's Structural and Demographic Business Statistics database provides indicators on business activity by size class, including entrepreneurship measures such as the birth rate of employer enterprises, survival rates, and growth rate of firms by size and year of incorporation. Other sources can measure aspects of the entrepreneurial ecosystem element inputs, such as the OECD SME and Entrepreneurship Financing Scoreboard, which provides relevant measures for finance inputs.

However, some aspects of entrepreneurial activity and/or entrepreneurial ecosystem elements are not adequately covered by official statistics. The diagnostics report favours the use of official data sources where available. For instance, venture capital is measured using OECD data. However, the diagnostic tool makes use of non-official sources where official data are not available for sufficient numbers of countries or do not measure closely enough the fundamental concept we seek to understand. When using non-official sources, we examine their validity in terms of their pertinence, provenance, representativeness across countries and correspondence to the key aspects of the concept needing to be measured. An aspect of this approach is to compare the distributions of the non-official data with distributions of related official data across countries and other variables.

Below we discuss this type of comparison in the case of two non-official sources, namely Crunchbase, a private, transaction-based database focused on venture capital investment, and GitHub which offers data on online activities of software developers.

Representativeness of Crunchbase data

Crunchbase is a commercial dataset that contains information on start-up numbers, venture capital investments, and other relevant variables. Crunchbase has emerged as a frequently used source for entrepreneurship research covering aspects of entrepreneurial ecosystems that are not well covered by other sources (Ferrati and Muffatto, 2020[24]; Dalle, Den Besten and Menon, 2017[25]).

Our diagnostic tool uses OECD data rather than Crunchbase as the main source for the venture capital investments indicator. It however relies on Crunchbase for data on number of serial entrepreneurs which are used in the Leadership element and the number of incubators and the number of mentors and coaches which are used in the Intermediate Services element. The number of startups and the share of female founders in the output and variation sections are also based on Crunchbase data.

Representativeness of Github software capability data

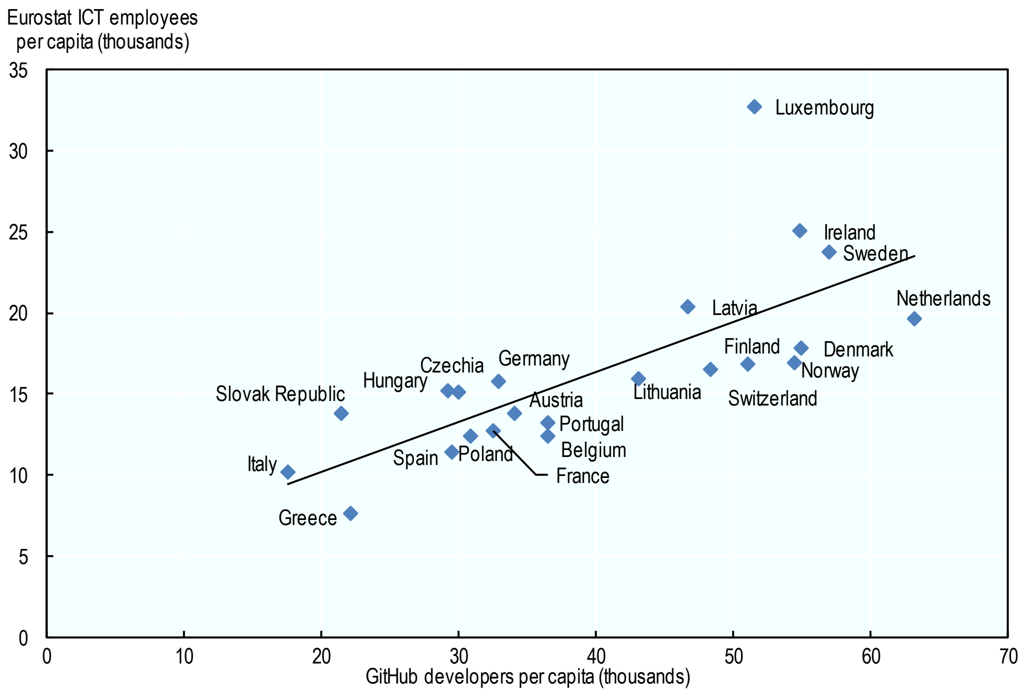
The diagnostic tool uses an indicator from GitHub, namely 'number of Github software uploads per thousand people', as measured by the 'number of times developers in a country uploaded code to GitHub'. This is used as a proxy for software capability. We also considered, but did not use, a second Github indicator 'Github developers per capita', as measured by 'number of software developers who have submitted new software through the Github platform'.

To our knowledge, there are no studies testing the potential bias in the GitHub data, and it has not been possible to locate a close comparator indicator from an alternative source on the scale of software development. Instead, we therefore make use of the second variable to test for potential bias. We compare the Github ‘number of software developers who have submitted new software through the Github platform’ with the indicator ‘Total ICT employees per capita’ from Eurostat for European countries.

Figure 3.1 shows that, at least in the European context, the GitHub indicator is highly correlated with Eurostat ICT employment measure. The GitHub indicator is, on average, on a higher scale. There are about 40 developers per thousand people in Europe, compared with about 16 ICT employees per thousand people. However, there is a consistent alignment between the number of ICT employees and the number of GitHub developers (in per capita terms) across the European countries. The only notable outlier is Luxembourg, where the GitHub indicator is significantly lower than the actual number of ICT employees per capita, relative to other countries. Although not providing a complete and robust test of GitHub indicators, this simple exercise provides some reassurance that these indicators are credible measures of software development.

Figure 3.1. Github developers track ICT employees closely in the European sample

Github developers and ICT employees per 1,000 people, European sample



Source: Github, Eurostat

Element indicators

Our diagnostics of the entrepreneurial ecosystem elements involves assembling groups of indicators capturing different aspects of the fundamental concept of each element, as discussed in chapter 2 above, and presenting both the individual indicators and composite element scores aimed at summarising the strength of the element in a country. The choice of indicators to be included in each element is driven by a desire to both have sufficient distinct indicators to capture each key part of the concept of the element, and a desire to avoid redundancy across indicators measuring the same aspect of a concept.

With this in mind, the selection of indicators for benchmarking the entrepreneurial ecosystem elements has been guided by the objective of constructing the most robust composite measure possible using the fewest indicators (Leendertse, Schrijvers and Stam, 2022[26]). Some elements are conceptually more complex than others, requiring more indicators. In addition, data availability varies by element. In some areas many suitable indicators are available, implying the need to make choices across similar measures, whereas in other areas few robust international comparative indicators exist, forcing them into the analysis if they meet our quality criteria.

To identify the indicators, we start off by examining the entrepreneurial ecosystems literature in terms of the types of indicators that have been used in other work and the rationale for this (e.g. (Leendertse, Schrijvers and Stam, 2022[26]; Stam and van de Ven, 2021[19]). In addition, we sought inputs from experts in the field, including, but not limited to, two international expert technical workshop at the OECD in December 2023 and in December 2024, which were convened to discuss the concept, measurement approach and potential indicators for this benchmarking exercise. Moreover, our methodology incorporates feedback received from delegates of the OECD Committee on SMEs and Entrepreneurship, which discussed a concept note on the benchmarking methodology at its 6th Session in April 2024 and on also discussed the first draft of this working paper and related Entrepreneurial Ecosystem Diagnostics report at its the 7th Session in November 2024.

In addition to theoretical and data availability considerations, to appropriately narrow down the list of indicators to remove redundancy and focus on the indicators with the most explanatory power. For this purpose, we conducted a Principal Components Analysis (PCA) and correlation analysis to make sure that selected indicators are relevant and add information. To respect parsimony and avoid multicollinearity, indicators that are not unique and are highly correlated with one another are excluded from the computation. Since moving average values will be used as inputs to compute the elements scores, PCA analysis is based on the 2020-2023 period which is computed as the moving 2020-2023 datapoints for each indicator. More details are provided in the Time periods section below.

In taking PCA results into account, a balance between internal consistency and information redundancy is considered. Indicators in each element should be correlated with the theoretical concept they contribute to measuring. We use PCA to statistically identify the key dimensions of the concept of each entrepreneurial ecosystem element that each indicator measures so as to focus down on a small number of indicators for each separate concept. We remove redundancy by removing indicators with lower explanatory capacity for each dimension.

In statistical terms, this corresponds to the principal component: a linear combination of indicators in one of the dimensional spaces defined by the set of selected indicator vectors. As a result, indicators are expected to have a high factor loading on the first principal component, which is the linear combination that can explain the highest share of the indicators' variance by itself. At the same time, however, composite indexes often include complementary concepts not correlated with one another which, together, contribute to measuring a complex concept. For instance, the indicator 'control of corruption' is weakly correlated with the indicator 'corporate taxation'. However, both corruption and taxation contribute to a conducive institutional setting for entrepreneurial activity and are therefore both relevant to the Institutions element. PCA in these cases can be used to identify unique dimensions that are not correlated with the first principal component but measure a second dimension of an element. In the example of Institutions, the quality of the institutional setting can be defined by multiple characteristics such as low corruption, fair levels of taxation, and other aspects. The results of PCA should thus be interpreted as a statistical guide to validate the selection of the indicators, especially when multiple indicators measuring similar concepts are available, and the PCA results are complemented with conceptual considerations when making the final choice of indicators to measure an element.

This section presents and discusses the rationale for the indicators selected within each entrepreneurial ecosystem element. As discussed above, there are two aspects of this. First, their connection to the theoretical concepts the elements are seeking to measure as discussed in the research literature and by expert work. Second, their independent explanatory power in terms of the aspects of the theoretical concept under consideration, as shown by the PCA and correlation analysis. Definitions and sources of each indicator used to compute elements' composite indexes are detailed in Tables A A.1- A A.10. In addition, Tables A A.11-A A.13 show the definitions of indicators considered but discarded.

Institutions

The literature shows that the traditions and institutions that determine how authority is exercised in a country impact economic development (Kaufmann, Kraay and Zoido-lobatón, 1999[27]; Hall and Jones, 1999[28]) and entrepreneurship (Estrin, Korosteleva and Mickiewicz, 2013[29]). The literature that studies the relationship between institutions and economic development partially overlaps with the literature on the role of institutions in promoting entrepreneurship. Focusing on the latter, multiple studies show that strong property rights enforcement, effective regulatory implementation and lack of corruption encourage entrepreneurial activity (Redford, 2020[30]; Ciccone and Papaioannou, 2021[31]; Estrin, Korosteleva and Mickiewicz, 2013[29]; Acs, Desai and Klapper, 2008[32]). The literature also indicates that additional aspect of this element concerns the functioning of the judicial system and the capacity of the public administration to enforce regulation (García-Posada and Mora-Sanguinetti, 2015[33]; Lichand and Soares, 2014[34]).

Based on this literature, the first indicators selected to build the Institutions element are the indicators 'control of corruption', taken from the Varieties of Democracy Project (Videm), the 'rule of law', taken from the World Bank's Worldwide Governance Indicator, and 'product market regulation', from the OECD. These three indicators capture by themselves alone a meaningful share of cross-country variation.

There is a relative abundance of institutional and public governance measures. It must be recognised, however, that these indicators share common weaknesses, such as subjectivity, weak theoretical grounding, measurement errors, and time comparability issues (Arndt and Oman, 2006[35]). Nonetheless, governance is an important enabler of entrepreneurship and must be measured, even if only through the use of proxies. In addition, many governance metrics are composite indicators which may contain overlapping and similar data within sub-indicators. Therefore, in selecting the three indicators above we excluded other indicators with potentially overlapping aspects, and we made sure that there is the least possible overlap between the selected metrics. It has to be noted that the Product Market Regulation index contains both aspects related to regulation and red tape as well as aspects related to competition and market access. For this reason, the Product Market Regulation index could have also been used as a component of the Markets element, however, given that competition regulation is under the direct supervision of public institutions, it has been decided to deploy it as an institutional measure.

A further aspect of the institutional setting that impacts entrepreneurial activity is taxation. Multiple studies point to a negative relation between high corporate taxes and entrepreneurship (e.g. (Djankov et al., 2010[36]; Langenmayr and Lester, 2018[37]; Gersbach, Schetter and Schneider, 2019[38]; Baliamoune-Lutz and Garello, 2012[39]). Based on this literature a measure of effective corporate tax rate is considered. One limitation of this metric is that start-ups and scale-ups may benefit from a variety of support programmes which may not be fully captured by the effective tax rate. An ideal measure would be the actual tax burden on start-ups and scale-ups, after discounting ad-hoc tax incentives. However, existing measures do not offer a specific and robust measurement of these issues, and the OECD's effective corporate tax rate remains the most general available measure of fiscal burden for conducting business activity, including entrepreneurial ventures.

There are additional aspects that would have been important to include in this element but were constrained but data availability. Notably, business regulatory aspects such as processes and costs to open, start and close a business play a significant role in stimulating or restricting entrepreneurship. However, the only source of international data on these aspects (the World Bank doing business database) has been discontinued. A new World Bank dataset measuring similar factors has recently been launched but it still covers too few OECD countries to be considered at this stage.

The Institution element is thus constructed using four indicators – corporate tax rate, rule of law, control of corruption and product market regulation index. The correlations among these indicators (Table A A.14) show a positive correlation between control of corruption and rule of law (80%), and a negative correlation between PMR index and both rule of law and control of corruption (53% and 65% respectively). These negative correlations are due to the fact that the PMR index is expressed as a measure of regulatory strictness. A higher PMR value indicates a worse regulatory landscape. These results are therefore in line expected synergies between low corruption, strong respect of the rule of low and good regulatory environments. The moderately high correlation between PMR index and other institutional quality variables highlights how the PMR index fits well the institutions element. When it comes to corporate tax rate, they are moderately correlated with the PMR index (32%) and not correlated with corruption (17%) and rule of law (5%).

PCA (Tables A A.15 and A A.16) shows that the Institutions element is composed of two factors: the taxation part and the institutional quality part. Corruption, rule of law, and product market regulation are highly correlated with the first principal component, but weakly or negatively correlated with the second component, taxation is highly correlated with the second principal component but weakly correlated with the first principal component.

Culture

A country's entrepreneurial culture is determined by multiple complementary aspects that must be measured with a heterogeneous set of indicators. Being abstract and linked to social norms, all these aspects can only be measured through surveys. Two established sources available for a large set of countries are the World Values Survey and the Global Entrepreneurship Monitor.

Trust is a first, well-established factor in the literature contributing to entrepreneurship levels and quality. The main channel through which trust benefits entrepreneurship is that in societies where people can trust each other, it is more likely that people exchange with others and take risk (Mickiewicz and Rebmann, 2020[40]).

A second cultural factor affecting entrepreneurship is the set of social norms and recognition that can drive people to try to become an entrepreneur. These social norms include the social status received by successful entrepreneurs and the desirability of a career in entrepreneurship (Nowiński and Haddoud, 2019[41]).

Two additional factors related to social norms and entrepreneurship are fear of failure and creativity. Fear to fail and the related stigma associated with a failed business may have a strong inhibiting effect on entrepreneurial activity (Cacciotti and Hayton, 2015[42])). When it comes to creativity, some studies have shown how educational curricula that tend to stimulate pupils' creativity can be used to raise entrepreneurial intentions (Yar, Wennberg and Berglund, 2008[43]). However, the available data on these aspects do not meet the selection principles outlined above. Hence after having tested these data, they were discarded, and these two factors are not measured in the framework.

The composite index for Culture uses three indicators measuring trust, status and desirability of an entrepreneurial career. Most of these indicators are weakly correlated with one another, highlighting the multidimensional nature of entrepreneurial culture. Only the status of entrepreneurs' indicator attains a correlation coefficient of about 55% with the trust indicators. All other correlation coefficients are below 30%, and in most cases non-significant (Table A A.17). The PCA also shows that two dimensions co-exist in the Culture element. Trust and status of an entrepreneurial career contribute to explain most of the variance, composing the first dimension. The desirability of an entrepreneurial career adda a second dimension to the Culture element (Table A A.18 and A A.19). The three indicators used to calculate the composite index measuring the Culture element are consistent to one another and capture two dimensions in the data.

Networks

The possibility of participating in business networks can encourage entrepreneurship by facilitating access to venture capital (Shane and Cable, 2002[44]), human resources, knowledge, and business contacts (e.g. suppliers or clients) (Greve and Salaff, 2003[45]). Entrepreneurs can also leverage networks to overcome size barriers. For example, start-ups or firms operating at a low scale cannot afford research equipment individually and must operate in networks to pool resources.

Indicators available in this space are scarce and cannot capture all channels through which the presence of established networks benefits entrepreneurial activity. However, two available indicators are the share of SMEs that cooperate on innovation activities, sourced from European Commission's European innovation scoreboard, and the extent to which firms collaborate with universities on R&D, obtained from the World Economic Forum's Executive Opinion Survey. The former indicator is a measure of the presence of established corporate networks for innovation, and although produced by a European institution, it covers all OECD countries. The latter is a measure to the existence of established systems that facilitate the conversion of academic research into commercial products, as well as the capacity and willingness of firms to work with the stakeholders in the ecosystem. Other potential indicators include measures of industrial networks and industry-science collaboration as well as networking events that start-ups can use to find partners or funds. However, these indicators are either only available for European countries or do not possess the data properties required and are therefore excluded.

The correlation analysis shows that the two indicators used for computing the Networks element are not strongly related with each other, with a correlation coefficient of about 50% (Table A A.20). The PCA confirms these results, showing that both components align well with the first principal component. These data however are not redundant. Using only one of the two indicators would explain less than 70% of the total variance (Tables A A.21 and A A.22). The indicators composing the Network element should thus be regarded as two reinforcing measures of networks for entrepreneurship.

Infrastructure

Transport and communications infrastructure enable entrepreneurship by facilitating exchanges of information, access to goods and services inputs, access to markets and access to labour. Commonly used metrics in this space are the quality of existing transport and telecommunications infrastructure.

An indicator capturing transportation is the transport infrastructure quality, obtained from the World Bank Logistic Performance Index database, which summarises the extent to which existing transport infrastructure is efficient and reliable. Other measures of transport network development such as roads and railroads per land area were considered but not used due to limited country coverage and inability to take differences in countries' geography (e.g. presence of mountains, islands, …) into account.

Telecommunications infrastructure is measured through fixed broadband subscription and mobile data usage, sourced from the OECD's Telecommunication database. Although mobile services are becoming a valid alternative to fixed connections, they still provide different types of telecommunication services, which should be available to entrepreneurs.

The correlation analysis among these indicators shows a limited overlap across selected measures. The highest correlation coefficient is found between fixed broadband subscriptions and transport infrastructure quality (Table A A.23). The principal component analysis confirms that the transport infrastructure and the fixed broadband indicators contribute to explain one component related to the availability of these infrastructure (Table A A.24 and A A.25). The mobile data use indicator instead provides unique information not captured by other indicators, with a limited overlap with fixed broadband subscriptions that captures the use intensity of these technologies.

Markets

Access to large, contestable markets is an important driver of entrepreneurial activity. The Markets element can be decomposed into t3ree factors. A measure of the domestic market size, a measure of foreign market access and a measure of competition. Available indicators that meet the OECD Statistical Quality Framework are limited in this space; hence our measurement of the Markets element uses proxies. In addition, the competition dimension of markets is strongly related to regulations, which is an area that pertains institutions as well as markets. To avoid double counting, competition and regulatory aspects of competition are measured in the Institutions element through the PMR indicator and are excluded from the Markets elements.

The domestic Market size can be proxied total GDP (expressed in Purchasing Power Parity terms), sourced from the OECD. GDP and total disposable income (which is simply the product of population and net disposable income per capita) were considered as possible alternatives. However, using GDP has the advantage to capture not only domestic consumer markets, but also business-to-business exchanges.

Foreign market size is partially measured by GDP (as it includes the difference between exports and imports). However, foreign market access should also consider potential extensions of foreign sales. To capture this aspect, the OECD's trade facilitation index is also used. This indicator measures the extent to which a country expedites the movement, release and clearance of goods at the border. In measuring market potential, indicators exist, but they are computed as potential exports to GDP. However, these measures are built on actual trade volumes rather than procedures, and, to avoid mixing inputs and outputs it was deemed more appropriate to measure potential obstacles that entrepreneurs encounter when they trade across borders rather than actual estimated export quantities.

The correlation analysis among these indicators shows a limited overlap across the selected measures. The correlation coefficient between GDP and trade facilitation index is about 23% (Table A A.26). The PCA shows that these two indicators fit the first principal component well, at that each indicator explains an important share of total variance (Tables A A.27 and A A.28). The first principal component captures domestic and international markets well and as long as competition is not measured within this element, the selected indicators are sufficient to capture the domestic and international market.

Finance

Access to finance is essential to both economic development and entrepreneurship (e.g. (Levine, 2005[46]). When it comes to entrepreneurship, however, new ventures are particularly sensitive to availability and access to credit and venture capital (Léon, 2019[47]; Keuschnigg, 2004[48]).

For the entrepreneurial ecosystem diagnostics, we measure the Finance element in terms of relative credit availability (Outstanding SME loans value, relative to a country's population), factoring and two measures of venture capital investments: early-stage and later-stage VC investments (relative to a country's population). The former two measures are sourced from the OECD's Financing SMEs and Entrepreneurs scoreboard and the latter two indicators are from the OECD's Entrepreneurship Financing Database. Together these measures capture the three main types of financing instruments available to grow a company: banking credit, equity and asset-based finance.

Another asset-based finance indicator (leasing) was also considered and collected from the OECD SME and Entrepreneurship Finance Scoreboard. However, data availability is limited to a sub-set of OECD countries. For this reason, it was excluded from the analysis.

When it comes to the venture capital indicator, we considered the possibility to use total venture capital investments rather than investments per capita. The intuition here is that there could be "critical mass" effects in venture capital supply. However, this is a new research area where consensus has not yet been achieved, and it has been deemed more prudent to scale venture capital investments to a measure of country size. Population has been seen as a more appropriate scaling factor than GDP to avoid normalisation disparities between advanced and middle-income economies.

The correlation analysis shows an expected high correlation between the two Venture Capital investment variables (about 97%), and very low correlations between venture capital and SME loans (less than 5%) and between factoring and SME loans (Table A A.29). These results show how these instruments are used differently in different countries. Notably, factoring and venture capital feature a negative correlation, albeit non-significant, which point at how countries that supply relatively more venture capital investments, tend to use fewer factoring instruments.

These results are confirmed by the PCA which shows that each of the three instrument types (equity, credit and asset-based finance) contribute to fund SMEs and entrepreneurs in different ways. (Tables A A.30 and AA.21). While equity plays a relatively large role, it can only explain 50% of the total variance by itself. Combining these indicators helps to provide a broader and more complete picture on the available financing options for SMEs and entrepreneurs.

Knowledge

Synergies between university, industry and government activities (the so-called triple helix system) are important drivers of knowledge-based entrepreneurship (Kim, Kim and Yang, 2012[49]). These synergies enable the emergence of new products by enabling the accumulation of knowledge and the translation of this scientific research knowledge into new commercial products. The presence of a large knowledge base is a manifestation of these mechanisms. For the diagnostics we use total R&D expenditure and patents per capita (both sourced from OECD statistics) to measure the amount of knowledge and related resources available in a country.

In addition to patents, other more intangible intellectual assets such as codes, data and software have recently emerged as an important source of service innovation. This is relevant not only for the small niche of start-ups that have developed on the base of software applications but more generally to larger numbers of innovative start-ups and scale-ups as a complementary knowledge asset type that entrepreneurs can leverage for venture start-up and growth. The increasing importance of investments in intangible intellectual assets has been documented by several authors (e.g. (Webster, 2023[50]; Corrado et al., 2021[51]) To represent these dynamics, an indicator tracking software uploads (sourced from Github) is included in the computation of the Knowledge element.

The pairwise correlation coefficients among the three selected indicators are between 49% and 68% (Table A A.32). Not surprisingly high R&D expenditure is correlated with patent applications (69%). However, the two variables do not overlap completely. Not all R&D converts into patents. Software development is a relatively unique indicator, with a correlation coefficient of about 50% with both public R&D expenditure and patent applications. The PCA confirms these results (Tables A A.33 and A A.34). Although all three variables are highly correlated with the first principal component, the software development indicator is highly correlated to the second principal component as well, conveying additional information.

Talent

There are two major components of the Talent element, the overall human capital base for the supply of skilled workers and potential entrepreneurs, and the share of the population with entrepreneurial skills.

A strong human capital base is conducive to entrepreneurial activity both because a better educated population is more prone to produce competent entrepreneurs and because the presence of a skilled talent pool eases the recruitment of staff for new ventures (Dimov, 2017[52]; Nguyen, Canh and Thanh, 2021[53]; Martin, McNally and Kay, 2013[54]).

Four measures are used to capture these different aspects of the talent base. The fist indicator is a measure of the overall human capital base in terms of the average formal education attainment by a country's population (mean years of schooling) sourced from UNESCO. A second indicator covers the skilled labour pool, measured in terms of overall OECD PISA scores (a combination of mathematics, reading and science scores), which capture the average competences acquired by the general population through the national education system. A third indicator tracks the availability of specific skills relevant for the development of new firms is IT skills. The proxy indicator to capture this aspect is the number of people who use the internet as a share of total population, obtained from the World Bank World Development Indicators database. The fourth indicator intends to pick up the level of entrepreneurial skills in the population. To this end, the indicator used is the percentage of adults who have the competences to start a business, sourced from the Global Entrepreneurship Monitor (GEM).

Other variables were considered in this space. A vocational training attainment indicator was tested in the attempt to capture the availability of personnel with practical technical experience as a human resource for start-ups. However, the indicator is negatively correlated with all output measures of entrepreneurship and has therefore not been retained. An alternative proxy measure of ICT-specific skills is the number of Github developers and the share of ICT technicians in employment. Intuitively, given the importance of digital skills in supporting start-ups and scale-ups, the presence of a significant number of workers with these skills can be considered particularly relevant for productive entrepreneurship. However, the indicator measuring the average IT skills in the general population rather than the incidence of IT-professional has been considered more appropriate. Entrepreneurs or aspiring entrepreneurs are not only those starting an IT-based start-up but can be anyone in the country, hence a broader ICT measure is better at capturing this general level of ICT capabilities among the population.

The correlation analysis for this element (Table A A.35) shows that, as expected, PISA scores are highly correlated with the mean years of schooling indicator (74%). A possible explanation is that high PISA proficiency scores tend to be associated with longer educational careers (Fischbach et al., 2013[55]). Mean years of schooling is also highly correlated with internet users (75%) as it makes intuitive sense that countries where on average people stay in school longer, tend also to acquire at least basic ICT skills. Less intuitively, the perceived entrepreneurial capabilities indicator is negatively correlated with all other indicators (PISA scores, internet users and mean years of schooling), pointing to a possible separation between entrepreneurial skills and other skills more relevant for potential start-up and scale-up employees.

The principal component analysis confirms these results (Table A A.36 and A A.37). The Talent composite score can be decomposed into an educational system principal component (highly correlated with PISA scores, internet users and mean years of schooling), and an entrepreneurial skills component. Both factors capture complementary talent necessary for founding, operating, and growing successful startups.

Leadership

Leaders who can coordinate an entrepreneurial ecosystem can play an important role in fostering entrepreneurship (Feldman and Zoller, 2012[56]). These leaders can bring the relevant stakeholders together, co-create a strategy for improving the entrepreneurial ecosystem, and making sure that actors are aligned and that the needed changes are implemented. This involves shared leadership for a common (entrepreneurial ecosystem) purpose. The measurement of these aspects, however, is challenging. There is a related literature on triple helix coalitions (Etzkowitz and Zhou, 2017[57]) and economic development partnerships (Olberding, 2002[58]) and more recently on entrepreneurial ecosystem orchestration (Harima, Harima and Freiling, 2024[59]) and entrepreneurial ecosystem catalyst organisations (Stam, Hendricksen and Van den Toren, 2023[60]). Few indicators that the meet the OECD Statistical Quality Framework standards are available in this space. The only proxy available is the number of serial entrepreneurs sourced from Crunchbase. Although this indicator does not fully capture all aspects related to leadership indicated by the literature, it does capture the extent to which there are relevant experts that can take leadership roles in the entrepreneurial ecosystem. The use of a single indicator to measure this element does not allow for correlation and principal component analysis. Notably a countries' number of serial entrepreneurs does not need to increase linearly with the population to inspire new entrepreneurs. At the same time, the presence of a critical mass of leaders is a necessary condition for them to set a vision and guide other emerging entrepreneurs.

Intermediate services

Entrepreneurs often encounter challenges in addressing legal, financial, accountancy, engineering, or regulatory issues. The availability of experts who can resolve these issues encourages entrepreneurship by reducing the need to acquire technical competences internally, hence reducing operational costs (Howells, 2006[61]; Clayton, Feldman and Lowe, 2018[62]). These services can be developed and offered by the business service industry. When this industry is not well developed, or when more specific sets of business support are needed, these services can be provided by entrepreneurial support organisations (Bergman and McMullen, 2022[63]).

Statistics in this domain are limited. However, three proxy measures can be used to capture the availability of experts, the availability of mentors or coaches and the availability other support mechanisms. The first aspect is measured through the share of employees in legal and accounting activities, management consultancy activities, architectural and engineering activities, technical testing and analysis, marketing, and other professional activities (as a share of total employees). This indicator is sourced from OECD statistics. The second aspect is measured in terms of number of coaches and mentors registered in the Crunchbase dataset. In many ecosystems coaching and mentoring programmes play an important role in helping entrepreneurs to grow their venture and/or improve SME practices. Therefore, a supportive ecosystem must offer a sufficient critical mass of experts who can be trainers or coaches. Similarly to leaders, there is no need that the amount of coaches and mentors growth with the population to support start-ups and scale-ups, hence the absolute value, rather than the population ratio of this indicator is used.

The third aspect is measured in terms of number of incubators and accelerators per capita, obtained from the Crunchbase dataset. The recent development of incubation and acceleration services has increasingly supported the development of start-ups and scale-ups, especially in high-tech domains. Through incubators and accelerators, entrepreneurs can access experts and other entrepreneurs who can provide services they need to develop their businesses. Incubators and accelerators may possibly display “critical mass” effects, yet, as the literature has not yet provided evidence on the extent of these effects on productive entrepreneurship, we scale this indicator by the country's population.

The correlation analysis and the PCA (Tables A A.38, A A.39 and A A.40) show that these three indicators are not overlapping and are rather complementary. The correlation coefficients between shares of technical employees and the other two variables in the element are about 40% indicating that these services are largely not interchangeable. Notably, the incubators indicator features a low correlation coefficient with the coaches and mentors' indicator (about 10%).

When combined into principal components, all three indicators align with the first principal component, indicating that they all fit well within the overall intermediate services concept and contribute to the supply of different types of support services needed by entrepreneurs. Among them, the availability of coaches and mentors' indicator is also highly correlated with the second component, implying some degree of uniqueness and capacity to explain an additional proportion of the total variance.

Output indicators

The selection of indicators of entrepreneurial ecosystem outputs has been guided by the requirement to capture different aspects of productive entrepreneurship. Productive entrepreneurship can be defined as entrepreneurial activity that generates significant wider economic and social benefits, through growth in productivity, value added, jobs or innovation (OECD, 2020[6]) (Baumol, Litan and Schramm, 2009[20]). However, we do not seek to focus only on the highest growth and most knowledge-intensive start-ups and scale-ups. Although these categories are of course of strong interest, we also see value in less dynamic or knowledge-intensive start-ups. The key issue is whether the enterprise can create innovation or sustainable employment for more than just the founder. At the same time, we seek to exclude what can be termed 'unproductive entrepreneurship', where there is little job creation, innovation, productivity gain or value creation for the economy. We class sole self-employment in this category, since it often does not lead to productive entrepreneurship in the sense of employing others or innovating (Stam, Van der Veen and Smetsers, 2019[64]).

Within productive entrepreneurship, there are different categories of start-up and scale-up activity, varying for example along the axis of degree of growth or degree of knowledge intensity. Different entrepreneurial ecosystems may be more or less conducive to different types of productive entrepreneurship activity. We therefore present a small dashboard of data on entrepreneurship outputs, picking up different aspects of what policy makers seek to promote.

Showing the data on entrepreneurship levels is important in enabling policy makers to monitor the comparative entrepreneurship performance of their countries internationally. They are also a key indicator of the overall health and nature of an entrepreneurial ecosystem. With appropriate follow up analyses, they can also be related back to the entrepreneurial ecosystem elements to help understand the relative importance of different interactions in the ecosystem.

Our dashboard includes measures of start-up and scale-up enterprises. These cover birth rates of employer enterprises, number of unicorns, and number of equity-based young firms. The dashboard also includes metrics on enterprise growth, including the employment share of young firms and the growth rate of start-ups. Another indicator covers entrepreneurial dynamism in the economy through churn rates (enterprise births plus enterprise deaths), which aims to pick up the level of creative destruction in an economy. The entrepreneurial outputs data also includes the survival rate of start-ups. Survival is an important aspect of entrepreneurial success and prospects for start-up survival are likely to vary with the quality of the entrepreneurial ecosystem.

Variation indicators

Our diagnostic tool also seeks to track the regional and social distribution of entrepreneurial activity within countries.

Measures of the regional distribution of entrepreneurship help show the extent to which entrepreneurial ecosystem conditions are relatively homogenous in a country, or whether entrepreneurship is concentrated in a few economic hubs. This has implications for the extent to which geographically differentiated entrepreneurship policies would appear to be merited in a country in order to promote levelling up. It also has implications for our degree of confidence in the broader relevance of the national averages shown in the diagnostics with respect to elements that could vary substantially over space. At the same time, it should be recognised that the variation measures we include in the Entrepreneurial Ecosystem Diagnostics report refer only to outputs, i.e. manifestations of the quality of ecosystem elements, and do not provide evidence directly on the regional quality of any ecosystem element.

We capture the geographical distribution of entrepreneurship within countries with one indicator of concentration/dispersion of entrepreneurial activity cities. It is the regional dispersion of start-ups from the Crunchbase data source. A second indicator was considered. The regional dispersion of enterprise birth rates density from the OECD Territorial Indicators dataset used to measure the dispersion across different TL2 level regions in each country. However, the data for this indicator have not been updated since 2016 and therefore discarded from the analysis.

The regional concentration of start-ups indicator focuses on cities, and – since tech firms represent a high share of start-ups in the Crunchbase data – it also tends to focus more on these types of firms. It is computed as the Herfindahl Hirschman Index, using the share of start-ups (0-5 years old) located in different cities within each country. The higher the value of this indicator, the more start-ups are concentrated in certain cities (in most cases the larger economic hubs in each country). Through this measure it is possible to capture whether tech-oriented start-ups cluster around a main city hub or emerge across multiple hubs in a country.

Social inclusion aspects of entrepreneurship are also important. There may be significant differences in productive entrepreneurship conditions for different segments of the population within a country. This can be measured along multiple axes, including gender, age and migration status of entrepreneurs and potential entrepreneurs. The OECD/European Union Missing Entrepreneurs report series shows that, in particular, in many countries, women, youth and foreign-born migrants lag behind in productive entrepreneurship activities, although the extent of the entrepreneurship gap varies by country and over time (OECD, 2023[65]). If entrepreneurial ecosystems are less conducive to certain parts of the population, this needs to be recognised by policy makers and appropriate actions developed to address group-specific entrepreneurial ecosystem bottlenecks. The degree of dispersion of entrepreneurship rates across social groups in a country helps policy makers to get an impression of the extent to which social inequality in entrepreneurship is an issue meriting further analysis and policy action. On the other hand, as with the regional dispersion, our diagnostic tool provides only output numbers and does not show differences by social group on the quality of each ecosystem element.

One of our measures is the OECD Missing Entrepreneurs rate. This estimates the numbers of additional entrepreneurs there would be if women, youth, seniors and immigrants created enterprises at the same rate as 30–49-year-old males, which is the population group with the highest entrepreneurship rate in all OECD countries, based on GEM data. A second measure gives the share of start-up founders and senior personnel that are women from Crunchbase data source. Data on entrepreneurial ecosystem variation are limited, especially when data quality parameters are taken into account, and we choose to focus on these two summary measures – one focused on the overall dispersion across several key social groups and the other on women, the single largest group of 'missing entrepreneurs' in the OECD area.

The definitions of all statistics composing the output and variation dashboard are detailed in Tables A A.41 and AA.42. A limitation of all statistics-based tools is that data are backward looking. Due to reporting and statistical production and quality control processes, there can be a lag of two or more years between the beginning of data collection and the publication of the related indicator. This limits the timeliness of statistics relatively to policy decisions and cannot be fully resolved. Nonetheless, quantitative benchmarking can provide significant value, first because most of the indicators used do not change rapidly over time. For instance, education data tend to change very slowly from year to year. In addition, producing time series for composite indices can provide a sense of direction and a trajectory for each element.

4 Normalisation, aggregation and missing values procedures for composite indices

This section discusses the process for creating composite indicators in the Entrepreneurial Ecosystem Diagnostics report. These were developed to provide a summary score for the entrepreneurial ecosystem elements for each country. These summary indicators offer a clear visualisation of the strengths and weaknesses of the different elements of an ecosystem at the country level relative to others and stand alongside the detailed sub-element scores. The development of the composite indices has involved the normalisation of indicators, checks for internal consistency, aggregation of sub-element indicators within elements, and dealing with missing values.

Time periods

The entrepreneurial ecosystem diagnostics tool has been produced retroactively for three periods: 2016-2020, 2018-2022 and 2020-2023, using moving averages of available data in each of these periods. The two-year interval has been chosen in order to allow for some variation between index instances. Several entrepreneurial ecosystem drivers take time to evolve, and two-year intervals give time for developments in the ecosystem to occur and become visible. The 2020-2023 period uses four years of data because 2024 statistics are incomplete for most indicators. The decision to stop back casting using 2016-2020 data is driven by the fact that data older than 2016 were considered too old and the number of indicators with missing value for the period 2014-2018 was considered too high to produce meaningful results.

The decision to use moving averages is motivated by addressing specific data issues with multi-source country-level panel data. First, the completeness and length of the time series varies considerably by indicator and country. It can be that some indicators are produced in different waves for different countries. For instance, data collection for countries A, B and C, could be conducted in 2015 and for countries D, E, and F in 2019. Second, newly developed indicators are only available on short time series that go back four to five years. Third, not all indicators are available for all years. For example, there could be a datapoint for one country in 2022, then a second datapoint in 2017, and no data in between. For another country the two datapoints available could be only 2021 and 2013.

As a result of these data structures, time series of the dataset required to compute the composite indices are patchy and uneven across countries and/or indicators. As shown in Figure A A.1, the dataset achieves its highest completion rate between 2018 and 2022 where over 72% of datapoints are available. The data completion rate is significantly lower before 2018 and after 2022. A more detailed discussion of missing values is presented in the next section.

Applying a moving average of all datapoints for 5-year time windows has been considered the most appropriate approach to address these issues. Its main advantages are that, first, and most importantly it allows to compute scores within similar time horizons for most countries. By imposing a 5-year window, it avoids that a country is assessed on a 2022 datapoint, while another country is assessed on as 2017 datapoint. Second, it smooths short-run volatility, capturing lasting performance differences across countries.

This approach, however, also has some limitations. First, there could be differences in the number of data points used to compute each moving average. For example, for an indicator, the moving average for country A can be computed using 5 data points, while for country B it can be computed using only 1 data point. This is however a less important issue than those highlighted above. Second, imposing a fixed 5-year windows to the data increases the number of missing values in the dataset. For example, if a country has data for the year 2017 but no data for the 2020-2023 period, there would be a missing value in the 2020-2023 period. More details on dealing with missing values are provided in the next section.

An alternative approach waws considered but discarded. It consists in taking the most recent year of data available, potentially going back as far as 2008. For example, for the 2023 period of the tool, the most recent datapoint available between 2023 and 2008 could be used, for the 2022 period the most recent datapoint between 2022 and 2008 could be used and for the 2020 period the most recent datapoint between 2020 and 2008 could be used. This method drastically reduces missing data but practically is implies that very old data can be used even for the most recent editions of the tool. For example, a country might de facto be assessed on 2023 data while another country might be assessed on 2015 data for the same variable. This limits the comparability across countries as well as comparability over time. Further it limits cross-elements comparability for the same country and period. This is because there could be some countries for which some elements scores are used using older data, and other elements using newer data.

While data censoring and moving averages are necessary to compute composite indices and for benchmarking output and variation indicators, where available, time series data were collected for all indicators for the longer period 2008-2023. In principles this data collection effort offers the potential for assessment of the long-run evolution of national entrepreneurial ecosystems and will be made available for researchers.

Addressing missing values

A significant empirical challenge for the construction of composite indices is the presence of missing values. Most of the existing composite indicators cannot rely on a complete, balanced panel datasets that include all datapoints for all countries and all years. In some cases, a whole time series for an indicator can be missing for one or more countries. This can be problematic because the presence of missing values can bias the calculation of composite indices, since a missing value in a composite indicator is implicitly imputed with the average of all non-missing variables in that indicator. This is an issue especially if a country’s true value for a missing value would be much higher (lower) than the average of the other indicators for which data are available.

Missing values is therefore an issue of particular importance for the computation of entrepreneurial ecosystem elements scores, since this is the part of the diagnostic tool where we develop the composite indices to summarize country performances.

Table A A.42 shows the count of missing values by indicator and year, and Table A A.43 shows the count of missing values by country and year, before applying imputation techniques. These table allows to visualise how some indicators and some countries present a higher incidence of missing values than others.

To address the presence of missing values we proceed in multiple steps. First, from the outset of the design of the diagnostics system, we have sought to minimise the extent of missing data by selecting indicators which have a large country coverage from reputable sources that update indicators regularly.

Nonetheless, there are several missing datapoints across the 38 countries and 29 indicators considered for calculating the entrepreneurial ecosystem elements.

Second, whenever there are gaps between two datapoints in the series, missing data are estimated using interpolation. For example, if there are data for 2021 and 2019, but not for 2020, the 2020 datapoint is estimated as the mid-point between the 2019 and 2021 data. In addition, as described in the previous section, a 5-year moving average is computed for each of the three periods for which data are available. This helps to reduce missing values compared to the approach of using a single datapoint available in a specific time window. As detailed in Table A A.43, after applying moving averages, for the 2020-2023 period, 94 missing values remain. This corresponds to about 8% of the total datapoints used.

Third, moving averages missing values are estimated using cross-country linear regression analysis, pooling the moving average values for the three periods together. Linear regressions use an external explanatory variable and countries fixed effects to predict the missing data points. This method is applied only applied to indicators that present at least a missing value and for which at least one good explanatory variable is available. It has not always been easy to find good left hand side variables for all indicators, which explains why it has not been possible to estimate every single missing data point with this method. There were thus indicators for which missing values remained. Table A A.45 lists the variables that required imputation and the main predictor.

Whenever missing values could not be imputed through linear regression, the geometric mean of non-missing values in an element is used to impute the missing indicators. This, however, occurs in very few cases. Post-application of imputation techniques a total of 9 missing values persists for the 2020-2023 period (0.8% of the dataset), and 5 missing values (0.4% of the dataset) for the 2018-2022 and the 2016-2020 periods of the tool.

Moreover, whenever more than one indicator within an element is either missing or imputed, the element score is not computed. This avoids constructing elements scores based on high shares of missing or estimated values.

A comparison of the baseline elements scores, which uses linear regression-based imputation and geometric means aggregation, and the alternative scenario which uses geometric means aggregation of non-imputed values are produced for the 2020-2023 period of the tool and described in the next chapter.

Two other alternative imputation approaches have been considered. The first alternative approach is the use of extrapolation. This assumes that a change observed in previous years continues at the same rhythm in the future. Our tests on the data show that this method does not provide reliable results. In addition, this method does not impute values for which no data are available throughout the time series.

The second alternative approach is the application of machine learning algorithms. More specifically, the missing forest algorithm was tested. Missing forest is a non-parametric entropy-based method which uses all indicators in the dataset to estimate the most likely missing value in an indicator given pattern in other indicators (Daniel J. Stekhoven and Peter Buehlmann, 2012[66]). We have tested this method by running the MissForest package provided by the software package R. However, the imputation appeared a black-box and the quality of imputation was not totally satisfactory in some cases.

Normalisation of indicators

To enable the aggregation of indicators with different ranges and units into a composite entrepreneurial ecosystem element score, all the variables within the element must be placed on a comparable scale. Multiple methodologies can be used to transform raw values into normalised scores, including standardisation (z-scores), Min-Max transformation, categorical scaling, distance to a reference value, among others.

In selecting the most appropriate normalisation methodology, a balance is needed between on the one hand maintaining the original data distribution to avoid distorting the results by applying a normalisation method that either "flattens" or "exacerbates" cross-country differences, and on the other hand, making sure that the presence of outliers does not cause the whole distribution of normalised scores to attain very low or very high values. For instance, if one country's score is 10 times higher than the second highest score in the distribution, normalising relative to the top value will result in very low normalised scores for all countries, except the one at the top of the distribution. A further factor to consider is the communication of results. Often, values are normalised to 0-100 scales to offer a simple interpretation of outcomes.

Among the most used methods, z-scores tend to better represent the underlying distribution. However, z-scores are weak to the presence of outliers and can be difficult to communicate, as they must be interpreted as the number of standard deviations a country's score is above or below the mean. The min-max transformation has the advantage of presenting results on a simple scale but can distort the distribution if the extreme values are not chosen correctly. A possible approach to address the selection of the minimum and the maximum values is to apply distribution-based identification of cut-off points. This approach consists in censoring the data using statistical formulas such as the 95th percentile (winsorisation) or the value corresponding to n standard deviations above/below the mean (clipping).

Our chosen methodology is the clipped min-max normalisation, cutting off 2 standard deviations from the mean. This approach allows us to maintain the distributional properties of each indicator, while addressing issues caused by the presence of outliers in a transparent way. This approach is implemented as follows. For each indicator, the minimum value is defined as the indicator's cross-section mean minus two times the indicator's standard deviation, and the maximum value is defined as the indicator's cross-section mean plus two times the indicator's standard deviation. Once these values are set, for each country i, indicator *k*, and period *t*, the normalised score is computed according to the formula:

$$norm_{ikt} = 100 * \frac{raw\_value_{ikt} - min_{kt}}{max_{kt} - min_{kt}}$$

where the raw value is the original raw data for a country, $min_{kt} = mean_{kt} - 2 * stdev_{kt}$ and $max_{kt} = mean_{kt} + 2 * stdev_{kt}$. The choice of using 2 standard deviations is arbitrary, yet it allows to maintain the normalised scores close to the original values but correcting for extreme outliers.

The formula transforms the data on a 0-100 scale. Whenever a country's raw value is below the minimum, a normalised score of 0 is assigned, and whenever a country's raw value is above the maximum, a normalised score of 100 is assigned.

There are cases where the normalisation formula must be flipped. Whenever an indicator's distribution is such that a high value corresponds to a negative outcome (e.g. taxation, debt), the normalisation in these cases is inverted as:

$$norm_{ikt} = 100 * \frac{raw\_value_{ikt} - max_{kt}}{min_{kt} - max_{kt}}$$

In these cases, whenever a country's raw value is below the minimum, a normalised score of 100 is assigned, and whenever a country's raw value is above the maximum, a normalised score of 0 is assigned.

Since the use of a normalisation method is not neutral, a comparison between z-score normalisation, winsorised min-max, and clipped min-max for two selected indicators is presented in Figures A A.1 and AA.2 of the Annex. These Figures compare the effect of alternative normalisation methods on countries' scores. This analysis has been conducted for all indicators in the Entrepreneurial Ecosystem Diagnostics report. The two examples presented describe the two main normalisation cases.

The first general case is whenever the distribution of an indicator is such that there are no extreme outliers (e.g. one or two countries with much higher or lower values than the rest of the sample). Figure A A.1

compares normalisation methods for the indicator “Fixed broadband” and represents a case where raw values do not feature extreme outliers at the top or bottom of the distribution. In these cases, the scores produced by the three normalisation methods are similar. The z-score normalisation tends to produce higher scores for countries throughout the distribution. While this outcome is closer to the actual distribution of the raw data, it can exacerbate differences between countries. Both min-max transformations tend to smooth the distribution slightly, with a difference between the winsorisation method and the 2 standard deviations clipping method. In the winsorisation case, the maximum is selected within the sample, which means that at least one country’s normalised score is 0 and at least one country’s normalised score is 100. In the clipping case, the maximum and minimum values can be defined outside the sample when all countries’ raw values are below the mean + 2 standard deviations, and/or above the mean - 2 standard deviations. This explains why in the clipping case, normalised scores at the top of the distribution are below the winsorisation case at the top of the distribution. This is a desirable feature of a normalisation method, as it reduces the number of instances where countries receive a 0 or a 100 score, compared to the winsorisation case.

The second general case where raw values include extreme outliers at the top or bottom of the distribution. Figure A A.2 compares normalisation methods for the indicator “Patents per capita”, which is an example of such a case. In these cases, the normalisation scores produced by the clipping method are significantly higher than those produced by the other two methods for almost all countries (except the very top value). The fact that the clipping method does not use within-sample minimum and maximum values explains this outcome and is the main reason why this method was used. As Figure A A.2 shows, whenever one raw value is well above all other values the normalised scores of all countries (except the top ones) are all very low. For example, using the winsorisation method for the air transport passenger indicator led the distribution to become almost binary for a significant part of the countries: 21 of the 38 countries for which data are available, obtain a score between 0 and 15. One country obtains a score of 78, and the country at the top a score of 100. This is not a desirable outcome as it over-penalises all countries in the middle of the distribution.

It is important to note that this paper builds composite indices for three periods, and normalisation scores are anchored to the 2020-2023 period’s values. This means that changes in elements score are due to countries performance changes relative to the top and bottom values in the 2020-2023 period. Changes in normalised scores across periods must thus be interpreted as relative change, respect to the 2020-2023’s OECD sample. Anchoring normalised scores to a particular year can encounter the issue of flattening the scores when all countries are improving rapidly. For instance, most countries tend to increase their mobile subscriptions over time, thus anchoring scores to a previous point in time (e.g. 2016) may lead all countries to have very high scores in 2023. This explains why it was chosen to anchor the scores to most recent data (the 2020-2023 moving average) rather than to the older data collected. In the future editions of the tool, it will be important to monitor the evolution of some indicators and possibly revise the anchoring points to make sure that the normalised scores continue to convey a meaningful information for all indicators.

Aggregation methods and weighting

There are differing numbers of indicators in the different entrepreneurial ecosystem elements. One element has just one indicator, but for the others there are up to five indicators. Where there is more than one indicator in an element, it is necessary to aggregate the different normalised indicator values into one score. There are different ways to do this, which will lead to different composite scores for countries.

Three aggregation methods have been considered for computing the element scores. The first method is the arithmetic mean. Denoting with x the normalised score of indicator k, in country i, at time t, values are aggregated according to the formula:

$$aritmethic\ mean\ score_{it} = \frac{1}{n}\sum_{k}^{n} x_{ikt}$$

This method assigns equal weights to all indicators that compose each element. It is the most frequently applied method to the development of composite indices (Nardo et al., 2005[67]). The main advantage of this method is its simplicity and ease of communication. It is also often perceived as the most "agnostic" way to aggregate different variables, especially when applying a specific weighting scheme is not possible or cannot be founded on solid empirical grounds. At the same time, assigning equal weights to all indicators can also be regarded as inappropriate if some indicators are relatively more important than others to the overall outcome. A further feature of arithmetic means is full compensability – i.e. a very low score on one indicator can be fully compensated by a high score on another indicator. This can be regarded as either a strength or a weakness of the approach depending on the situation. If the absence of a sufficiently high score on a sub-element is expected to penalise the overall result, then it is better not to adopt an aggregation method with full compensability. Taking a hypothetical example, this might be the case, for example, if within the Culture element an absence of trust in others creates a bottleneck for entrepreneurship within the Culture factors, even if there are otherwise good perceptions of entrepreneurship as a career choice and status of entrepreneurs in the society. In this hypothetical case, high scores on the other sub-element indicators should not lead to a high overall score and the arithmetic mean would not be the best measure. On the other hand, in some cases, full compensability is a desirable feature. For example, in the Infrastructure element, a low score on availability of fixed broadband might be rightly compensated for by a high score on mobile broadband. If this were the case, then the arithmetic mean would be the right aggregation measure.

A second method is the geometric mean, a non-linear aggregation method. Denoting with x the normalised score of indicator k, of country i, at time t, and with n the number of indicators composing the element, the aggregated scores are computed according to the formula:

$$geomean\ score_{it} = e^{\frac{1}{n}\sum_{k}^{n}\ln(1+x_{ikt})}$$

The main advantage of geometric mean aggregation is that it does not compensate a low value on one indicator with a high value on another indicator. This is suitable where the degree of compensability across variables within a composite indicator is not high. In most cases, it cannot be assumed that one indicator is a perfect substitute for another one (Munda and Nardo, 2009[68]), which reflects the efforts often made by those developing composite indicators to include a battery of measures covering different aspects of an issue, which can all have an independent influence on overall outcomes. This is one of the main reasons why the United Nations Development Programme (UNDP) has produced the Human Development Index using geometric means (UNDP, 2015[69]).

Limiting the possibility to compensate a low performance on one indicator with a higher performance on another is an attractive feature for benchmarking methodologies. The fact that indicators are non-perfect substitutes is a particularly binding condition for ecosystems analysis. In the case of the entrepreneurial ecosystems concept, it is appropriate to assume that an incomplete ecosystem can hinder entrepreneurship even if other conditions are good. For example, entrepreneurs may not respond to additional incubation services if there is limited venture capital availability in a country.

The use of geometric means shares common features with "penalty for bottlenecks (PFB) algorithms" approaches (Acs, Rappai and Szerb, 2012[70]). These authors discuss the problem of compensability of the variables in composite index construction and provide an algorithm to identify the weakest link in the ecosystem and reduce the index score depending on the extent of these weak links. The use of a geometric mean also penalises low scores, but to a lesser extent than PBF approaches. Nonetheless, because the geometric aggregation is non-linear, a low score on one indicator must be balanced with much higher scores on other sub-indicators to attain the same aggregate score as a country that attains relatively high scores on all indicators. This feature counterbalances some of limitations of the geometric mean which include complexity, difficulty of communication to non-technical audiences, and, as with the arithmetic mean, absence of a weighting scheme that quantifies the relative importance of each indicator. In addition, a technical limitation is that the geometric mean cannot be computed whenever an indicator's normalised score equals 0. This issue can be overcome by adding a value of one to each normalised score before applying logarithms.

The third aggregation method considered in this paper is the weighted average. Denoting with x the normalised score of indicator k, in country i, at time t, with n the number of indicators composing the element, and with w the weight of each indicator, values are aggregated according to the formula:

$$weighted\ average\ score_{it} = \frac{1}{n}\sum_{k}^{n} w_{ikt}\, x_{ikt}$$

The main advantage of this method is the possibility of assigning a higher weight to indicators that are considered relatively more important to an outcome. The challenge is to identify a robust methodology to determine such a weighting scheme. Two approaches are commonly used. One method is to use evidence from existing empirical literature and/or experts' opinions to convert research insight into a concrete weighting scheme. A second approach is to use PCA to derive weights through factor loadings. In this paper, we tested the second approach by computing a weighted average using PCA, using the factor loadings of the first two principal components. This is to allow a fair representation of indicators that measure different aspects of an element, which by construction would feature a low factor loading (and thus a low weight) on the first principal component, but a high factor loading on the second principal component. The resulting weights are listed in table A A.46.

However, we choose not to go forward with this aggregation method. This mainly reflects the lack of a strong enough theoretical logic to assign the weights. PCA-based weights can provide a scheme related to the importance that these indicators play in explaining the input variance, but an ideal weighting scheme should assess the relative importance of the indicator in explaining the aggregate concept. We do not have a theoretical method for this. Furthermore, whenever a composite index is produced to measure a concept that cannot be measured by any other indicator, it is not possible to find an appropriate variable to calibrate the estimation of weights through regression analysis.

Overall, having considered the three main options, our choice of aggregation is the geometric mean. This reflects the theory indicating that not allowing for full compensability is a considerable advantage. With respect to the weighted average scores, lacking a theory to determine the relative importance of indicators into each element, the advantage of presenting simple, easily reproductible results is deemed as more important than applying a data-driven weighting scheme.

We undertake sensitivity analysis to compare the results obtained with geometric means and the results obtained with arithmetic means, which are reported in section 5. The scores are similar for both aggregation methods for 8 out of the 9 elements which have more than one indicator. However, the country rankings alter. Conceptually, the geometric mean provides the better approach and the more reliable rankings.

We do not aggregate the ten elements further into a single index score. Although computing a single index may provide an easily comprehensible snapshot of the overall conditions for entrepreneurship in a country, such score is likely to convey an overly simplified depiction of the complexity of entrepreneurial ecosystems and may divert the attention from the elements, which are more informative metrics for a policy diagnosis than a single aggregate score.

Internal consistency

Building high-quality composite indices includes checking that indicators are selected in a non-arbitrary manner with good control and awareness of the interrelationships between selected indicators (Nardo et al., 2005[67]). However, building composite indices often requires striking a balance between using indicators that are correlated with one another and using indicators that capture different concepts and add new information. Using only indicators that are highly correlated with one another leads to redundancy that may double-reward or double-penalise countries.

The selection of variables was discussed in Chapter 3, which explained how indicators were tested through Principal Component Analysis. This section aims at testing the internal consistency of each element based on the indicators selected in the previous step. The exercise conducted in this section is not used to select indicators, but to transparently report internal consistency scores of each element, where internal consistency can be defined as the extent to which indicators in a composite index contribute to explaining the same concept. By construction, however, elements that combine different concepts are expected to carry a lower internal consistency, and those that measure a more homogenous concept are expected to carry a higher internal consistency. There can be a trade-off between internal consistency and the construction of multi-dimensional indices. Having prioritised conceptual considerations, leading to a number of multi-dimension indices, the results presented in this section simply quantify how conceptually homogenous each element is. It can be viewed as a complementary informative test to the PCA analysis.

The Cronbach alpha statistic provides a common approach to testing whether a composite index has internal consistency. To do so, it uses covariances across indicators to assess the extent to which the sum of indicators' covariances explain a large share of a cross-country variation among the selected indicators. A high alpha coefficient is found when the mean covariance of the indicators is high relative to the mean variance of the indicators and indicate that all indicators within an index point in the same direction.

Denoting with k the count of indicators in each element, with c the mean covariance between indicators ($c = \frac{\sum_i^k \sum_{j-1}^k cov_{ij}}{\sum_1^k (n_j - 1)}$, where n indicates the count of covariances) and with v the mean indicators' variance ($v = \frac{\sum_1^k var_i}{k}$), the Cronbach alpha of each element is:

$$\alpha = \frac{k * c}{v + (k-1) * c}$$

The Cronbach's alpha is interpreted as a measure of internal consistency. By construction, the alpha score ranges between 0 and 1, where zero indicates that all indicators are entirely independent, and 1 indicates that all indicators are perfectly correlated. Neither of the two extremes are appropriate for a composite index. Often, a value of 0.7 is used as a rule of thumb for defining high internal consistency. However, when a composite index includes multiple dimensions, measuring all the relevant dimensions comes at the cost of lower internal consistency. In these cases, some dimensions may not be correlated with the others, even if together they capture the complexity of the composite element they measure.

The computed Cronbach's alpha statistics are reported in Table A A.47. They show that while some elements achieve a high internal statistical consistency, others do not. This is partially explained by the fact that some elements are more complex than others. On the one hand, the Talent and Knowledge elements display high internal consistency because they capture few and consistent concepts. All indicators used to compute the Talent element are measures of population skills and all indicators used to compute the Knowledge element are related to intangible assets. On the other hand, the Network and Finance elements feature the two lowest Cronbach's alpha (0.36 and 0.38 respectively). This is due partially because the capture at least two dimensions, but also partially because the available indicators are statistically uncorrelated. For instance, in the case of Finance, countries that perform very well on Venture Capital measures perform significantly less well on SME loans and Factoring, which underscore differences in financial instruments used by SMEs in different countries. In the case of Network, the correlation between the only two available indicator is about 50%, which explains why the alpha score is relatively low. In this case, however, there are few statistics available, and it has been necessary to use proxies.

Overall, the Cronbach's alpha analysis shows that there is good internal consistency in the elements where the PCA shows that the data can be condensed into one principal component which has high explanation of total variance. In other words, there is good internal consistency of the selected indicators within the elements that measure single concepts. Where there is lower internal consistency, this is not a concern because this relates to the multi-dimensional nature of the elements. This can be read in combination with the PCA reported in chapter 3, corroborating the findings there.

5 Results of sensitivity and robustness tests

Baseline results

The baseline results for each of the ten elements are computed using clipped min-max normalization, geometric mean aggregation and without imputing missing values. Moving averages of 5-year windows were used in each iteration. To avoid that the presence of many missing values in one element distorts the distribution of the elements' scores, whenever there is more than one indicator with missing values or imputed values for an element, in a country, that element score is not computed for that country.

Table A A.48 presents the overall composite scores for all ten elements for all 38 OECD countries using 2020-2023 moving averages for all indicators. Although it is not possible a priori to predict a country's score on an element, there are formed expectation on results based on prior knowledge about countries' entrepreneurship characteristics and performance. Most of the results are in line with the expectations. For instance, most Nordic countries attain a high score on the institution's element, with Ireland also achieving a high score due to the combination of relatively strong government effectiveness, limited corruption, and attractive corporate taxation. Similarly, on the Networks' element the countries that attain the highest scores are those highly active in connecting start-ups domestically and internationally (e.g., the United States, United Kingdom, Finland). Other results are less intuitive on a superficial glance but offer a comprehensive representation of the element's complexity. For instance, on the Finance element, the United States' score is lower than expected. This is partially due to the fact that although the United States is a well-known Venture Capital Hub, in per capita term, its venture capital investments are at par with Israel and relatively less distant from other smaller, dynamic countries (e.g. Canada, Estonia). In addition, for SMEs, based on preliminary data, the use of SME loans and factoring in the United States is somewhat less intensively used than in many European countries. Similarly, countries which achieve high credit or factoring scores, such as Netherlands achieve overall higher Finance element scores than expected. While the use of geometric means limits the possibility to compensate low scores with high scores, extremely strong performances on one indicator can affect the element score.

The countries' scores distribution across the ten elements reveal that in the two elements (networks and leadership) countries attain the highest minimum score, indicating that among OECD countries, these two elements are the relatively more advanced in supporting productive entrepreneurship in at least some countries. Leadership, however, is the element displaying the highest cross-country variance, indicating that although all countries are active in supporting leaders, there is a large gap between countries at the top and those at the top of the distribution. These results are robust to the exclusion of the two lowest performers. Further, the Talent and Finance elements feature the lowest maximum scores among the OECD countries across the ten entrepreneurial ecosystem elements, indicating that multiple countries can improve of these aspects.

Cross-correlations of elements' aggregate show that almost all elements interact with each other, with about 40% of coefficients above 60% (Table A A.49). Notably, two elements feature correlation coefficients above 80% with other elements. The Infrastructure element displays coefficients above 80% with the

Knowledge, the Institutions element displays a coefficient above 80% with the Talent element. Infrastructure, Knowledge, and Talent tend to grow with national incomes, and good institutions and education (which constitute a significant portion of the Talent element) tend to reinforce each-other. On the opposite front, the Institutions element is weakly correlated to the Leadership element. This may be counterintuitive as one would expect to find more serial entrepreneurs in countries with regulations and governance. However, good institutions are clearly not enough to generate successful entrepreneurs. In addition, Leadership is measured with a proxy due to limited data availability, which could contribute to explaining this outcome. The other two elements that display low correlation are Markets and Culture, as indeed, there is no obvious direct connection between entrepreneurial culture and size or foreign market access of a country.

The correlations presented in Table A A.49 tend to be somewhat lower than regional-focused entrepreneurial ecosystem analysis (e.g. (Leendertse, Schrijvers and Stam, 2022[26]), which may suggest that ecosystem interactions are stronger at regional than national level. From another perspective, however, although there could be tighter connections among elements at sub-national level, there are also positive correlations at national level, and exploring these interdependences, albeit less tight is relevant for national entrepreneurship policies.

Aggregation sensitivity analysis

In this section the element scores are computed using three alternative aggregation methods (arithmetic mean, geometric mean, and weighted average). In the weighted average case, the weights are obtained from the PCA by averaging the factor loadings of each indicator on the first two principal components and rescaling them on a 0-1 scale. The results are presented in Tables A A.50, A A.51, A A.52.

The results show that element scores are robust to different aggregation methods. The distribution of scores remains fairly similar across the three specifications for most elements. The main differences across methods are in line with expectations. First, the scores produced by geometric means tend to be lower for all countries and all elements. This is explained by the fact that geometric means allow for less compensation across indicators, hence a low performance on one indicator reduces aggregate scores more than other method. Second, higher levels of sample variance are attained using geometric means, compared to other methods. This is due to the fact that limiting the possibility to compensate low scores with high scores tend to penalise weaker performers, expanding the gap between countries that tend to register high scores on most indicators and countries that present more weak spots. Third, the low-compensability effect of geometric means is more visible in elements that include more heterogenous concepts. For instance, in the Institutions element, geometric mean scores tend to penalise high corporate tax rates significantly more than arithmetic mean scores.

Imputation sensitivity analysis

In this section the baseline scores of all ten elements (which used missing values) are compared the scores computed that does not include imputed values. In both cases, the calculations are based on geometric means, using the 2020-2023 period (the moving average of 2020-2023 data). The results are presented in Tables A A.53 and A A.54.

Before comparing aggregate results, it is worth noting that imputation quality varies by indicator. In many cases the predict values or linear regression is closer to the actual value (for countries that have data available), in other cases the forecast error is larger. The uncertainty in imputation quality has led to decision of not using imputed values in the computation of elements score whenever there is more than 1 imputed or missing value among the indicators used to compute each indicator. This means that the number of elements for which a score is not computed is similar among the two scenarios.

Elements scores in the baseline scenario, which includes imputed values, and the alternative scenario without imputed values are similar in most cases. Importantly, the fact that many imputed values are not used when more than one variable per element is missing or imputed, does not fully capture the potential impact on elements scores if imputed values were used more widely.

Looking at elements' scores individually, in five elements (Institutions, Infrastructure, Markets, Knowledge, Leadership) there are no missing values, hence imputation does not apply. In addition, for one element (Intermediate services), it was not possible to find a regressor for the indicator 'Technical employment'. In both these cases elements scores between the imputation and non-imputation scenarios are identical by construction. In one element (Finance), using imputed values lead to marginally higher scores for most countries, relative to the no-imputed value scenario. In the Culture and Networks elements, the use of missing values impacts the scores of several countries. Most scores are between 1 and 10 points lower in these elements when imputed values are used, relatively to the case without imputed values.

The most impacted element by the presence of missing values is Talent, where using imputing values or not change several countries scores downwards, and in few cases upwards. New Zealand and Türkiye are the two countries whose scores are the most affected by the addition of imputed values in the Talent element. New Zealand's score is reduced by the fact that the estimated score for the indicator Perceived entrepreneurial capabilities is lower than the scores of other indicators in the element. Türkiye's score is reduced by the fact that the estimated score for the indicator Mean years of schooling is lower than the scores of other indicators in the element. These two examples show how missing values can impact overall score if a country's performance on the missing values is not in line with the other indicators of the element.

More generally, as shown in Table A A.44, Costa Rica, is the country featuring most missing values (9 out of 29 indicators), followed by Luxembourg, Iceland, and New Zealand (7 out of 29). Most of these missing indicators however are concentrated in specific elements, hence for Costa Rica, the scores of the Culture, Networks, and Finance elements are not computed. The only elements where adding imputed values changes the score is Talent, where the score is lower when imputed values are used. In the cases of Iceland, Culture and Finance scores are not computed, while using imputed values reduce Networks and Talent scores. In the case of Luxembourg, Culture, Finance, and Talent scores are not computed, while using imputed values reduces Networks scores. For New Zealand, Culture, Networks and Finance scores are not computed. As mentioned in the previous paragraph, Talent is the main negatively impacted element for New Zealand, when imputed values are used.

Time series results

In this section element scores are computed and compared across periods. It is important to recognise that the normalisation is anchored to the values of the 2020-2023 period. For instance, the maximum and the minimum value of each indicator are set equal to the (mean +/ – 2*st.dev) of the 2020-2023 moving average sample for both the 2016-2020 and 2018-2022 periods. This means that changes in countries performances are relative to constant numbers, hence it is possible to interpret normalised scores as progress over time. An alternative approach could have been to change the minimum and maximum scores in each period, in order to reflect the data distribution of each period. However, this would have made results more difficult to interpret as they would have been dependent on changes of all other countries.

Figures A A.2 – A A.12 show the variation in country scores over time for each of the ten elements, using the baseline computation methodology (geometric mean of 5-year moving average scores, with imputation of missing values).

The results show that scores move slowly over time for most indicators, adding to the robustness of the methodology. The fact that 5-years moving averages were used to compute these scores contribute to reduce variability over time. That being said, four elements (Talent, Markets, Infrastructure and Culture) display the most variance over time for most countries. With respect to Talent, Markets and Infrastructure, most countries show an improvement in their scores.

This can be explained by the fact that for these elements most indicators are an upwards trend in OECD countries. Notably, Talent scores are driven by general improvements in educational attainment and internet users, Markets scores are driven by both GDP growth and improving trade facilitation practices, while Infrastructure scores are mainly driven by constantly increasing ICT use (both mobile and fixed broadband). The Culture element variance is mainly built on survey-based indicators. While cultural element can only by captured but survey-based methodologies, responses can partially reflect general economic sentiment, which, however, do reflect actual changes in short-time conditions of entrepreneurial activity and play an important role in engaging with productive entrepreneurship ventures. On the opposite front, the Finance and the Knowledge element scores are stable over time. In the case of the Knowledge element, most underlying indicators reflect structural features of economies that do no change rapidly in relatively short periods of time. More specifically, cross-countries differences in R&D expenditure per GDP and patents in per capita have not varied significantly in OECD countries, over the period considered. The Github software uploads indicator, within the Knowledge element, instead, features some time variance, but most countries scores increase by relatively small amounts, which are not sufficient to drive visible changes in the Knowledge element scores. In the case of the Finance element, the indicators composing the element tend to be dominated by few countries that achieve high scores, while the rest of the distribution achieve similar, but much lower raw values. As a result, changes in the lower-value part of the distribution are not fully reflected after normalisation because relatively to the minimum and maximum extremes, the changes are not significant. In addition, the use of moving average contributes to reduce fluctuations over time.

Output indicators and robustness tests

In addition to the indicators used to compute the entrepreneurial ecosystem elements, the tool provides a dashboard of output indicators and variation indicators. The former enables tracking of the extent of productive entrepreneurship in each country, the latter the extent of regional and social distribution of productive entrepreneurship. Measurement of productive entrepreneurship outcomes include nascent entrepreneurs, growth expectations of start-ups, unicorns, birth, churn and growth rates of employer enterprises as well as survival rates. Measurements of entrepreneurial ecosystem variation include measures of regional concentration, share of women entrepreneurs and a measure of missing entrepreneurs. The complete list and definition of output and variation variables is provided in Table A A.51.

The raw values of these variables are reported in Tables A A.55 , A A.56, and A A.57. In addition, although these variables are not aggregated nor used to compute composite indices, a normalised value (ranging between 0-100) is calculated, using the clipped min-max normalisation describe in section 3. Normalised scores of all outcome and variation variables are produced to facilitate country comparisons but are not reported in this paper.

Output indicators can be used to test the capacity of each element to contribute to entrepreneurship outcomes. Showing that elements scores are correlated with output metrics should provide empirical preliminary evidence that computed scores are relevant in propelling productive entrepreneurship. A series of scatterplots, representing two-way regressions between selected output variables and all computed entrepreneurship ecosystem elements, are provided in Figures A A.14 and A A.15. The selected output measures are the birth rate of employer' enterprise and the number of equity-based young firms per thousand population registered every year in Crunchbase. Both measures are imperfect measures of productive entrepreneurship, however these are the closest output variables available. As Figure A A.15 shows, all computed elements scores are positively correlated with the number of equity based young firm per population. The correlations between elements and the birth rates of employers' enterprise (Figure A A.14) are weaker and vary significantly by element. The Knowledge element for instance is strongly positively correlated while other elements are weakly correlated with the birth rate of employer's enterprise. A possible explanation of this result is that birth rate of employer enterprises is a more general output measure of entrepreneurship capturing all types of new ventures, including those in traditional sectors and those with lower innovative content. It is also a measure of entrepreneurship kick-start, rather than growth and development. Many new-born firms exit the market a few years later because uncapable to stand the competition or not productive enough. The elements instead have been built to measure productive entrepreneurship drivers and better align with data that include a higher share of highly productive start-ups. While the limitations of both output measures must be acknowledged, the high correlations found between the elements scores and the number equity-based young firms per population indicate that the elements are adequate to measure inputs into ecosystem generating productive entrepreneurship ventures.

6 Conclusions and potential future developments

Conclusions

This paper discusses the methodology used to develop the OECD's Entrepreneurial Ecosystem Diagnostics report, which aims to support the assessment and monitoring of entrepreneurial ecosystems at national level in OECD member countries through a set of systematic, reliable, and periodically updated indicators and composite indices. Our approach builds on existing frameworks and methodologies but proposes a novel methodology and data aggregation.

The paper has described this methodology. It has provided rationales on indicator selection, normalisation, aggregation and missing data imputation. It has provided a correlation analysis of all variables used for the computation of the composite indices. It has also provided a battery of robustness tests on alternative outcomes using different aggregation methods and imputation methods.

After having considered the alternatives, ten composite indices – each measuring a key entrepreneurial ecosystem element – have been produced using clipped min-max normalisation and geometric means. Data were averaged across five-years windows, and indicator-specific imputation techniques were applied, using linear regression methods. Three iterations of these indices were computed retroactively with a window of two years between one iteration and the previous one. The most recent iteration uses moving averages of 2020-2023 data, and the oldest iteration uses moving average of 2016-2020 data for all indicators.

Despite considering many options, there remain challenges to entrepreneurial ecosystem measurement. These include the availability of data to precisely capture entrepreneurial ecosystem elements and determining the appropriate weights of different indicators in the composite indices. No perfect response exists to these challenges. Imputation methods have been applied and tested but should be further refined in future iterations of the tool. However, this paper has sought to provide full transparency on the approaches we have taken and the logic for the approaches.

Future development directions

The Entrepreneurial Ecosystem Diagnostics report is seen as a first edition. Further refinements and extensions can be considered for future editions. The following are among the potential areas for future development of the diagnostic tool.

Data refinements

The indicators used to produce the current version of the diagnostic tool have emerged from an extensive data research, consultations and feedback from stakeholders, including international policy makers and experts. Datasets, however, constantly improve, and there may be opportunities to exploit new sources and new types of data. Future editions of the Entrepreneurial Ecosystem Diagnostics report should therefore make sure to reflect the most recent data available. Furthermore, governments are encouraged to work with the OECD to improve the international data sets available in this field.

Additional countries

While the Entrepreneurial Ecosystem Diagnostics report only covers OECD countries there is a good logic to extend the diagnostics to cover other countries for which similar data are available. It is also possible to adapt the indicators or methodology to better suit other groups of countries.

Sub-national data

The current version of the entrepreneurial ecosystems diagnostics covers only the national level. There are a number of justifications for this:

- Several aspects of entrepreneurial ecosystems, in particular the regulation of labour, capital, and knowledge, are common across a country.
- National policymakers are responsible for developing the ecosystem at national level. They need guidance on priorities for intervention.
- An important challenge in developing a comprehensive diagnostics analysis at regional level is the availability of comparable data for a large share of OECD countries. Regional-level comparable statistics are currently available only for some countries, mostly in the European Union.

However, some authors argue that the entrepreneurial ecosystems concept is most powerful at the regional level, stressing that agglomeration effects can play an important role in entrepreneurial outcomes (Cavallo et al., 2020[71]) and that many entrepreneurial ecosystem conditions vary strongly between regions within countries, such as culture, availability of finance, levels of leadership and levels of talent, as well as entrepreneurship levels themselves (Sternberg, 2009[72]; Fritsch, 2013[73]). On the same lines, (Fischer et al., 2022[74]) argues that there are merits in measuring entrepreneurial ecosystems at all spatial levels.

Future work could seek to include a regional level of analysis for the countries and variables for which this would be feasible. A major constraint in building sub-national diagnostics instruments, however, is data availability. International statistics capturing many of the relevant elements of entrepreneurial ecosystems for all 38 OECD economies at sub-national level are often lacking. Future regional analysis would need to address these data constraints, limit the analysis to countries where adequate sub-national data would be available, or focus on summary measures of regional conditions such as core-periphery or urban-rural distinctions.

Benchmarking within clubs of countries

One of the issues for policy makers interested in benchmarking entrepreneurship outputs and conditions is the choice of peer countries with which to make the comparison. In the first version of the diagnostic tool, we are leaving the selection of countries to benchmark with to the reader. Their default option might be to compare their country to the top performers or to the average performers, or potentially to neighbour countries. Often the objective would be to select comparator countries that perform better but are not too different. In particular, the most important lessons might come from the policy settings in countries that have somewhat similar institutional and economic conditions.

An approach to the presentation of the data could be developed that seeks to identify a small number of clubs of countries with similar institutional or economic conditions or entrepreneurship rates, distinguishing for example between more liberal and more managed economies, countries with high or middle income levels, countries with high and low entrepreneurship rates, or countries with a more knowledge-intensive weighted entrepreneurship or a more general type of start-up and. The opportunities for learning may be greater within these and the comparisons with other countries more telling.

The creation of clubs could stem from statistical analyses including Principal Components Analysis (PCA), Cluster Analysis and Factor Analysis, from which similar groups of countries can be identified on entrepreneurial ecosystem conditions. Alternatively, it could be developed based on geographical considerations, e.g. having a Latin America group, North America group, European Union group etc.

More work could be undertaken to consider the presentation of results benchmarked against for clubs in addition to against OECD countries averages. In this view, future work should better define which group of countries (or “clubs”) should be used to benchmark ecosystems against. This would support learnings among countries that encounter similar challenges in promoting entrepreneurship or that share similar structural characteristics.

Analysis of relationships in entrepreneurial ecosystems

One important but still under-researched question is the relative importance of each of the ten input elements in driving entrepreneurship outputs. This is a challenging research question because different elements may play a different role in different contexts (e.g. income level), in different points in time (e.g. some entrepreneurship drivers that were important in 1990 might be less important today), and might be interlinked with other elements. Disentangling these aspects represent an ambitious but interesting future area of research.

The data that have been collected together in the OECD entrepreneurial ecosystem diagnostics work can be a basis for econometric analyses to extend existing research knowledge on the empirical values of the linkages in the system. On the one hand, this could cover the correlations between elements and outputs, seeking to identify the strongest relationships in different contexts. In addition, it could investigate the synergies or interlinkages in the ecosystem, and the extent to which there are penalties for bottlenecks. The weightings of elements in the diagnostic tool could potentially eventually start to reflect this type of knowledge.

References

Acs, Z., S. Desai and L. Klapper (2008), "What does "entrepreneurship" data really show?", *Small Business Economics*, Vol. 31/3, https://doi.org/10.1007/s11187-008-9137-7. [32]

Acs, Z., G. Rappai and L. Szerb (2012), "Index-Building in a System of Interdependent Variables: The Penalty for Bottleneck", *SSRN Electronic Journal*, https://doi.org/10.2139/ssrn.1945346. [70]

Ahmad, N. and A. Hoffman (2008), "A Framework for Addressing and Measuring Entrepreneurship: OECD Statistics Directorate Working Paper", *STD/DOC (2008)* January. [1]

Arndt, C. and C. Oman (2006), *Uses and abuses of governance indicators*. [35]

Audretsch, D., M. Cruz and J. Torres (2022), "Revisiting Entrepreneurial Ecosystems", *Finance, Competitiveness and Innovation Global Practice* November. [7]

Baliamoune-Lutz, M. and P. Garello (2012), "Tax Structure and Entrepreneurship", *SSRN Electronic Journal*, https://doi.org/10.2139/ssrn.2068495. [39]

Baumol, W., R. Litan and C. Schramm (2009), *Good capitalism, bad capitalism, and the economics of growth and prosperity*, https://doi.org/10.32418/rfs.2009.256.1996. [20]

Bergman, B. and J. McMullen (2022), "Helping Entrepreneurs Help Themselves: A Review and Relational Research Agenda on Entrepreneurial Support Organizations", *Entrepreneurship: Theory and Practice*, Vol. 46/3, https://doi.org/10.1177/10422587211028736. [63]

Bosma, N. et al. (2018), "Institutions, entrepreneurship, and economic growth in Europe", *Small Business Economics*, Vol. 51/2, https://doi.org/10.1007/s11187-018-0012-x. [2]

Brown, R. and C. Mason (2017), "Looking inside the spiky bits: a critical review and conceptualisation of entrepreneurial ecosystems", *Small Business Economics*, Vol. 49/1, https://doi.org/10.1007/s11187-017-9865-7. [10]

Cacciotti, G. and J. Hayton (2015), "Fear and entrepreneurship: A review and research agenda", *International Journal of Management Reviews*, Vol. 17/2, https://doi.org/10.1111/ijmr.12052. [42]

Cavallo, A. et al. (2020), "Agglomeration dynamics of innovative start-ups in Italy beyond the industrial district era", *International Entrepreneurship and Management Journal*, Vol. 16/1, https://doi.org/10.1007/s11365-018-0521-8. [71]

Ciccone, A. and E. Papaioannou (2021), "Red Tape and Delayed Entry", *SSRN Electronic* [31]

Journal, https://doi.org/10.2139/ssrn.989601.

Clayton, P., M. Feldman and N. Lowe (2018), *Behind the scenes: Intermediary organizations that facilitate science commercialization through entrepreneurship*, https://doi.org/10.5465/amp.2016.0133. [62]

Corrado, C. et al. (2021), "Intangible Capital and Growth in Advanced Economies: Measurement Methods and Comparative Results", *SSRN Electronic Journal*, https://doi.org/10.2139/ssrn.2114913. [51]

Dalle, J., M. Den Besten and C. Menon (2017), "Using Crunchbase for economic and managerial research Using Crunchbase for Economic and Managerial Research", *OECD Science, Technology and Industry Working Papers*, Vol. No. 2017/2. [25]

Daniel J. Stekhoven and Peter Buehlmann (2012), *missForest: Nonparametric Missing Value Imputation using Random Forest. R package version 1.5.*. [66]

Dimov, D. (2017), "Towards a qualitative understanding of human capital in entrepreneurship research"*, International Journal of Entrepreneurial Behaviour and Research*, https://doi.org/10.1108/IJEBR-01-2016-0016. [52]

Djankov, S. et al. (2010), "The effect of corporate taxes on investment and entrepreneurship", *American Economic Journal: Macroeconomics*, Vol. 2/3, https://doi.org/10.1257/mac.2.3.31. [36]

Estrin, S., J. Korosteleva and T. Mickiewicz (2013), "Which institutions encourage entrepreneurial growth aspirations?", *Journal of Business Venturing*, Vol. 28/4, https://doi.org/10.1016/j.jbusvent.2012.05.001. [29]

Etzkowitz, H. and C. Zhou (2017), *The triple helix: University-industry-government innovation and entrepreneurship*, https://doi.org/10.4324/9781315620183. [57]

Feld, B. (2012), *Startup Communities: Building an Entrepreneurial Ecosystem in Your City*. [12]

Feldman, M. and T. Zoller (2012), "Dealmakers in Place: Social Capital Connections in Regional Entrepreneurial Economies", *Regional Studies*, Vol. 46/1, https://doi.org/10.1080/00343404.2011.607808. [56]

Ferrati, F. and M. Muffatto (2020), *Using crunchbase for research in entrepreneurship: Data content and structure*, https://doi.org/10.34190/ERM.20.120. [24]

Fischbach, A. et al. (2013), "PISA proficiency scores predict educational outcomes", *Learning and Individual Differences*, Vol. 24, https://doi.org/10.1016/j.lindif.2012.10.012. [55]

Fischer, B. et al. (2022), "Spatial features of entrepreneurial ecosystems", *Journal of Business Research*, Vol. 147, https://doi.org/10.1016/j.jbusres.2022.04.018. [74]

Fritsch, M. (2013), *New business formation and regional development: A survey and assessment of the evidence*, https://doi.org/10.1561/0300000043. [73]

García-Posada, M. and J. Mora-Sanguinetti (2015), "Entrepreneurship and enforcement institutions: disaggregated evidence for Spain", *European Journal of Law and Economics*, Vol. 40/1, https://doi.org/10.1007/s10657-014-9470-z. [33]

Gersbach, H., U. Schetter and M. Schneider (2019), "Taxation, Innovation and [38]

Entrepreneurship”, *Economic Journal*, Vol. 129/620, https://doi.org/10.1111/ecoj.12588.

Greve, A. and J. Salaff (2003), “Social Networks and Entrepreneurship”, *Entrepreneurship Theory and Practice*, Vol. 28/1, https://doi.org/10.1111/1540-8520.00029. [45]

Hall, R. and C. Jones (1999), “Why do some countries produce so much more output per worker than others?”, *Quarterly Journal of Economics*, Vol. 114/1, https://doi.org/10.1162/003355399555954. [28]

Harima, A., J. Harima and J. Freiling (2024), “Ecosystem Orchestration: Unpacking the Leadership Capabilities of Anchor Organizations in Nascent Entrepreneurial Ecosystems”, Sage Journals. [59]

Howells, J. (2006), “Intermediation and the role of intermediaries in innovation”, *Research Policy*, Vol. 35/5, https://doi.org/10.1016/j.respol.2006.03.005. [61]

Huggins, R. et al. (2024), “Entrepreneurial Ecosystems in Cities and Regions: Emergence, Evolution, and Future”, University of Edinburgh. [9]

Isenberg, D. (2010), “The big idea: How to start an entrepreneurial revolution”, *Harvard Business Review*, Vol. 88/6. [8]

Kaufmann, D., A. Kraay and P. Zoido-lobatón (1999), “Governance Matters October 1999”, *World Bank Policy Research Working Paper* August. [27]

Keuschnigg, C. (2004), “Venture capital backed growth”, *Journal of Economic Growth*, Vol. 9/2, https://doi.org/10.1023/B:JOEG.0000031428.35711.fc. [48]

Kim, Y., W. Kim and T. Yang (2012), “The effect of the triple helix system and habitat on regional entrepreneurship: Empirical evidence from the U.S.”, *Research Policy*, Vol. 41/1, https://doi.org/10.1016/j.respol.2011.08.003. [49]

Kolev, J. et al. (2023), “Of Academics and Creative Destruction: Startup Advantage in the Process of Innovation”, *Academy of Management Proceedings*, Vol. 2023/1, https://doi.org/10.5465/amproc.2023.15844abstract. [4]

Langenmayr, D. and R. Lester (2018), *Taxation and corporate risk-taking*, https://doi.org/10.2308/accr-51872. [37]

Leendertse, J., M. Schrijvers and E. Stam (2022), “Measure Twice, Cut Once: Entrepreneurial Ecosystem Metrics”, *Research Policy*, Vol. 51/9, https://doi.org/10.1016/j.respol.2021.104336. [26]

Léon, F. (2019), “Long-term finance and entrepreneurship”, *Economic Systems*, Vol. 43/2, https://doi.org/10.1016/j.ecosys.2018.10.004. [47]

Levine, R. (2005), “Finance Growth Theory Evidence”, *Nber Working Paper Series* September. [46]

Lichand, G. and R. Soares (2014), “Access to justice and entrepreneurship: Evidence from Brazil’s special civil tribunals”, *Journal of Law and Economics*, Vol. 57/2, https://doi.org/10.1086/675087. [34]

Martin, B., J. McNally and M. Kay (2013), “Examining the formation of human capital in entrepreneurship: A meta-analysis of entrepreneurship education outcomes”, *Journal of* [54]

Business Venturing, Vol. 28/2, https://doi.org/10.1016/j.jbusvent.2012.03.002.

Mason, Colin; Brown, R. (2014), "Entrepreneurial Ecosystems and Growth Oriented Entrepreneurship", *Oecd*. [14]

Mason, C. and R. Harrison (2006), "After the exit: Acquisitions, entrepreneurial recycling and regional economic development", *Regional Studies*, Vol. 40/1, https://doi.org/10.1080/00343400500450059. [22]

Mickiewicz, T. and A. Rebmann (2020), "Entrepreneurship as trust", *Foundations and Trends in Entrepreneurship*, Vol. 16/3, https://doi.org/10.1561/0300000088. [40]

Munda, G. and M. Nardo (2009), "Noncompensatory/nonlinear composite indicators for ranking countries: A defensible setting", *Applied Economics*, Vol. 41/12, https://doi.org/10.1080/00036840601019364. [68]

Nardo, M. et al. (2005), "Handbook on Constructing Composite Indicators: Methodology and User Guide, OECD Statistics Working Papers", *Handbook on Constructing Composite Indicators: Methodology and User Guide*. [67]

Nguyen, B., N. Canh and S. Thanh (2021), "Institutions, Human Capital and Entrepreneurship Density", *Journal of the Knowledge Economy*, Vol. 12/3, https://doi.org/10.1007/s13132-020-00666-w. [53]

Nowiński, W. and M. Haddoud (2019), "The role of inspiring role models in enhancing entrepreneurial intention", *Journal of Business Research*, Vol. 96, https://doi.org/10.1016/j.jbusres.2018.11.005. [41]

OECD (2023), *SME and Entrepreneurship Outlook 2023*. [3]

OECD (2023), *The Missing Entrepreneurs 2023 Report, Policies for Inclusive Entrepreneurship and Self-Employment*. [65]

OECD (2022), *Financing Growth and Turning Data into Business: Helping SMEs Scale Up*, OECD Studies on SMEs and Entrepreneurship, OECD Publishing. [21]

OECD (2021), *Understanding Firm Growth: Helping SMEs Scale Up*, OECD Studies on SMEs and Entrepreneurship, OECD Publishing, Paris, https://doi.org/10.1787/fc60b04c-en. [5]

OECD (2020), *International Compendium of Entrepreneurship Policies, OECD Studies on SMEs and Entrepreneurship*. [6]

OECD (2012), "Quality Framework and Guidelines for OECD Statistical Activities, Version 2011/1", Statistics Directorate, OECD. [23]

Olberding, J. (2002), "Diving into the "Third Waves" of Regional Governance and Economic Development Strategies: A Study of Regional Partnerships for Economic Development in U.S. Metropolitan Areas", *Economic Development Quarterly*, Vol. 16/3, https://doi.org/10.1177/089124240201600305. [58]

Qian, H. and Z. Acs (2023), *Entrepreneurial Ecosystems and Economic Development Policy*, https://doi.org/10.1177/08912424221142853. [11]

Queissner, M., L. Stolz and M. Weiss (2022), "A Meta-Analysis of Entrepreneurial Ecosystems and their Effects on Entrepreneurial Activity", *Academy of Management Proceedings*, [16] Vol. 2022/1, https://doi.org/10.5465/ambpp.2022.14432abstract.

Redford, A. (2020), "Property rights, entrepreneurship, and economic development", *Review of Austrian Economics*, Vol. 33/1-2, https://doi.org/10.1007/s11138-019-00485-6. [30]

Shane, S. and D. Cable (2002), "Network ties, reputation, and the financing of new ventures", *Management Science*, Vol. 48/3, https://doi.org/10.1287/mnsc.48.3.364.7731. [44]

Spigel, B., F. Kitagawa and C. Mason (2020), "A manifesto for researching entrepreneurial ecosystems", *Local Economy*, Vol. 35/5, https://doi.org/10.1177/0269094220959052. [13]

Stam, E. (2015), "Entrepreneurial Ecosystems and Regional Policy: A Sympathetic Critique", *European Planning Studies*, Vol. 23/9, https://doi.org/10.1080/09654313.2015.1061484. [17]

Stam, E., T. Hendricksen and J. Van den Toren (2023), "Entrepreneurial ecosystem catalysts: A comparative study on the nature and effectiveness of a new type of organization.", Utrecht University / Birch Consultants / Tilburg University. [60]

Stam, E. and B. Spiegel (2018), *Entrepreneurial Ecosystems, in The SAGE Handbook of Small Business and Entrepreneurship*, SAGE Publications Ltd, 1 Oliver's Yard, 55 City Road London EC1Y 1SP , https://doi.org/10.4135/9781473984080. [18]

Stam, E. and A. van de Ven (2021), "Entrepreneurial ecosystem elements", *Small Business Economics*, Vol. 56/2, https://doi.org/10.1007/s11187-019-00270-6. [19]

Stam, E., M. Van der Veen and D. Smetsers (2019), "Economische groei vraagt om meer dan zzp'ers", *ESB, 104(4774), 278-280*. [64]

Sternberg, R. (2009), "Regional dimensions of entrepreneurship", *Foundations and Trends in Entrepreneurship*, Vol. 5/4, https://doi.org/10.1561/0300000024. [72]

UNDP (2015), *Training Material for Producing National Human Development Reports*. [69]

Webster, E. (2023), *The Economics of Intangible Investment*, https://doi.org/10.4337/9781782543770. [50]

Wurth, B., E. Stam and B. Spigel (2023), "Entrepreneurial Ecosystem Mechanisms", *Foundations and Trends in Entrepreneurship*, Vol. 19/3, https://doi.org/10.1561/0300000089. [15]

Yar, D., W. Wennberg and H. Berglund (2008), "Creativity in entrepreneurship education", *Journal of Small Business and Enterprise Development*, Vol. 15/2, https://doi.org/10.1108/14626000810871691. [43]

Annex A. Tables and Figures

Table A A.1. Institutions element, indicators description

List of indicators used to compute the Institutions element

Element	Indicator (units)	Description	Source
1.Institutions	Rule of law, 0-100 best	Index scaled 0-100 built by combining eight questions. i. Extent to which violent crime poses a significant problem for government and/or business (0 no problem - 4 significant problem). ii. Extent to which organised crime likely to be a problem for government and/or business? (0 no problem - 4 significant problem). iii. Extent to which extent to there is a risk that the legal process/the courts can be interfered with or distorted to serve particular interests (0 very low - 4 very high likelihood). iv. Extent to which there is a risk that contract rights will not be enforced (0 minimal risk - 4 high risk). v. Extent to which the judicial process is speedy and efficient (0 very speedy, 4 very slow). vi. Extent to which there is a risk of expropriation of foreign assets (0 no risk -4 high risk). vii. Intellectual property protection (0 very good - 4 very low protection).viii. Private property rights (0 very high, 4 very low guarantee). Respondents are experts from a network of over 500 correspondents, reviewed for consistency by panels of regional experts.	Economist Intelligence Unit (EIU) accessed via World Bank - Worldwide Governance Indicators
	Control of corruption index, 0-100 low incidence	Index of corruption based on population survey questions about frequency and spread of corruption.	Varieties of Democracy Project (Videm) accessed via World Bank - Worldwide Governance Indicators
	Product Market Regulation, Index 0-6 stringent	"Product Market Regulation Index. It measures the degree to which policies promote or inhibit competition in areas of the product market where competition is viable. A higher value indicates a higher level of regulatory stringency. It is computed through multiple aggregation layers. The overall index is computed as the average of Distortions Induced by State Involvement sub-index and the Barriers to Domestic and Foreign Entry sub-index. The Distortions Induced by State Involvement sub-index includes Quality and Scope of Public Ownership, Governance of SOEs, Retail Price Controls and Regulation, Involvement in Business Operation in Network Sectors, Involvement in Business Operations in Services Sectors, Public Procurement, Assessment of Impact on Competition, Interaction with Stakeholders. The Barriers to Domestic and Foreign Entry sub-index includes Administrative Requirements for LLCs and POES, Communication and Simplification of Administrative and Regulatory Burden, Barriers to entry in Service Sectors, Barriers to entry in Network Sectors, Barriers to FDI, Barriers to Trade Facilitation, Tariff Barriers. Each measure captures de jure regulation: they reflect the status of existing laws and regulations. They are based on a questionnaire compiled by national authorities and vetted by OECD experts.	OECD

Table A A.2. Culture element, indicators description

List of indicators used to compute the Culture element

Element	Indicator (units)	Description	Source
2.Culture	Entrepreneurship as a good career choice, % 18-64 pop.	Percentage of 18-64 population who agree with the statement that in their country, most people consider starting a business as a desirable career choice.	Global Entrepreneurship Monitor (GEM)
	High status to successful entrepreneurs, % 18-64 pop.	Percentage of 18-64 population who agree with the statement that in their country, successful entrepreneurs receive high status.	Global Entrepreneurship Monitor (GEM)
	Trust in others, % respondents	Share of respondents who believe that most people can be trusted.	World Value Survey (WVS)

Table A A.3. Networks element, indicators description

List of indicators used to compute the Networks element

Element	Indicator (units)	Description	Source
3.Networks	SMEs collaborating on innovation, % total SMEs	SMEs with innovation cooperation activities as a share of all SMEs. The small and medium-sized enterprises (SMEs) with innovation cooperation activities include all enterprises that had any cooperation agreements on innovation activities with other enterprises or institutions in the three years of the survey period. It corresponds to the indicator "3.2.1 Innovative SMEs collaborating with others" in the EU scoreboard.	European Commission - European innovation scoreboard
	University-business collaboration, 1-7 best	Average score across respondents who answered the question: in your country, to what extent do business and universities collaborate on research and development (R&D)? [1 = Not at all; 7 = To a great extent].	World Economic Forum

Table A A.4. Infrastructure element, indicators description

List of indicators used to compute the Infrastructure element

Element	Indicator (units)	Description	Source
4.Infrastructure	Mobile data use, Gb per subscrip./month	Mobile data use, Gigabits per subscription per month.	OECD - Broadband and telecom databases
	Fix broadband, subs. per 100 pop.	Total fixed broadband subscriptions per 100 population. Fixed broadband technologies corresponds to DSL, cable modem, fiber-to-the-home and other fixed technologies (such as broadband over power-line and leased lines).	OECD - Telecommunications database
	Transport infrastructure quality, 1-5 high	Logistics performance index: Quality of trade and transport-related infrastructure (1=low to 5=high).	World Bank - Logistic Performance Index (LPI)

Table A A.5. Markets element, indicators description

List of indicators used to compute the Markets element

Element	Indicator (units)	Description	Source
5.Markets	Gross domestic product, PPP$	Gross domestic product, expenditure approach, expressed in Purchasing Power Parity (PPP - international dollar).	OECD - Annual GDP and

	million		components
	Trade facilitation index, 0-2 best	Average trade facilitation performance computed as the average of the following aspects of border procedures: A. Information availability, B. Involvement of the trade community, C. Advance ruling, D. Appeal procedures, E. Fees and charges, F. Documents, G. Automation, H. Procedures, I. Internal border agency cooperation, J. External border agency cooperation, K. Governance and impartiality.	OECD - Trade Facilitation Indicators

Table A A.6. Finance element, indicators description

List of indicators used to compute the Finance element

Element	Indicator (units)	Description	Source
6.Finance	Early-stage VC investment, USD per capita	Total venture capital investments is the sum of seed, start-up and other early stage venture capital investments, in current USD, divided by the total population (USD dollars per person).	OECD Entrepreneurship Financing Database
	Later-stage VC investment, USD per capita	Later stage venture capital investments, in current USD, divided by the total population (USD dollars per person). Growth stage investments is defined as total investments minus seed, start-up and other early stage investments.	OECD Entrepreneurship Financing Database
	Outstanding SME loans, thousands USD per capita	Outstanding loans to SMEs divided by population.	OECD - Financing SMEs and Entrepreneurs: An OECD Scoreboard
	Factoring, thousands USD per capita	Total value of factoring contracts, divided by population.	OECD - Financing SMEs and Entrepreneurs: An OECD Scoreboard

Table A A.7. Knowledge element, indicators description

List of indicators used to compute the Knowledge element

Element	Indicator (units)	Description	Source
7.Knowledge	Patents, per million pop.	Number of patents filed at triadic patent families per capita.	OECD - Main Science and Technology Indicators
	R&D expenditure, % GDP	Gross Domestic Expenditure on R&D, % GDP.	OECD - Main Science and Technology Indicators
	GitHub software uploads, per thousand people	Number of times developers in a country uploaded code to GitHub per thousand people.	GitHub

Table A A.8. Talent element, indicators description

List of indicators used to compute the Talent element

Element	Indicator (units)	Description	Source
8.Talent	Mean years of schooling, years	Average number of completed years of education of a country's population aged 25 years and older, excluding years spent repeating individual grade. In general, the indicator’s value denotes the level of skills and competencies of a country's population, which could be seen as a proxy of both the	UNESCO

		quantitative and qualitative aspects of the stock of human capital. A relative high value indicates great shares of the adult population according to the highest level of education attained or completed and reflects a performing educational system. Caution is required when using this indicator for cross-country comparison. Countries do not always classify degrees and qualifications at the same ISCED levels, even if they are received at roughly the same age or after a similar number of years of schooling. Also, certain educational programmes and study courses cannot be easily classified according to ISCED levels. It is also important to note that this indicator is based on education levels attained or completed, in terms of years of schooling, and do not necessarily reveal the quality of the education (e.g. learning achievement or other outcomes).	
	Pisa, score	Average of Math, Reading, and Science of Pisa test scores.	OECD - PISA
	Internet users, % pop.	Individuals who have used the internet as a share of the total population. Internet users are individuals who have used the Internet (from any location) in the last 3 months. The Internet can be used via a computer, mobile phone, personal digital assistant, games machine, digital TV etc.	World Bank, World Development Indicators
	Perceived entrepreneurial capabilities, % 18-64 pop.	Percentage of 18-64 population who believe they have the required skills and knowledge to start a business.	Global Entrepreneurship Monitor (GEM)

Table A A.9. Leadership element, indicators description

List of indicators used to compute the Leadership element

Element	Indicator (units)	Description	Source
9.Leadership	Serial entrepreneurs, unit count	Number of people described as serial entrepreneurs in the crunchbase's individuals database.	Crunchbase

Table A A.10. Intermediate services element, indicators description

List of indicators used to compute the Intermediate services element

Element	Indicator (units)	Description	Source
10.Intermediate services	Incubators, per million pop.	Total count of active incubators, accelerators, co-working spaces, entrepreneurship programmes, and startup competition programmes, divided by the total population.	Crunchbase and OECD
	Coaches, unit count	Number of people described as mentors or coaches in the crunchbase's individuals database.	Crunchbase
	Technical employment, % total employment	Proxy measure for availability of experts in technical domains. It is measured in terms of employment in professional, scientific and technical activities as a share of total employment. Professional, scientific and technical activities correspond to the section M of UN's international standard industrial classification of all economic activities (ISEC, rev.4) and includes: (Division 69) Legal and accounting activities, (Division 70) Activities of head offices; management consultancy activities, (Division 71) Architectural and engineering activities; technical testing and analysis, (Division 72) Scientific research and development, (Division 73) Advertising and market research, (Division 74) Other professional, scientific and technical activities (e.g. design, photographic activities, other).	OECD

Table A A.11. Indicators considered but discarded 1/3

Element	Indicator	Description	Source	Rationale for exclusion
1.Institutions	SME income tax rate, % taxable income	SME Combined targeted corporate income tax rate	OECD	Too few countries covered
	Civil justice index, 0-100 best	The civil justice composite index is computed as the simple average of the following 7 variables: i. People can access and afford civil justice, ii. Civil justice is free of discrimination, iii. Civil justice is free of corruption, iv. Civil justice is free of improper government influence, v. Civil justice is not subject to unreasonable delay, vi. Civil justice is effectively enforced, vii. Alternative dispute resolution mechanisms are accessible, impartial, and effective. Each of these variables in turn, is a combination of one or more survey questions.	World Justice Project (WJP)	Too few countries covered
	Regulatory enforcement index, 0-100 best	The regulatory enforcement composite index is computed as the simple average of the following 5 variables: i. Government regulations are effectively enforced, ii. Government regulations are applied and enforced without improper influence, iii. Administrative proceedings are conducted without unreasonable delay, iv. Due process is respected in administrative proceedings, v. The government does not expropriate without lawful process and adequate compensation.	World Justice Project (WJP)	Too few countries covered
	SME implied tax subsidy rates, index	Implied tax subsidy rates on R&D expenditures of SME under a profitable profit scenario. The tax subsidy rate is defined as 1 minus the B-index, a measure of the before-tax income needed by a “representative” firm to break even on one additional monetary unit of R&D outlay (Warda, 2001). As tax component of the user cost of R&D, the B-Index is is directly linked to measures of effective marginal tax rates.	OECD - R&D Tax Incentives database	Too few countries covered
	Absence of corruption index, 0-100 best	The absence of corruption composite index is computed as the simple average of the following 4 variables: i. Government officials in the executive branch do not use public office for private gain, ii. Government officials in the judicial branch do not use public office for private gain, iii. Government officials in the police and the military do not use public office for private gain, iv. Government officials in the legislative branch do not use public office for private gain.	World Justice Project (WJP)	Too few countries covered
	Access to civil justice, 0-100 best	This indicator is a combination of survey questions that assess if the general population is aware of civil justice remedies, can afford legal representation, can access courts, how they assess procedural complexity to access civil justice. It corresponds to the sub-indicator "People can access and afford civil justice" of the Civil justice sub-index, from the WJP's Rule of law index.	World Justice Project (WJP)	Too few countries covered
	Government effectiveness, 0-100 best	Index scaled 0-100 built by combining two questions. i. How pervasive is red tape. ii. To what degree do vested interests/cronyism distort decision-making in the public and/or private sectors? Respondents are experts from a network of over 500 correspondents, reviewed for consistency by panels of regional experts.	Economist Intelligence Unit (EIU) accessed via World Bank - Worldwide Governance Indicators	Too few countries covered

Table A A.12. Indicators considered but discarded 2/3

Element	Indicator	Description	Source	Rationale for exclusion
2. Culture	Creative thinking PISA score, index	Average score obtained by students on the creative thinking PISA test, index	OECD	Too few countries covered
	Fear of failure, % 18-64 pop.	Percentage of the 18-64 population who agree that they see good opportunities but would not start a business for fear it might fail.	Global Entrepreneurship Monitor (GEM)	Volatile results
	Openness to technology, % respondents	Share of respondents who believe that in the future, more emphasis on technology is a good thing	World Value Survey (WVS)	Too few countries covered
	Propensity to imagination, % respondents	Share of respondents who believe that imagination is an important quality for a child	World Value Survey (WVS)	Conceptual miss-match
3. Networks	Participants to business events, per million pop.	Total number of all participants to business events that took place in a country in a given year as registered in Crunchbase. All types of participants are counted. It is divided by the total population.	Crunchbase	Conceptual miss-match
	Public-private co-patenting, % patents	Share of co-inventions (simple patent families) developed jointly by public-private partnerships across all technologies, as a share of total patents.	WIPO	Conceptual miss-match
	International co-patents, % co-inventions	Share of co-inventions (simple patent families) developed jointly by at least two inventors from different countries across all technologies. They are calculated based on inventor country - integer counts by country of residence of the inventors, and the first filing date worldwide, under the Paris Convention. The priority date is considered to be closest to the actual date of invention.	OECD - STI Micro-data Lab: Intellectual Property Database	Conceptual miss-match
	Domestic co-patents, % co-inventions	Share of co-inventions (simple patent families) developed jointly by at least two inventors from the same country, across all technologies. They are calculated based on inventor country - integer counts by country of residence of the inventors, and the first filing date worldwide, under the Paris Convention. The priority date is considered to be closest to the actual date of invention.	OECD - STI Micro-data Lab: Intellectual Property Database	Conceptual miss-match
4. Infrastructure	Density of rail network, Km per Sq. Mt.	Total Kilometres of rail roads divided by the country's surface area	OECD transport infrastructure indicators	Conceptual miss-match
	Density of road network, Km per Sq. Mt.	Total Kilometres of roads divided by the country's surface area	OECD transport infrastructure indicators	Conceptual miss-match
	Air transport, passengers per capita	Air passengers carried include both domestic and international aircraft passengers of air carriers registered in the country. It is divided by the country's population.	World Bank - World Development Indicators	Conceptual miss-match
	Mobile broadband, subs. per 100 pop.	Total mobile broadband subscriptions per 100 population. Mobile broadband subscriptions are mobile subscriptions that advertise data speeds of 256 kbit/s or greater.	OECD - Telecommunications database	Better alternative available
	Firms with 100 Mb/s	Percentage of enterprises with at least 10 employees with a fixed broadband	OECD ICT Access and Usage by Businesses	Too few countries

	connection, % firms	connection with download speed at least 100 Mbit/s	database	covered

Table A A.13. Indicators considered but discarded 3/3

Element	Indicator	Description	Source	Rationale for exclusion
5. Markets	Net disposable income, PPP, per capita	Net disposable income per capita, PPP-adjusted US dollars, chain linked volume.	Per person	Conceptual miss-match
	Population, million	Country population expressed in millions.	Thousands persons	Conceptual miss-match
	Public procurement expenditure, million $ per capita	Total public procurement expenditure (USD million) by all levels of public administration.	PPP million	Too few countries covered
6. Finance	Interest rate spread, % large	Interest rate spread between interests applied to SME loans and interests applied to large company loans, expressed as a share of the interest rates paid by large companies.	OECD Scoreboard	Too few countries covered
	Leasing, USD thousands per capita	Total value of factoring expressed in thousands USD per capita.	OECD Scoreboard	Too few countries covered
7. Knowledge	Citable documents, per million pop.	Number of citable documents (articles, reviews and conference papers) published per million people.	Per million people	Conceptual miss-match
8. Talent	Tertiary education attainment, % working age pop.	Share of adults 25-64 years old who have attained at least tertiary education degree.	% adult population	Conceptual miss-match
	ICT technicians, % total employees	Percentage of ICT professional and technicians in total employment. It is computed as the sum of Professionals (ISCO-08) and Technicians and associate professionals (ISCO-08) in the Information and communication sector (Economic activity J. (ISIC-Rev.4)), divided by all employees in all sectors.	International Labour Organisation (ILO) - ILOSTAT database	Too few countries covered
	Adults TVET attainment, %	Share of population (both genders) aged 25-64 years old who have attained upper secondary vocational training education.	OECD	Conceptual miss-match
	PIAAC scores	Measure of adults' proficiency in literacy, numeracy and problem solving.	OECD	Too few countries covered
	Number of developer accounts on GitHub, per million pop.	Number of developer accounts on GitHub in a given economy. This count excludes users that are bots or otherwise flagged as "spammy" within internal systems. A three-year moving average is applied to the original data.	GitHub	Conceptual miss-match
9.Leadership	-	-	-	-
10.Intermediate services	-	-	-	-

Table A A.14. Institutions element, correlations among indicators used

Correlation coefficients among indicators used to compute the institutions element

	Rule of law	Corp. tax	PMR index	Contr. Corruption
Rule of law	1			
Corp. tax	0.0555	1		
PMR index	-0.5344*	0.3233*	1	
Contr. Corruption	0.8081*	0.1676	-0.6494*	1

Table A A.15. Institutions element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	2.33542	0.5839	0.5839
Comp2	1.18176	0.2954	0.8793
Comp3	0.351887	0.088	0.9673

Table A A.16. Institutions element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
Rule of law	0.5825	0.1557
Corp. tax	-0.0292	0.8929
PMR index	-0.5367	0.3641
Contr. Corruption	0.6098	0.2145

Table A A.17. Culture element, correlations among indicators used

Correlation coefficients among indicators used to compute the culture element

	Entrep. Status	Entrep. Carrer	Trust
Entrep. Status	1		
Entrep. Carrer	0.2747	1	
Trust	0.5606*	-0.1053	1

Table A A.18. Culture element , PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	1.58598	0.5287	0.5287
Comp2	1.08113	0.3604	0.889
Comp3	0.332897	0.111	1

Table A A.19. Culture element, PCA factor loadings

Correlation of indicators with the principal components

	Comp1	Comp2
Entrep. Status	0.7247	0.1092
Entrep. Carrer	0.2146	0.9025
Trust	0.6547	-0.4167

Table A A.20. Networks element, correlations among indicators used

Correlation coefficients among indicators used to compute the networks element

	SME collab.	uni-business collab.
SME collab.	1	
uni-business collab.	0.5318*	1

Table A A.21. Networks element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	1.53184	0.7659	0.7659
Comp2	0.468157	0.2341	1

Table A A.22. Networks factor loadings

Correlation of indicators with the principal components

	Comp1	Comp2
SME collab.	0.7071	0.7071
Unversityi-business collab.	0.7071	-0.7071

Table A A.23. Infrastructure element, correlations among indicators used

Correlation coefficients among indicators used to compute the infrastructure element

	Mobile Data	Fixed Bband subs.	Transport quality
Mobile Data	1		
Fixed Bband subs.	-0.1354	1	
Transport quality	0.0564	0.7500*	1

Table A A.24. Infrastructure element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	1.75425	0.5847	0.5847
Comp2	1.01964	0.3399	0.9246
Comp3	0.226111	0.0754	1

Table A A.25. Infrastructure element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
Mobile Data	-0.0751	0.982
Fixed Bband subs.	0.7099	-0.0693
Transport quality	0.7003	0.1755

Table A A.26. Markets element, correlations among indicators used

Correlation coefficients among indicators used to compute the markets element

	Trade facilitation	GDP PPP
Trade facilitation	1	
GDP PPP	0.2326	1

Table A A.27. Markets element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	1.23262	0.6163	0.6163
Comp2	0.767385	0.3837	1

Table A A.28. Markets element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
GDP PPP	0.7071	-0.7071
Trade facilitation	0.7071	0.7071

Table A A.29. Finance element, correlations among indicators used

Correlation coefficients among indicators used to compute the finance element

	VC early investment	VC later investment	SME loans	Factoring
VC early investment	1			
VC later investment	0.9794*	1		
SME loans	0.0436	0.0099	1	
Factoring	-0.1692	0.1854	-0.016	1

Table A A.30. Finance element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	2.05628	0.5141	0.5141
Comp2	1.01458	0.2536	0.7677
Comp3	0.924268	0.2311	0.9988

Table A A.31. Finance element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
VC early investment	0.6865	-0.0852
VC later investment	0.6874	-0.1097
SME loans	0.0447	0.9053
Factoring	-0.2327	-0.4014

Table A A.32. Knowledge element, correlations among indicators used

Correlation coefficients among indicators used to compute the knowledge element

	R&D expenditure	GitHub software upload	Patents
R&D expenditure	1		
GitHub software upload	0.5212*	1	
Patents	0.6977*	0.4942*	1

Table A A.33. Knowledge element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	2.13759	0.7125	0.7125
Comp2	0.562727	0.1876	0.9001

Table A A.34. Knowledge element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
R&D expenditure	0.6065	-0.3036
GitHub software upload	0.5281	0.8447
Patents	0.5943	-0.4408

Table A A.35. Talent element, correlations among indicators used

Correlation coefficients among indicators used to compute the markets element

	Pisa score	Entrep. Skills	Mean years school	Internet users
Pisa score	1			
Entrep. Skills	-0.5374*	1		
Mean years school	0.7482*	-0.3706	1	
Internet users	0.5903*	-0.2058	0.7525*	1

Table A A.36. Talent element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	2.83399	0.7085	0.7085
Comp2	0.820745	0.2052	0.9137
Comp3	0.225344	0.0563	0.97

Table A A.37. Talent element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
Pisa score	0.567	-0.099
Mean years school	0.536	0.2689
Internet users	0.5024	0.4514
Entrep. Skills	-0.3725	0.8451

Table A A.38. Intermediate services element, correlations among indicators used

Correlation coefficients among indicators used to compute the intermediate services element

	Incubators	Technical employment	Coach/Mentors
Incubators	1		
Technical employment	0.4621*	1	
Coach/Mentors	0.1067	0.4161*	1

Table A A.39. Intermediate services element, PCA eigenvalues

Principal components' eigenvalues

Component	Eigenvalue	Proportion	Cumulative
Comp1	1.65092	0.5503	0.5503
Comp2	0.943184	0.3144	0.8647
Comp3	0.405896	0.1353	1

Table A A.40. Intermediate services element, PCA factor loadings

Correlation of indicators with the principal components

Variable	Comp1	Comp2
Incubators	0.5332	-0.6685
Coach/Mentors	0.4884	0.7437
Technical employment	0.6908	-0.0098

Table A A.41. Entrepreneurship outputs and variation indicators

List of output variable complementing the entrepreneurial ecosystem elements

Element	Label	Description	Source
	Birth rate of	New firm creation among employer firms (i.e. with at least one	OECD - SDBS

Output	employer enterprises, % business pop.	employee), as a proportion of active business population.	business demography
	Medium and high-growth enterprises, %	Rate of medium and high-growth enterprises (10%+ growth based on employment) as a proportion of active business population.	OECD - SDBS business demography
	Equity-based young firms, per million pop.	Number of companies aged 0-5 years old, divided by the population. Since the Crunchbase dataset is skewed towards equity-based firms, the indicator better represents equity-based firms that are 0-5 years old.	Crunchbase, OECD
	Unicorns, per million pop.	Number of private companies with a valuation over USD 1 billion, divided by the population.	CB Insights
	Enterprise churn rate, % business pop.	Sum of births and deaths of employer enterprises (firms with at least 1 employee) as a proportion of active business population.	OECD - SDBS business demography
	2-year-old employer enterprises, % business population	2-year-old employer enterprises as a proportion of active business population.	OECD - SDBS business demography
	3-year survival rate., % new employer enterp.	3-year survival rate of employer enterprises, as a proportion of new employer enterprises.	OECD - SDBS business demography
	Expectation to create jobs, % entrepreneurs	Percentage of those involved in early-stage entrepreneurial activity who expect to create 6 or more jobs in 5 years.	Global Entrepreneurship Monitor (GEM)
Variation	Geographical dispersion of start-ups, 0-100 high conc.	Herfindahl Hirschman Index, calculated using the share of equity-based start-ups (0-5 years old) located in different cities within a country. A higher value of this indicator implies a higher concentration of start-ups in the main cities.	Crunchbase
	Regional dispersion of enterp. birth, st dev.	Standard deviation of birth employer firms' density (standard deviation of births rates percentages across TL2 regions, divided by 1 000 population, within a country). Based on total economy aggregate values (industry, construction, and business services).	OECD - SDBS business demography
	Women founders, % founders	Female founders as a share of total founders. Founders are defined as people whose job description refers to CEO, founder, co-founder, member of the board or partner. To avoid that extremely small samples drive results, the women founders share is computed only if the total number of entrepreneurs recorded in Crunchbase is at least 25.	Crunchbase
	Missing entrepreneurs' rate, % early stage entrepreneurs	Size of estimated "missing entrepreneurs" group divided by the population all early-stage entrepreneurs (missing entrepreneurs calculated as the additional number of entrepreneurs there would be if women, youth, seniors and immigrants created businesses at the same rate as males 30-49 years old).	OECD Missing Entrepreneurs publication series

Table A A.42. Missing values count by indicator and year

Count of missing of moving average values by indicator and year

Indicator	2016	2017	2018	2019	2020	2021	2022	2023
Gross domestic product, PPP$ million	0	0	0	0	0	0	0	0
Rule of law, 0-100 best	0	0	0	0	0	0	0	0
Serial entrepreneurs, unit count	0	0	0	0	0	0	0	0
Coaches, unit count	0	0	0	0	0	0	0	0
Control of corruption index, 0-100 low incidence	1	1	1	1	1	0	0	0
University-business collaboration, 1-7 best	0	1	0	2	1	1	1	3
Internet users, % pop.	0	0	0	0	0	0	0	10

Effective tax rate, % taxable income	38	0	0	0	0	0	0	0
Fix broadband, subs. per 100 pop.	0	0	0	0	0	0	0	38
Incubators, per million pop.	0	0	0	0	0	0	0	38
R&D expenditure, % GDP	3	0	4	2	4	2	8	37
Technical employment, % total employment	4	4	4	4	6	4	5	32
Patents, per million pop.	0	0	0	0	0	0	38	38
Factoring, thousands USD per capita	5	5	5	5	5	5	11	38
Outstanding SME loans, thousands USD per capita	8	8	7	6	5	5	6	38
Perceived entrepreneurial capabilities, % 18-64 pop.	10	13	15	13	17	12	13	13
Early-stage VC investment, USD per capita	10	10	11	10	10	10	10	38
Mobile data use, Gb per subscrip./month	38	38	38	0	0	0	0	0
Later-stage VC investment, USD per capita	12	10	12	12	10	10	10	38
Entrepreneurship as a good career choice, % 18-64 pop.	11	13	15	13	23	18	16	15
High status to successful entrepreneurs, % 18-64 pop.	11	13	15	13	23	19	16	15
GitHub software uploads, per thousand people	38	38	38	38	0	0	0	38
Transport infrastructure quality, 1-5 high	0	38	0	38	38	38	0	38
Trade facilitation index, 0-2 best	38	1	38	0	38	38	0	38
SMEs collaborating on innovation, % total SMEs	38	38	7	38	7	38	9	38
Product Market Regulation, Index 0-6 stringent	38	38	0	38	38	38	38	0
Pisa, score	38	38	0	38	38	38	1	38
Mean years of schooling, years	38	19	24	22	18	34	36	38

Table A A.43. Missing values count by country and year

Count of missing of moving average values by country and year

country	2016	2017	2018	2019	2020	2021	2022	2023
Australia	12	9	9	7	12	12	10	19
Austria	12	11	5	10	7	11	4	19
Belgium	12	10	8	10	9	10	7	19
Canada	10	9	7	8	6	9	6	16
Switzerland	10	7	5	6	8	7	7	15
Chile	12	10	7	10	11	11	8	16
Colombia	11	10	6	8	8	10	9	17
Costa Rica	16	15	12	15	13	14	13	19
Czechia	13	10	9	9	9	10	6	19
Germany	10	8	3	7	6	7	4	16
Denmark	13	11	8	11	9	11	8	19
Spain	9	7	4	7	5	7	4	16
Estonia	9	7	6	10	8	10	7	16
Finland	10	10	9	10	9	9	8	18
France	9	7	4	9	9	7	4	16
United Kingdom	9	7	5	7	6	6	4	17
Greece	9	7	5	7	5	7	5	16
Hungary	9	11	7	10	8	7	4	16
Ireland	9	7	5	7	9	7	9	20
Iceland	16	15	12	13	12	14	12	19
Israel	9	8	6	7	7	7	5	18
Italy	9	8	4	7	7	7	7	15
Japan	14	10	7	8	10	9	6	20
Korea	11	10	6	9	8	9	7	16
Lithuania	13	10	8	11	8	10	4	16
Luxembourg	12	11	8	11	11	11	9	18

Latvia	9	8	8	6	6	6	4	16
Mexico	14	9	10	10	13	13	8	16
Netherlands	9	7	4	7	7	9	5	18
Norway	12	10	7	7	9	10	5	17
New Zealand	16	14	13	13	12	13	12	20
Poland	9	7	5	7	5	7	4	16
Portugal	9	10	7	6	7	10	7	19
Slovak Republic	10	7	5	7	6	7	3	16
Slovenia	10	6	6	7	6	7	4	16
Sweden	9	6	5	6	8	9	7	16
Türkiye	12	14	7	12	12	10	10	19
United States	10	7	6	8	6	7	6	17

Table A A.44. Missing values post application of moving averages

Country	Total	Institut.	Culture	Network	Infrastr.	Markets	Finance	Know ledge	Talent	Leaders hip	Interm. Serv.
Australia	6	-	2	-	-	-	3	-	1	-	-
Austria	1	-	-	-	-	-	-	-	1	-	-
Belgium	5	-	3	-	-	-	-	-	2	-	-
Canada	1	-	-	-	-	-	-	-	-	-	1
Chile	4	-	-	-	-	-	2	-	1	-	1
Colombia	3	-	-	1	-	-	2	-	-	-	-
Costa Rica	9	-	3	1	-	-	4	-	1	-	-
Czechia	4	-	2	-	-	-	-	-	2	-	-
Denmark	4	-	2	-	-	-	1	-	1	-	-
Estonia	0	-	-	-	-	-	-	-	-	-	-
Finland	3	-	2	-	-	-	-	-	1	-	-
France	1	-	-	-	-	-	-	-	1	-	-
Germany	0	-	-	-	-	-	-	-	-	-	-
Greece	0	-	-	-	-	-	-	-	-	-	-
Hungary	0	-	-	-	-	-	-	-	-	-	-
Iceland	7	-	2	-	-	-	4	-	1	-	-
Ireland	2	-	1	-	-	-	-	-	1	-	-
Israel	3	-	1	1	-	-	-	-	1	-	-
Italy	0	-	-	-	-	-	-	-	-	-	-
Japan	2	-	-	-	-	-	2	-	-	-	-
Korea	3	-	-	-	-	-	2	-	1	-	-
Latvia	1	-	-	-	-	-	-	-	1	-	-
Lithuania	0	-	-	-	-	-	-	-	-	-	-
Luxembourg	7	-	3	-	-	-	2	-	2	-	-
Mexico	4	-	-	1	-	-	2	1	-	-	-
Netherlands	2	-	2	-	-	-	-	-	-	-	-
New Zealand	7	-	2	1	-	-	3	-	1	-	-
Norway	2	-	-	1	-	-	-	-	1	-	-
Poland	0	-	-	-	-	-	-	-	-	-	-
Portugal	3	-	2	-	-	-	-	-	1	-	-
Slovak Republic	1	-	-	-	-	-	-	-	1	-	-
Slovenia	1	-	-	-	-	-	-	-	1	-	-
Spain	0	-	-	-	-	-	-	-	-	-	-
Sweden	1	-	-	-	-	-	-	-	1	-	-
Switzerland	2	-	-	1	-	-	-	-	1	-	-
Türkiye	4	-	-	-	-	-	2	-	1	-	1

United Kingdom	0	-	-	-	-	-	-	-	-	-	-
United States	1	-	-	-	-	-	-	-	-	-	1

Table A A.45. List of variables imputed and the main estimator used for linear regression estimation

Variable	Main estimator	Main estimator source	Note
Entrepreneurship as a good career choice	Self employment rate	OECD	1 datapoint could not be estimated
High status to successful entrepreneurs.	Self employment rate	OECD	1 datapoint could not be estimated
Perceived entrepreneurial capabilities	Self employment rate	OECD	1 datapoint could not be estimated
Trust in others	Control of corruption	V-Dem	
Venture capital early-stage investment	VC funding rounds	Crunbase	Outliers (ISR USA) excluded from the regression
Venture capital growth investment	VC funding rounds	Crunbase	Outliers (ISR USA) excluded from the regression
Outstanding SME loans	Domestic credit to GDP	World Bank	
Factoring	Domestic credit to GDP	World Bank	
Mean years of schooling	Share of the adult population with tertiary education	OECD	
University-business collaboration	SMEs collaborating on innovation	Eurostat	Only 1 country estimated with this method
SMEs collaborating on innovation	University-business collaboration	World Economic Forum	Only 1 country estimated with this method
Technical employment	No estimator available	-	4 missing values
Pisa score	No estimator available	-	1 missing values
R&D expenditure	No estimator available	-	1 missing values

Table A A.46. Weighting scheme, alternative specification

Element	Variable	Weight
1.Institutions	Rule of law, 0-100 best	22%
1.Institutions	Effective tax rate, % taxable income	27%
1.Institutions	Product Market Regulation, Index 0-6 stringent	27%
1.Institutions	Control of corruption index, 0-100 low incidence	24%
2.Culture	Entrepreneurship as a good career choice, % 18-64 pop.	41%
2.Culture	High status to successful entrepreneurs, % 18-64 pop.	25%
2.Culture	Trust in others, % respondents	34%
3.Networks	SMEs collaborating on innovation, % total SMEs	50%
3.Networks	University-business collaboration, 1-7 best	50%
4.Infrastructure	Fix broadband, subs. per 100 pop.	29%
4.Infrastructure	Transport infrastructure quality, 1-5 high	32%
4.Infrastructure	Mobile data use, Gb per subscrip./month	39%
5.Markets	Gross domestic product, PPP$ million	50%
5.Markets	Trade facilitation index, 0-2 best	50%
6.Finance	Outstanding SME loans, thousands USD per capita	37%
6.Finance	Factoring, thousands USD per capita	10%
6.Finance	Early-stage VC investment, USD per capita	26%
6.Finance	Later-stage VC investment, USD per capita	27%
7.Knowledge	Patents, per million pop.	31%
7.Knowledge	R&D expenditure, % GDP	27%

7.Knowledge	GitHub software uploads, per thousand people	41%
8.Talent	Perceived entrepreneurial capabilities, % 18-64 pop.	34%
8.Talent	Mean years of schooling, years	21%
8.Talent	Pisa, score	19%
8.Talent	Internet users, % pop.	26%
9.Leadership	Serial entrepreneurs, unit count	100%
10.Intermediate services	Coaches, unit count	38%
10.Intermediate services	Incubators, per million pop.	39%
10.Intermediate services	Technical employment, % total employment	23%

Note : Weights derived from PCA analysis

Table A A.47. Elemements' internal statistical consistency

Crobach's alpha of all element

Element	K-Alpha	Interpretation
Institutions	0.51	Moderate internal consistency
Networks	0.18	Limited internal consistency
Culture	0.42	Moderate internal consistency
Infrastructure	0.60	Moderate internal consistency
Finance	0.28	Limited internal consistency
Knowledge	0.82	High internal consistency
Markets	0.44	Moderate internal consistency
Talent	0.81	High internal consistency
Leadrship	-	One indicator
Interediate services	0.41	Moderate internal consistency

Table A A.48. Elements' scores by country, baseline

Aggregate scores(0-100 scale) of the ten elements for the 38 countries using geometric mean aggregation

Country	Institutions	Culture	Networks	Infrastruct.	Markets	Finance	Knowledge	Talent	Leadership	Interm. Serv.
Australia	40.3	.	49.9	55.0	68.0	.	43.2	64.0	73.6	65.9
Austria	49.6	43.0	49.3	59.8	51.9	50.8	57.0	55.4	47.9	54.8
Belgium	48.1	.	63.2	52.1	54.2	58.4	57.2	.	41.3	50.3
Canada	53.4	70.0	61.2	52.9	64.5	44.9	51.6	69.1	90.9	74.2
Chile	50.4	38.8	25.9	17.3	21.0	.	18.1	35.0	30.4	32.8
Colombia	6.6	14.7	.	5.3	25.1	.	17.8	3.0	37.5	17.7
Costa Rica	9.6	.	.	5.7	15.0	.	19.7	2.2	23.6	14.9
Czechia	51.7	.	44.9	32.3	36.0	39.7	42.2	.	27.7	36.2
Denmark	67.9	.	51.5	71.9	51.3	56.6	69.2	69.4	51.5	50.7
Estonia	65.3	43.7	44.8	51.4	24.1	45.6	48.9	61.3	33.8	48.6
Finland	68.1	.	65.4	71.5	51.0	57.6	72.1	52.9	60.2	61.1
France	51.0	27.1	40.4	63.5	70.3	44.1	48.9	42.0	70.0	51.5
Germany	46.9	53.1	50.8	59.9	69.2	49.5	60.2	52.1	84.0	49.3
Greece	31.8	37.3	6.7	46.4	26.3	38.2	29.4	27.7	32.7	34.4
Hungary	18.1	40.7	30.9	35.7	12.4	36.7	33.7	38.5	25.4	33.2
Iceland	56.3	.	50.8	57.2	4.4	.	65.2	62.6	7.0	46.9
Ireland	76.2	46.0	54.2	45.2	53.7	54.1	47.5	56.3	47.9	64.4
Israel	42.0	61.9	.	39.4	12.0	65.3	86.1	40.7	79.1	78.7
Italy	36.6	18.3	44.3	51.1	53.8	47.7	30.2	27.9	56.0	46.2

Japan	37.2	10.4	49.0	52.5	80.6	.	54.8	20.7	49.5	27.2
Korea	44.6	55.9	49.6	66.0	79.1	.	81.2	73.6	45.0	36.1
Latvia	54.8	28.0	27.5	39.5	20.7	36.6	30.9	57.6	29.7	27.2
Lithuania	62.5	43.3	42.9	44.4	33.9	37.3	35.8	50.0	17.2	30.1
Luxembourg	42.7	.	46.9	47.6	32.1	.	39.0	.	33.8	52.0
Mexico	4.6	28.3	.	6.9	35.3	.	14.1	3.7	50.1	29.3
Netherlands	60.1	.	57.8	55.3	70.9	57.3	64.6	51.5	55.2	69.1
New Zealand	48.6	.	.	44.4	45.9	.	44.7	49.0	37.1	50.7
Norway	66.0	82.0	.	59.3	59.0	52.6	58.5	57.1	53.4	43.9
Poland	58.6	28.7	17.4	29.4	63.3	41.3	34.6	53.3	49.5	32.6
Portugal	36.4	.	35.3	45.9	53.5	38.7	37.5	27.2	38.8	41.9
Slovak Republic	43.7	23.6	15.2	34.5	30.1	38.7	26.6	45.2	18.4	21.6
Slovenia	51.1	57.2	35.8	43.5	30.6	39.4	40.0	58.0	26.9	37.9
Spain	48.6	23.6	25.1	47.8	66.4	50.9	35.0	44.6	68.4	51.8
Sweden	70.1	67.1	54.6	69.7	60.9	46.7	79.7	57.8	62.9	67.9
Switzerland	63.0	33.7	.	77.0	55.0	52.2	87.3	62.5	68.4	76.6
Türkiye	3.0	39.8	23.1	25.2	31.4	.	21.7	14.8	45.3	33.4
United Kingdom	70.9	69.2	67.3	51.5	73.6	41.0	57.1	64.7	97.5	83.0
United States	52.6	67.0	95.3	54.0	86.8	59.3	65.5	70.6	100.0	79.8

Table A A.49. Correlations across elements' scores

Pairwise correlation coefficients among the ten elements aggregate scores (geometric mean)

	Institutions	Culture	Networks	Infrastructure	Markets	Finance	Knowledge	Talent	Leadership	Int. Serv.
Institutions	1									
Culture	0.4583*	1								
Networks	0.4469*	0.6149*	`1							
Infrastructure	0.6910*	0.3940*	0.5883*	1						
Markets	0.3598*	0.2021	0.5249*	0.5389*	1					
Finance	0.296	0.3448	0.6736*	0.4844*	0.3363	1				
Knowledge	0.6062*	0.5526*	0.7339*	0.8086*	0.4164*	0.7447*	1			
Talent	0.8202*	0.6346*	0.5660*	0.7434*	0.3858*	0.2855	0.6666*	1		
Leadership	0.233	0.4388*	0.5676*	0.3913*	0.6993*	0.5225*	0.4598*	0.3189	1	
Int. Serv.	0.5617*	0.6040*	0.7680*	0.6239*	0.4641*	0.6585*	0.7020*	0.6113*	0.7520*	1

Table A A.50. Elements' scores by country, sensitivity analysis (1/3)

Aggregate scores(0-100 scale) of the ten elements for the 38 countries using three alternative aggregation methods

Element	Institutions			Culture			Networks		
Aggregation	Artimetic mean	Geometric mean	Weighted average	Artimetic mean	Geometric mean	Weighted average	Artimetic mean	Geometric mean	Weighted average
Australia	49.0	40.3	47.4	.	.	.	50.1	49.9	50.1
Austria	51.9	49.6	50.6	48.9	43.0	45.0	50.1	49.3	50.1
Belgium	49.5	48.1	48.8	.	.	.	64.0	63.2	64.0
Canada	55.0	53.4	53.9	70.1	70.0	69.9	62.6	61.2	62.6
Chile	50.8	50.4	50.9	47.1	38.8	50.2	28.6	25.9	28.6
Colombia	9.6	6.6	10.2	19.2	14.7	20.6	.	.	.
Costa Rica	18.2	9.6	17.5	.	.	.	.	.	.
Czechia	52.6	51.7	53.4	.	.	.	45.8	44.9	45.8
Denmark	68.3	67.9	67.9	.	.	.	53.9	51.5	53.9

Estonia	66.2	65.3	67.1	43.9	43.7	43.5	45.6	44.8	45.6
Finland	68.5	68.1	68.0	.	.	.	66.1	65.4	66.1
France	54.3	51.0	53.5	36.1	27.1	40.5	40.4	40.4	40.4
Germany	53.0	46.9	51.8	55.3	53.1	52.4	53.2	50.8	53.2
Greece	35.4	31.8	36.7	43.3	37.3	45.0	22.5	6.7	22.5
Hungary	41.5	18.1	43.3	41.9	40.7	43.8	30.9	30.9	30.9
Iceland	57.7	56.3	57.0	.	.	.	50.8	50.8	50.8
Ireland	77.5	76.2	78.7	46.4	46.0	47.2	55.5	54.2	55.5
Israel	43.3	42.0	43.6	63.1	61.9	60.9	.	.	.
Italy	45.6	36.6	47.4	28.9	18.3	32.3	44.9	44.3	44.9
Japan	42.3	37.2	40.9	23.8	10.4	22.2	49.3	49.0	49.3
Korea	45.5	44.6	45.0	60.5	55.9	55.9	50.9	49.6	50.9
Latvia	55.7	54.8	56.4	28.9	28.0	30.2	27.5	27.5	27.5
Lithuania	65.4	62.5	67.0	47.1	43.3	50.8	43.3	42.9	43.3
Luxembourg	48.2	42.7	46.6	.	.	.	49.6	46.9	49.6
Mexico	10.5	4.6	11.3	32.5	28.3	34.6	.	.	.
Netherlands	62.3	60.1	61.8	.	.	.	61.1	57.8	61.1
New Zealand	54.4	48.6	52.8	.	.	.	.	.	.
Norway	67.0	66.0	66.4	84.3	82.0	81.6	.	.	.
Poland	60.8	58.6	62.1	28.8	28.7	28.7	18.7	17.4	18.7
Portugal	39.1	36.4	38.0	.	.	.	36.4	35.3	36.4
Slovak Republic	44.5	43.7	45.2	23.6	23.6	23.9	18.5	15.2	18.5
Slovenia	54.2	51.1	55.1	62.5	57.2	60.4	35.8	35.8	35.8
Spain	50.1	48.6	50.4	26.0	23.6	27.2	25.2	25.1	25.2
Sweden	71.1	70.1	70.8	68.7	67.1	67.9	56.6	54.6	56.6
Switzerland	63.7	63.0	63.2	47.9	33.7	44.4	.	.	.
Türkiye	19.1	3.0	20.9	46.6	39.8	47.2	23.4	23.1	23.4
United Kingdom	71.3	70.9	71.8	69.5	69.2	68.8	67.3	67.3	67.3
United States	54.1	52.6	53.2	68.1	67.0	68.7	95.4	95.3	95.4

Table A A.51. Elements' scores by country, sensitivity analysis (2/3)

Aggregate scores(0-100 scale) of the ten elements for the 38 countries using three alternative aggregation methods

Element	Infrastructure			Markets			Finance		
Country	Artimetic mean	Geometric mean	Weighted average	Artimetic mean	Geometric mean	Weighted average	Artimetic mean	Geometric mean	Weighted average
Australia	56.3	55.0	55.6	68.1	68.0	68.1	.	.	.
Austria	65.6	59.8	68.8	52.2	51.9	52.2	52.8	50.8	49.5
Belgium	57.6	52.1	55.0	54.3	54.2	54.3	62.5	58.4	55.8
Canada	58.6	52.9	56.0	64.7	64.5	64.7	44.4	44.9	48.4
Chile	27.1	17.3	30.1	28.1	21.0	28.1	.	.	.
Colombia	11.0	5.3	12.5	33.5	25.1	33.5	.	.	.
Costa Rica	11.3	5.7	12.5	15.9	15.0	15.9	.	.	.
Czechia	35.3	32.3	34.2	37.0	36.0	37.0	41.1	39.7	40.0
Denmark	72.2	71.9	71.4	52.4	51.3	52.4	55.9	56.6	55.3
Estonia	53.6	51.4	55.1	37.5	24.1	37.5	44.4	45.6	44.5
Finland	74.9	71.5	77.6	53.6	51.0	53.6	58.8	57.6	54.7
France	64.9	63.5	63.3	70.8	70.3	70.8	45.7	44.1	46.9
Germany	65.6	59.9	62.9	70.7	69.2	70.7	51.8	49.5	46.8
Greece	48.9	46.4	47.0	28.5	26.3	28.5	39.9	38.2	39.6
Hungary	37.8	35.7	37.6	22.0	12.4	22.0	38.2	36.7	38.1
Iceland	59.0	57.2	60.2	9.5	4.4	9.5	.	.	.

Ireland	46.2	45.2	47.5	54.1	53.7	54.1	57.5	54.1	49.5
Israel	40.1	39.4	40.7	23.1	12.0	23.1	60.5	65.3	76.0
Italy	51.8	51.1	52.4	56.9	53.8	56.9	51.6	47.7	44.3
Japan	55.8	52.5	54.5	80.7	80.6	80.7	.	.	.
Korea	66.9	66.0	65.6	79.3	79.1	79.3	.	.	.
Latvia	49.9	39.5	54.2	25.2	20.7	25.2	38.3	36.6	37.9
Lithuania	50.7	44.4	54.0	37.1	33.9	37.1	38.8	37.3	38.7
Luxembourg	48.4	47.6	47.3	42.0	32.1	42.0	.	.	.
Mexico	10.9	6.9	12.2	45.4	35.3	45.4	.	.	.
Netherlands	61.4	55.3	58.6	71.9	70.9	71.9	60.4	57.3	53.1
New Zealand	47.1	44.4	45.5	48.5	45.9	48.5	.	.	.
Norway	61.3	59.3	59.4	61.1	59.0	61.1	54.4	52.6	50.9
Poland	32.8	29.4	34.4	63.3	63.3	63.3	43.3	41.3	39.6
Portugal	48.6	45.9	46.7	55.1	53.5	55.1	40.6	38.7	41.4
Slovak Republic	35.2	34.5	34.9	30.1	30.1	30.1	40.3	38.7	39.7
Slovenia	43.7	43.5	44.2	35.7	30.6	35.7	41.0	39.4	39.3
Spain	48.6	47.8	47.8	66.5	66.4	66.5	54.5	50.9	47.6
Sweden	70.1	69.7	69.7	62.3	60.9	62.3	47.4	46.7	52.0
Switzerland	78.2	77.0	76.6	55.2	55.0	55.2	55.9	52.2	65.9
Türkiye	30.6	25.2	32.5	44.0	31.4	44.0	.	.	.
United Kingdom	53.7	51.5	51.9	73.8	73.6	73.8	42.3	41.0	43.1
United States	55.1	54.0	54.0	87.6	86.8	87.6	67.7	59.3	70.2

Table A A.52. Element scores by country, sensitivity analysis (3/3)

Aggregate scores(0-100 scale) of the ten elements for the 38 countries using three alternative aggregation methods

Element	Knowledge			Talent			Leadership			. Interm. services		
Country	Artim. mean	Geom. mean	Weight. Avg.	Artim. mean	Geom. mean	Weight. Avg.	Artim. mean	Geom. mean	Weight. Avg.	Artim. mean	Geom. mean	Weight. Avg.
Australia	43.4	43.2	44.1	64.1	64.0	.	73.6	73.6	73.6	67.2	65.9	117.1
Austria	57.9	57.0	56.2	55.5	55.4	55.8	47.9	47.9	47.9	55.9	54.8	82.6
Belgium	58.5	57.2	56.6	.	.	.	41.3	41.3	41.3	50.4	50.3	81.1
Canada	53.7	51.6	56.4	69.4	69.1	68.4	90.9	90.9	90.9	75.9	74.2	.
Chile	19.3	18.1	19.0	46.5	35.0	54.0	30.4	30.4	30.4	33.2	32.8	.
Colombia	19.0	17.8	18.9	20.8	3.0	28.2	37.5	37.5	37.5	23.7	17.7	54.0
Costa Rica	20.9	19.7	21.1	5.5	2.2	5.7	23.6	23.6	23.6	16.2	14.9	23.7
Czechia	42.7	42.2	43.0	.	.	.	27.7	27.7	27.7	36.3	36.2	60.6
Denmark	69.4	69.2	70.4	69.9	69.4	71.2	51.5	51.5	51.5	50.9	50.7	81.3
Estonia	52.0	48.9	55.1	63.1	61.3	59.9	33.8	33.8	33.8	52.8	48.6	80.0
Finland	72.3	72.1	73.4	55.0	52.9	52.1	60.2	60.2	60.2	61.8	61.1	93.0
France	48.9	48.9	48.5	42.6	42.0	42.4	70.0	70.0	70.0	53.0	51.5	92.9
Germany	61.0	60.2	59.3	55.7	52.1	52.3	84.0	84.0	84.0	52.9	49.3	106.2
Greece	30.0	29.4	29.0	32.0	27.7	34.8	32.7	32.7	32.7	35.2	34.4	56.5
Hungary	33.9	33.7	33.5	40.9	38.5	38.3	25.4	25.4	25.4	33.4	33.2	53.6
Iceland	66.7	65.2	69.0	67.7	62.6	70.8	7.0	7.0	7.0	58.4	46.9	70.7
Ireland	51.0	47.5	54.2	57.8	56.3	58.0	47.9	47.9	47.9	64.6	64.4	100.3
Israel	86.7	86.1	85.0	44.5	40.7	42.0	79.1	79.1	79.1	79.0	78.7	122.3
Italy	31.6	30.2	30.2	33.8	27.9	34.8	56.0	56.0	56.0	48.0	46.2	82.0
Japan	65.4	54.8	60.7	46.0	20.7	38.7	49.5	49.5	49.5	29.3	27.2	59.3
Korea	82.2	81.2	80.2	74.2	73.6	72.2	45.0	45.0	45.0	36.3	36.1	62.9
Latvia	32.2	30.9	33.9	57.8	57.6	57.7	29.7	29.7	29.7	28.7	27.2	42.7
Lithuania	36.7	35.8	38.2	51.4	50.0	50.6	17.2	17.2	17.2	34.0	30.1	43.5

Luxembourg	40.3	39.0	40.8	.	.	.	33.8	33.8	33.8	59.5	52.0	78.8
Mexico	18.0	14.1	.	23.3	3.7	31.4	50.1	50.1	50.1	31.8	29.3	65.2
Netherlands	65.5	64.6	67.4	52.5	51.5	50.9	55.2	55.2	55.2	70.2	69.1	109.6
New Zealand	46.0	44.7	47.9	54.4	49.0	50.6	37.1	37.1	37.1	52.1	50.7	76.6
Norway	62.1	58.5	66.0	58.8	57.1	57.8	53.4	53.4	53.4	44.6	43.9	65.2
Poland	34.8	34.6	35.2	55.3	53.3	54.6	49.5	49.5	49.5	34.6	32.6	68.5
Portugal	37.8	37.5	38.0	35.7	27.2	38.1	38.8	38.8	38.8	42.1	41.9	68.5
Slovak Republic	26.8	26.6	26.5	46.1	45.2	46.2	18.4	18.4	18.4	22.3	21.6	33.6
Slovenia	40.6	40.0	40.0	58.9	58.0	60.3	26.9	26.9	26.9	40.2	37.9	53.0
Spain	35.0	35.0	35.0	47.4	44.6	48.6	68.4	68.4	68.4	52.9	51.8	96.4
Sweden	79.8	79.7	80.2	58.2	57.8	57.5	62.9	62.9	62.9	70.7	67.9	103.0
Switzerland	88.1	87.3	88.9	64.0	62.5	61.6	68.4	68.4	68.4	77.6	76.6	115.1
Türkiye	24.7	21.7	22.8	31.0	14.8	35.3	45.3	45.3	45.3	35.5	33.4	.
United Kingdom	57.6	57.1	57.4	64.9	64.7	64.1	97.5	97.5	97.5	84.9	83.0	145.6
United States	66.0	65.5	65.1	71.0	70.6	71.8	100.0	100.0	100.0	81.9	79.8	.

Table A A.53. Elements scores by country, imputation robustness (1/2)

Aggregate scores of the ten elements by country using geometric aggregation imputated and non-imputed indicators

Country	Institutions		Culture		Networks		Infrastructure		Markets	
	WITH imp.	No imp.	WITH imp.	No imp.	WITH imp.	No imp.	WITH imp.	No imp.	WITH imp.	No imp.
Australia	40.3	40.3	.	.	49.9	53.9	55.0	55.0	68.0	68.0
Austria	49.6	49.6	43.0	44.5	49.3	52.5	59.8	59.8	51.9	51.9
Belgium	48.1	48.1	.	.	63.2	69.1	52.1	52.1	54.2	54.2
Canada	53.4	53.4	70.0	70.4	61.2	66.4	52.9	52.9	64.5	64.5
Chile	50.4	50.4	38.8	41.7	25.9	28.0	17.3	17.3	21.0	21.0
Colombia	6.6	6.6	14.7	19.1	.	.	5.3	5.3	25.1	25.1
Costa Rica	9.6	9.6	.	.	.	.	5.7	5.7	15.0	15.0
Czechia	51.7	51.7	.	.	44.9	47.2	32.3	32.3	36.0	36.0
Denmark	67.9	67.9	.	.	51.5	54.3	71.9	71.9	51.3	51.3
Estonia	65.3	65.3	43.7	45.5	44.8	49.3	51.4	51.4	24.1	24.1
Finland	68.1	68.1	.	.	65.4	71.9	71.5	71.5	51.0	51.0
France	51.0	51.0	27.1	33.4	40.4	42.8	63.5	63.5	70.3	70.3
Germany	46.9	46.9	53.1	54.1	50.8	53.4	59.9	59.9	69.2	69.2
Greece	31.8	31.8	37.3	40.0	6.7	7.2	46.4	46.4	26.3	26.3
Hungary	18.1	18.1	40.7	43.3	30.9	31.9	35.7	35.7	12.4	12.4
Iceland	56.3	56.3	.	.	50.8	55.2	57.2	57.2	4.4	4.4
Ireland	76.2	76.2	46.0	46.5	54.2	58.1	45.2	45.2	53.7	53.7
Israel	42.0	42.0	61.9	68.2	.	.	39.4	39.4	12.0	12.0
Italy	36.6	36.6	18.3	26.9	44.3	46.7	51.1	51.1	53.8	53.8
Japan	37.2	37.2	10.4	11.1	49.0	52.8	52.5	52.5	80.6	80.6
Korea	44.6	44.6	55.9	56.5	49.6	52.6	66.0	66.0	79.1	79.1
Latvia	54.8	54.8	28.0	31.4	27.5	27.5	39.5	39.5	20.7	20.7
Lithuania	62.5	62.5	43.3	46.3	42.9	45.2	44.4	44.4	33.9	33.9
Luxembourg	42.7	42.7	.	.	46.9	48.5	47.6	47.6	32.1	32.1
Mexico	4.6	4.6	28.3	31.6	.	.	6.9	6.9	35.3	35.3
Netherlands	60.1	60.1	.	.	57.8	61.5	55.3	55.3	70.9	70.9
New Zealand	48.6	48.6	.	.	.	.	44.4	44.4	45.9	45.9
Norway	66.0	66.0	82.0	81.8	.	.	59.3	59.3	59.0	59.0

Poland	58.6	58.6	28.7	31.4	17.4	17.6	29.4	29.4	63.3	63.3
Portugal	36.4	36.4	.	.	35.3	35.5	45.9	45.9	53.5	53.5
Slovak Republic	43.7	43.7	23.6	26.7	15.2	16.0	34.5	34.5	30.1	30.1
Slovenia	51.1	51.1	57.2	58.2	35.8	37.7	43.5	43.5	30.6	30.6
Spain	48.6	48.6	23.6	27.7	25.1	25.4	47.8	47.8	66.4	66.4
Sweden	70.1	70.1	67.1	67.6	54.6	58.2	69.7	69.7	60.9	60.9
Switzerland	63.0	63.0	33.7	36.6	.	.	77.0	77.0	55.0	55.0
Türkiye	3.0	3.0	39.8	42.3	23.1	23.3	25.2	25.2	31.4	31.4
United Kingdom	70.9	70.9	69.2	69.5	67.3	74.9	51.5	51.5	73.6	73.6
United States	52.6	52.6	67.0	67.6	95.3	95.0	54.0	54.0	86.8	86.8

Table A A.54. Elements scores by country, imputation robustness (2/2)

Aggregate scores of the ten elements by country using geometric aggregation imputated and non-imputed indicators

Country	Finance		Knowledge		Talent		Leadership		. Interm. services	
	WITH imp.	No imp.	WITH imp.	No imp.	WITH imp.	No imp.	WITH imp.	No imp.	WITH imp.	No imp.
Australia	.	.	43.2	43.2	64.0	64.0	73.6	73.6	65.9	65.9
Austria	50.8	49.9	57.0	57.0	55.4	56.9	47.9	47.9	54.8	54.8
Belgium	58.4	57.8	57.2	57.2	.	.	41.3	41.3	50.3	50.3
Canada	44.9	44.9	51.6	51.6	69.1	68.4	90.9	90.9	74.2	74.2
Chile	.	.	18.1	18.1	35.0	39.7	30.4	30.4	32.8	32.8
Colombia	.	.	17.8	17.8	3.0	4.3	37.5	37.5	17.7	17.7
Costa Rica	.	.	19.7	19.7	2.2	4.2	23.6	23.6	14.9	14.9
Czechia	39.7	39.7	42.2	42.2	.	.	27.7	27.7	36.2	36.2
Denmark	56.6	54.6	69.2	69.2	69.4	67.2	51.5	51.5	50.7	50.7
Estonia	45.6	45.4	48.9	48.9	61.3	60.5	33.8	33.8	48.6	48.6
Finland	57.6	56.4	72.1	72.1	52.9	49.7	60.2	60.2	61.1	61.1
France	44.1	43.8	48.9	48.9	42.0	42.6	70.0	70.0	51.5	51.5
Germany	49.5	48.9	60.2	60.2	52.1	50.8	84.0	84.0	49.3	49.3
Greece	38.2	38.4	29.4	29.4	27.7	28.1	32.7	32.7	34.4	34.4
Hungary	36.7	37.0	33.7	33.7	38.5	37.6	25.4	25.4	33.2	33.2
Iceland	.	.	65.2	65.2	62.6	57.3	7.0	7.0	46.9	46.9
Ireland	54.1	53.3	47.5	47.5	56.3	64.0	47.9	47.9	64.4	64.4
Israel	65.3	65.7	86.1	86.1	40.7	33.0	79.1	79.1	78.7	78.7
Italy	47.7	47.2	30.2	30.2	27.9	28.7	56.0	56.0	46.2	46.2
Japan	.	.	54.8	54.8	20.7	20.7	49.5	49.5	27.2	27.2
Korea	.	.	81.2	81.2	73.6	72.9	45.0	45.0	36.1	36.1
Latvia	36.6	36.9	30.9	30.9	57.6	54.7	29.7	29.7	27.2	27.2
Lithuania	37.3	37.6	35.8	35.8	50.0	49.4	17.2	17.2	30.1	30.1
Luxembourg	.	.	39.0	39.0	.	.	33.8	33.8	52.0	52.0
Mexico	.	.	14.1	14.1	3.7	5.4	50.1	50.1	29.3	29.3
Netherlands	57.3	56.7	64.6	64.6	51.5	51.0	55.2	55.2	69.1	69.1
New Zealand	.	.	44.7	44.7	49.0	65.5	37.1	37.1	50.7	50.7
Norway	52.6	51.8	58.5	58.5	57.1	54.8	53.4	53.4	43.9	43.9
Poland	41.3	41.2	34.6	34.6	53.3	52.9	49.5	49.5	32.6	32.6
Portugal	38.7	38.8	37.5	37.5	27.2	25.2	38.8	38.8	41.9	41.9
Slovak Republic	38.7	38.8	26.6	26.6	45.2	40.5	18.4	18.4	21.6	21.6
Slovenia	39.4	39.4	40.0	40.0	58.0	57.6	26.9	26.9	37.9	37.9
Spain	50.9	50.1	35.0	35.0	44.6	46.2	68.4	68.4	51.8	51.8
Sweden	46.7	46.5	79.7	79.7	57.8	58.0	62.9	62.9	67.9	67.9
Switzerland	52.2	52.2	87.3	87.3	62.5	57.8	68.4	68.4	76.6	76.6
Türkiye	.	.	21.7	21.7	14.8	36.2	45.3	45.3	33.4	33.4

United Kingdom	41.0	41.1	57.1	57.1	64.7	64.1	97.5	97.5	83.0	83.0
United States	59.3	59.8	65.5	65.5	70.6	70.1	100.0	100.0	79.8	79.8

Table A A.55. Entrepreneurship output indicators 1/2

Output variables raw values (moving average 2020-2023)

Country	Birth rate of employer enterprises, % business pop.	Equity-based young firms, per million pop.	Unicorns, per million pop.	Enterprise churn rate, % business pop.
Australia	.	0.5	0.1	.
Austria	7.0	0.1	0.1	13.60
Belgium	3.5	0.2	0.1	4.30
Canada	.	0.4	0.2	.
Chile	.	0.1	0.0	.
Colombia	17.6	0.0	0.0	.
Costa Rica	5.2	0.0	0.0	10.23
Czechia	6.6	0.1	0.0	14.60
Denmark	12.2	0.4	0.1	13.90
Estonia	13.1	1.1	0.2	25.80
Finland	12.0	0.3	0.2	41.50
France	10.5	0.2	0.1	22.40
Germany	7.6	0.2	0.1	15.80
Greece	9.6	0.1	0.1	.
Hungary	9.7	0.1	0.0	29.00
Iceland	12.5	0.4	0.0	17.60
Ireland	3.8	0.4	0.4	5.20
Israel	11.1	0.5	0.7	19.00
Italy	7.4	0.2	0.0	15.30
Japan	.	0.1	0.0	.
Korea	14.6	0.1	0.0	.
Latvia	7.4	0.2	0.0	11.10
Lithuania	7.8	0.3	0.1	13.10
Luxembourg	.	0.5	0.0	.
Mexico	.	0.0	0.0	.
Netherlands	8.5	0.8	0.1	16.80
New Zealand	12.3	0.3	0.0	.
Norway	7.6	0.3	0.2	13.30
Poland	9.7	0.1	0.0	19.20
Portugal	8.2	0.2	0.0	16.20
Slovak Republic	6.4	0.1	0.0	12.50
Slovenia	7.4	0.1	0.0	16.40
Spain	8.5	0.2	0.0	17.90
Sweden	11.3	0.4	0.1	21.50
Switzerland	.	0.6	0.2	.
Türkiye	14.0	0.0	0.0	.
United Kingdom	.	0.5	0.2	.
United States	.	0.5	0.5	.

Table A A.56. Entrepreneurship output indicators 2/2

Output variables raw values (moving average 2020-2023)

Country	Medium and high-growth enterprises, % total	3-year survival rate., % new employer enterprises	Expectation to create jobs, % entrepreneurs	2-year-old employer enterprises, % business population
Australia	.	.	.	.
Austria	7.3	55.3	8.1	1.7
Belgium	6.9	75.6	.	1.1
Canada	.	.	19.6	.
Chile	.	.	32.9	.
Colombia	2.9	38.3	37.0	5.0
Costa Rica	14.4	.	.	.
Czechia	8.8	61.6	.	1.8
Denmark	11.6	22.9	.	1.7
Estonia	9.5	49.1	19.1	2.7
Finland	12.5	54.7	10.9	10.7
France	8.6	54.8	24.6	2.9
Germany	8.3	64.2	18.3	1.4
Greece	.	44.2	15.5	3.4
Hungary	10.4	19.7	15.6	2.8
Iceland	6.9	44.9	.	2.0
Ireland	.	81.9	31.5	1.3
Israel	7.8	56.0	17.5	.
Italy	9.5	53.1	14.8	2.8
Japan	.	.	21.1	.
Korea	9.0	.	27.4	.
Latvia	9.5	67.4	28.2	3.1
Lithuania	10.9	50.7	23.5	2.9
Luxembourg	.	.	31.3	.
Mexico	.	.	25.6	.
Netherlands	12.5	59.1	18.6	2.3
New Zealand	10.3	51.0	.	3.1
Norway	9.1	45.0	23.9	1.2
Poland	9.8	35.2	18.4	3.1
Portugal	11.6	65.1	.	3.0
Slovak Republic	8.6	56.2	19.9	2.8
Slovenia	11.0	54.7	19.3	2.2
Spain	11.3	53.0	10.0	2.7
Sweden	15.7	88.7	13.9	1.8
Switzerland	.	.	14.5	.
Türkiye	12.3	48.2	53.8	5.8
United Kingdom	.	.	20.0	.
United States	.	.	27.1	.

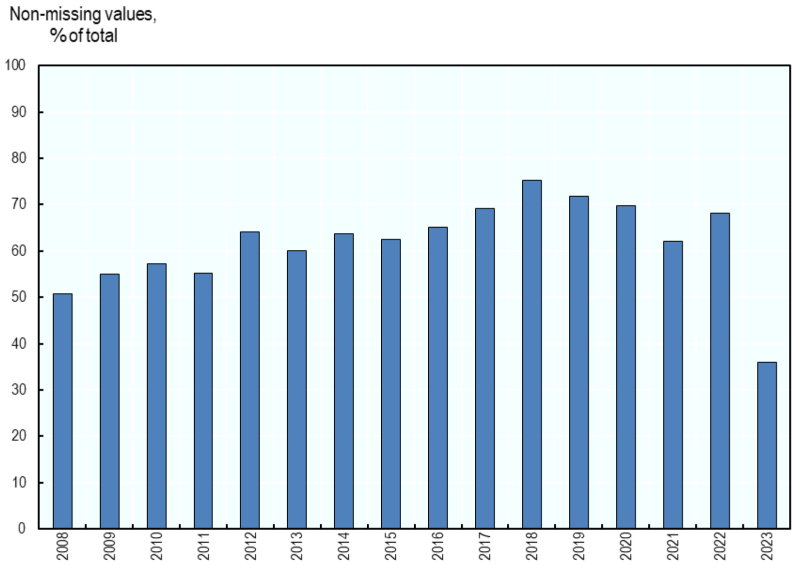
Table A A.57. Entrepreneurship variation indicators

Variation variables raw values (moving average 2020-2023)

Country	Geographical dispersion of start-ups, 0-100 high conc.	Missing entrepreneurs' rate, % early stage entrepreneurs	Women founders, % founders
Australia	10.4	41.5	22.0
Austria	31.4	42.1	.

Belgium	6.0	.	.
Canada	10.5	26.8	23.1
Chile	45.8	26.9	.
Colombia	33.5	11.1	.
Costa Rica	12.6	.	.
Czechia	25.0	.	.
Denmark	23.8	.	.
Estonia	82.7	52.0	.
Finland	31.0	48.0	.
France	17.9	29.2	26.4
Germany	8.4	39.5	17.4
Greece	28.6	14.6	.
Hungary	39.4	41.1	.
Iceland	75.3	.	.
Ireland	40.2	23.2	23.6
Israel	26.6	33.4	17.5
Italy	6.5	81.9	9.4
Japan	25.4	64.7	.
Korea	17.0	41.0	.
Latvia	76.7	53.6	.
Lithuania	66.1	46.3	.
Luxembourg	46.2	28.7	.
Mexico	7.6	21.0	19.2
Netherlands	5.3	21.5	12.1
New Zealand	30.7	.	.
Norway	29.0	54.7	.
Poland	19.9	37.1	.
Portugal	14.2	31.9	.
Slovak Republic	46.7	42.8	.
Slovenia	41.2	54.7	.
Spain	13.7	28.8	17.3
Sweden	21.9	25.5	18.4
Switzerland	6.9	38.7	13.7
Türkiye	17.2	43.9	.
United Kingdom	29.2	45.5	25.3
United States	1.4	34.5	26.7

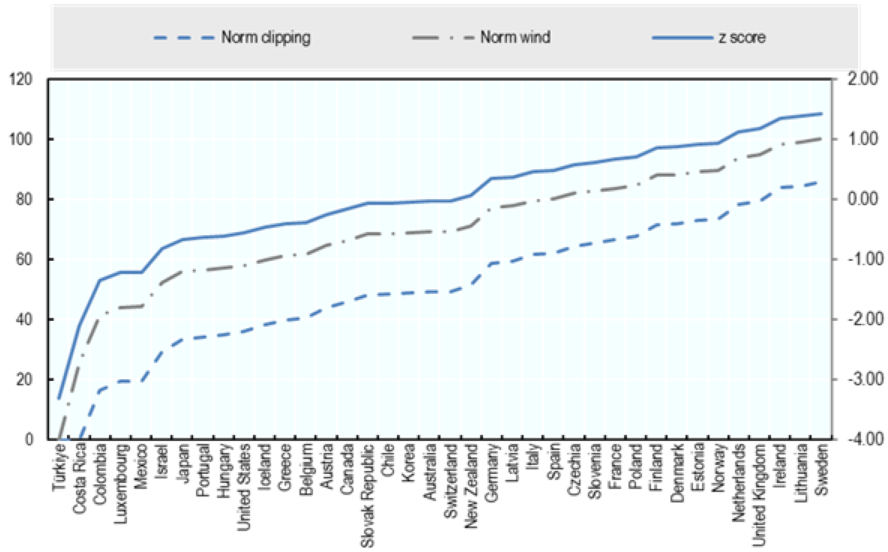
Figure A A.1. Total sample completion rate by year



Note: Analysis based on raw values.

Figure A A.2. Alternative normalisation methods, case 1

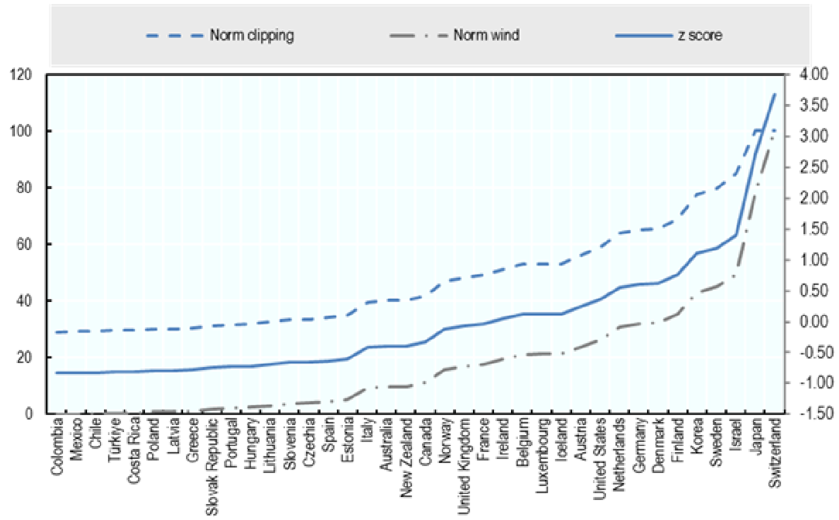
Distribution of normalised scores for the “Product Market Regulation” indicator, using three methods



Note : Norm clipping corresponds to a min-max transformation of the moving average data setting the minimum score to the sample mean minus two times the standard deviation and the maximum score to the sample mean plus two times the standard deviation; Norm wins corresponds to a min-max transformation of the moving average data setting the minimum score to the sample’s fifth percentile and the maximum score to the sample’ 95th percentile. Z score corresponds to the normalisation of the moving average data (value minus sample mean divided by the standard deviation).

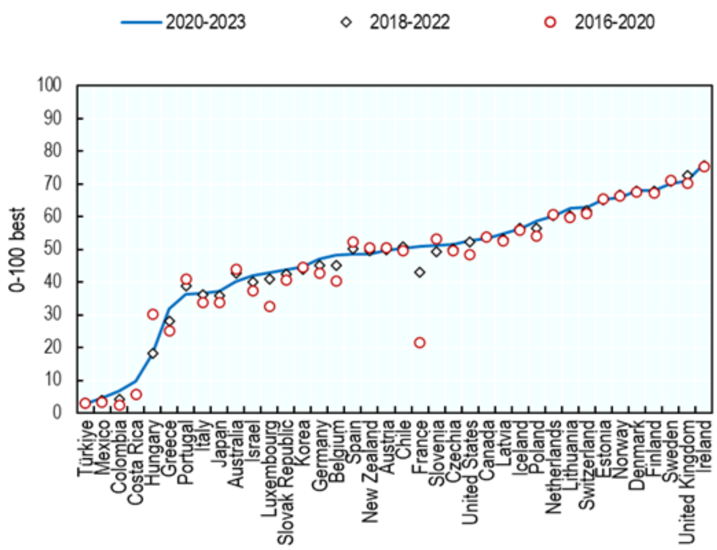
Figure A A.3. Alternative normalisation methods, case 2

Distribution of normalised scores for the “Patents per capita” indicator, using three methods



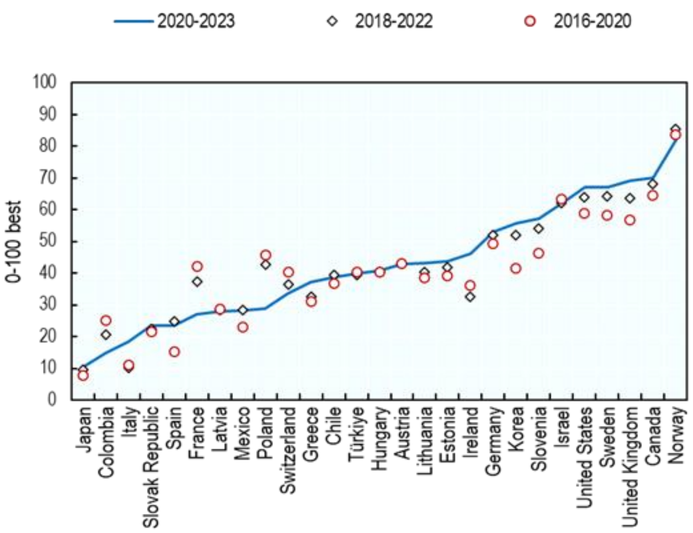
Note : Norm clipping corresponds to a min-max transformation of the moving average data setting the minimum score to the sample mean minus two times the standard deviation and the maximum score to the sample mean plus two times the standard deviation; Norm wins corresponds to a min-max transformation of the moving average data setting the minimum score to the sample’s fifth percentile and the maximum score to the sample’ 95th percentile. Z score corresponds to the normalisation of the moving average data (value minus sample mean divided by the standard deviation).

Figure A A.4. Institutions element’s scores over time by country



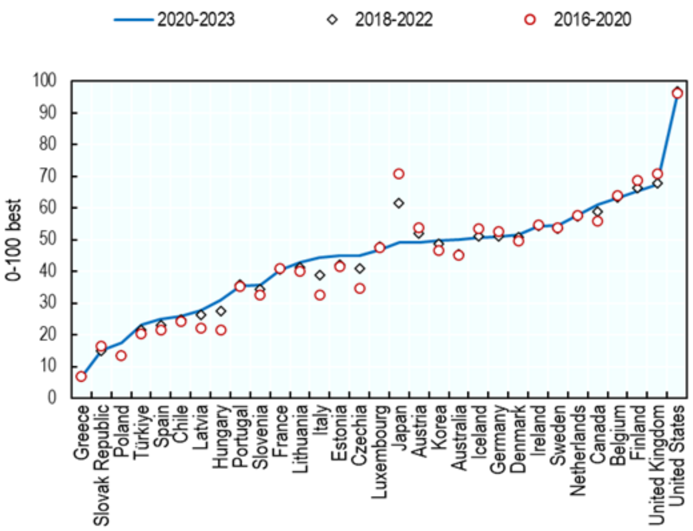
Note: These scores are computed as the geometric mean for countries across the following indicators: i). Rule of law (crime, enforcement of contract, legal process/courts transparency, speed of judicial processes, risk of expropriation of foreign assets, intellecatual property protection, private property rights, source: World Bank – Worldwide Governance Indicators; ii.) Control of corruption index, Source: Vi-dem Project; iii.) Effective tax rate, % taxable income, Source: OECD - Corporate Tax Statistics Databas; iv. Product Market Regulation Index, Source: OECD. Before aggregation, data are normalised using a min-max transformations where the max/min are equalt to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023’s data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.5. Culture element's scores over time by country



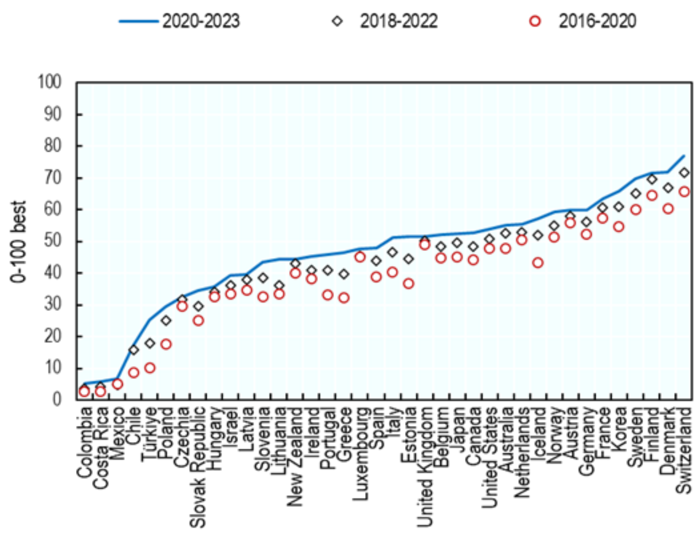
Note: These scores are computed as the geometric mean for countries across the following indicators: i). Percentage of 18-64 population who consider starting a business as a desirable career choice, Source: Global Entrepreneurship Monitor (GEM); ii.) Percentage of 18-64 population who agree that successful entrepreneurs receive high status, Source: Global Entrepreneurship Monitor (GEM); iii) Share of people who believe that most people can be trusted, Source: World Value Survey (WVS). Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.6. Networks element's scores over time by country



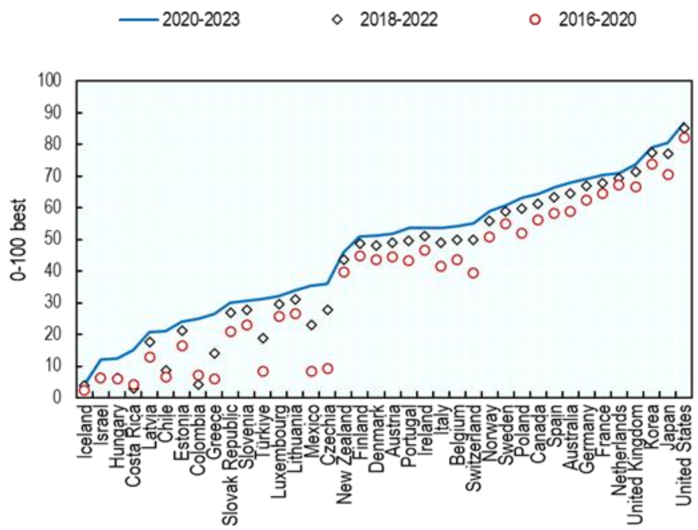
Note: These scores are computed as the geometric mean for countries across the following indicators: i.) Share of SMEs with innovation cooperation activities, Source: European Commission - European innovation scoreboard; ii.) Extent to which businesses collaborate with universities on R&D, Source: World Economic Forum, Executive Opinion Survey. Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.7. Infrastructure element's scores over time by country



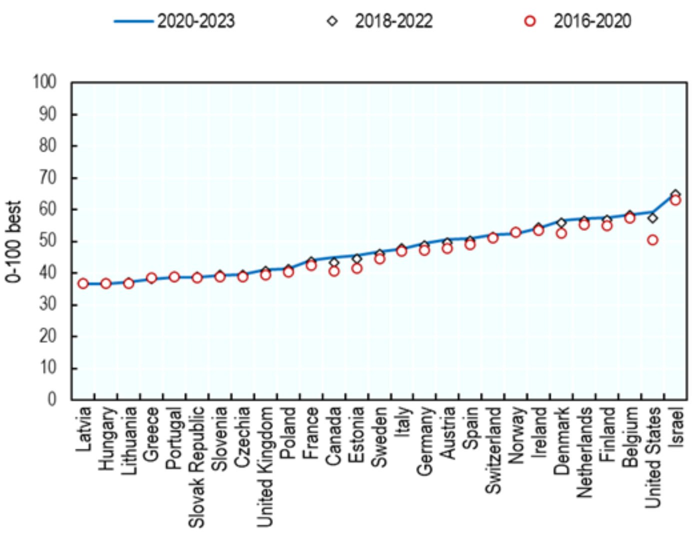
Note: These scores are computed as the geometric mean for countries across the following indicators: i.) Fix broadband, subscriptions per 100 population, Source: OECD - Telecommunications database, iii.) Mobile data use, Gb/per subscription/month, Source: OECD – Broadband and telecommunications database; iv.) Transport infrastructure quality index, Source: World Bank - Logistic Performance Index (LPI). Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.8. Markets element's scores over time by country



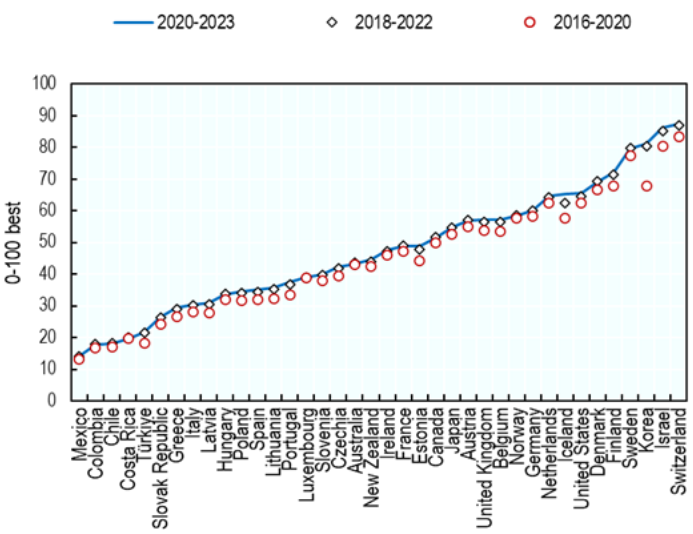
Note: These scores are computed as the geometric mean for countries across the following indicators: i.) GDP, PPP, Source: OECD, ii.) Total population, Source: OECD - Historical Population Data, ii.) Trade facilitation index, Source: OECD - Trade Facilitation Indicators. GDP data are transformed to logarithms before normalisation. Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.9. Finance element's scores over time by country



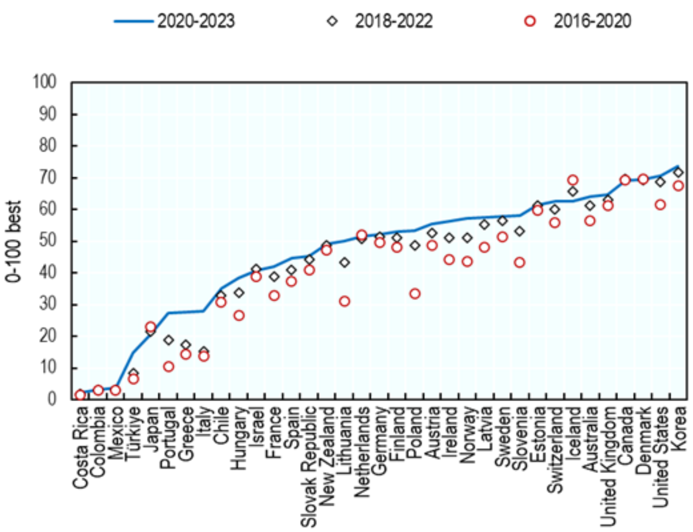
Note: These scores are computed as the geometric mean for countries across the following indicators: i. Seed VC investment per thousand pop. Source: OECD Entrepreneurship Financing Database; ii. Growth start-up VC investments per thousand pop., Source: OECD Entrepreneurship Financing Database; Factoring, thousands USD per capita; iii. Outstanding loans to SMEs, thousands USD per capita, Source: OECD Financing SMEs and Entrepreneurs: An OECD Scoreboard, iv. Factoring value, USD per capita, Source: OECD Financing SMEs and Entrepreneurs: An OECD Scoreboard. Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.10. Knowledge element's scores over time by country



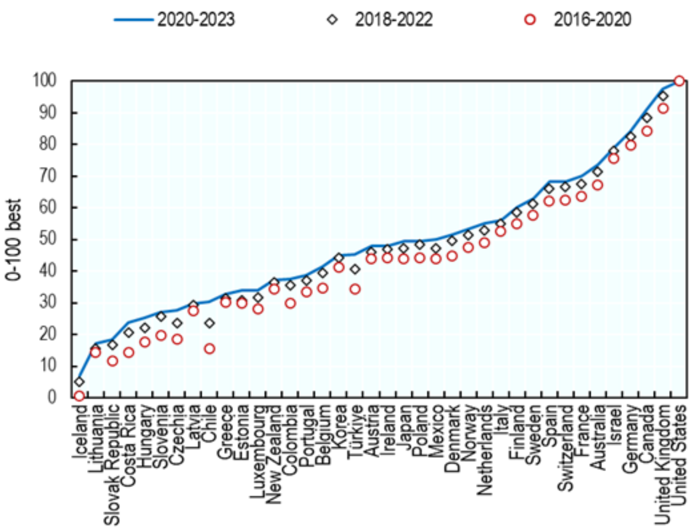
Note: These scores are computed as the geometric mean for countries across the following indicators: i.) Patents, per million population, Source: OECD - Main Science and Technology Indicators, ii.) R&D expenditure, % GDP, Source: OECD - Main Science and Technology Indicators; iii.) GitHub software uploads, per thousand people, Source: GitHub. Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.11. Talent element's scores over time by country



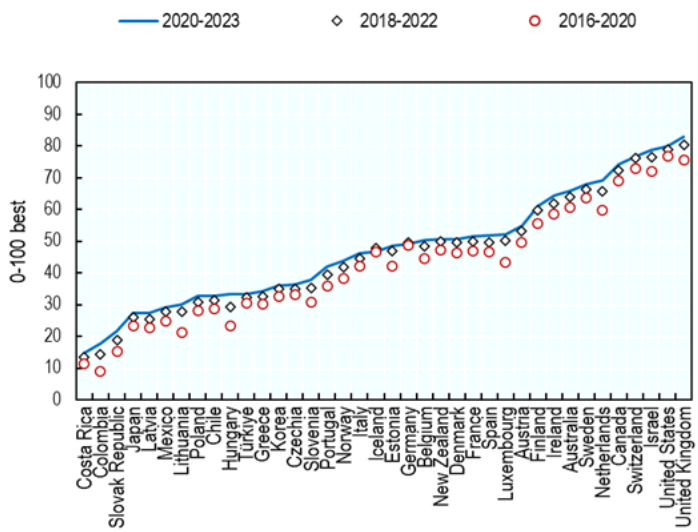
Note: These scores are computed as the geometric mean for countries across the following indicators: i.) Mean years of schooling, Source: UNESCO, ii.) Average of Math, Reading, and Science Pisa scores, Source: OECD; iii.) Percentage of 18-64 population who believe they have the required skills and knowledge to start a business, Source: Global Entrepreneurship Monitor (GEM); iv) Internet users, % of the population, Source: World Bank, World Development Indicators. Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.12. Leadership element's scores over time by country



Note: These scores correspond to the normalised values of the number of serial entrepreneurs total count. Source: Crunchbase and OECD. Values are transformed to logarithms before normalisation. Data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

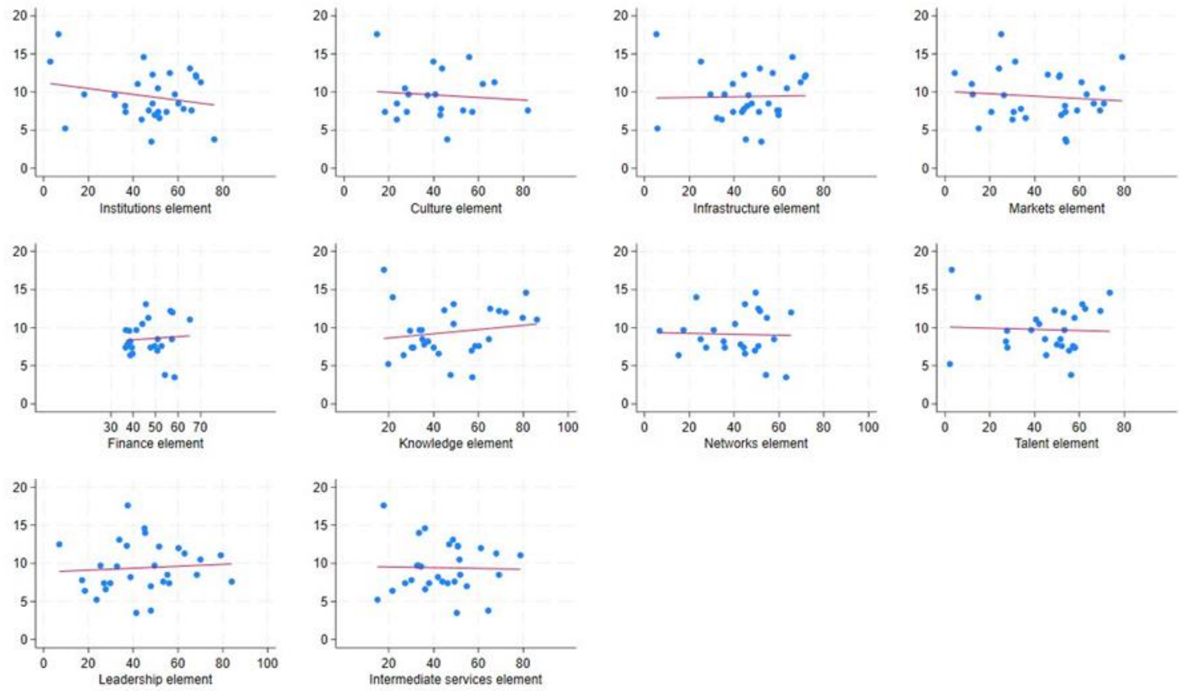
Figure A A.13. Intermediate Services element's scores over time by country



Note: These scores are computed as the geometric mean for countries across the following indicators: i.) Number of incubators, accelerators and start-up support programmes per capita, Source: Crunchbase and OECD, ii.) Technical employment, % total employment, Source: OECD; iii. Coaches and mentors total count. Source: Crunchbase and OECD. Coaches and mentors values are transformed to logarithms before normalisation. Before aggregation, data are normalised using a min-max transformations where the max/min are equal to the sample mean +/- 2*sample standards deviations, relative to the average of data from the 2020-2023 period. 2016-2020 and 2018-2022 scores are anchored to the 2020-2023's data and must be interpreted as relative performance the 2020-2023 period.

Figure A A.14. Elements (inputs) correlations with enterprise birth rate (output)

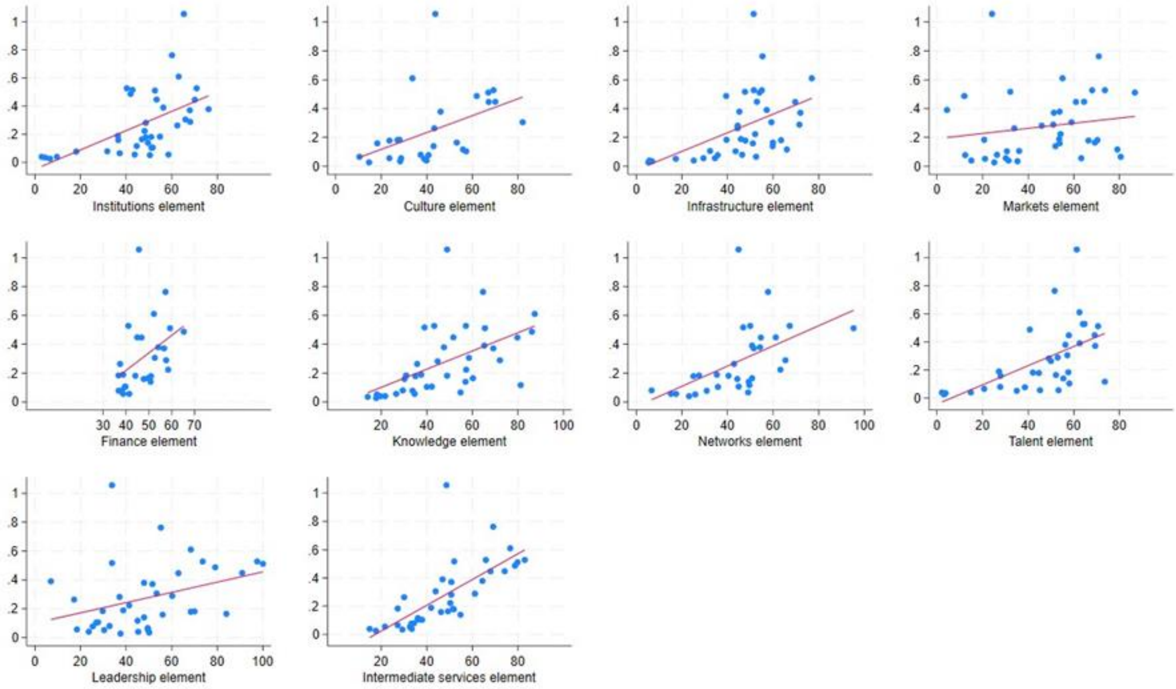
Two-ways correlations of elements scores (geometric mean aggregation, horizontal axis) with birth rate of employer enterprises (vertical axis)



Note: The enterprise birth rate is the number of new firm creation among employer firms (i.e. with at least one employee), as a proportion of active business population.
Source: OECD

Figure A A.15. Elements (inputs) correlations with equity-based young firms per capita (output)

Two-ways correlations of elements scores (geometric mean aggregation – horizontal axis) with equity-based young firm per capita (vertical axis)



Note: Equity-based start-up enteprises per capita are the number of companies newly added to the Crunchbase dataset during the previous 5 years devided by the population.
Source: OECD and Crunchbase

Sign up for our Newsletter

www.oecd.org/cfe/

Follow us on

OECD_Local | oecd-local

COGITO

oecdcogito.blog/

OECD